Call: HORIZON-CL6-2021-CIRCBIO-01
(Circular economy and bioeconomy sectors)

Topic: HORIZON-CL6-2021-CIRCBIO-01-04

Type of Action: HORIZON-IA

Proposal number: 101059639

Proposal acronym: WHITECYCLE

Type of Model Grant Agreement: HORIZON Action Grant Budget-Based

Table of contents

Section	Title	Action
1	General information	
2	Participants	
3	Budget	
4	Ethics and security	
5	Other questions	

Proposal ID **101059639**

Acronym **WHITECYCLE**

1 - General information

Fields marked * are mandatory to fill.

Topic	HORIZON-CL6-2021-CIRCBIO-01-04	Type of Action	HORIZON-IA
Call	HORIZON-CL6-2021-CIRCBIO-01	Type of Model Grant Agreement	HORIZON-AG

Acronym	WHITECYCLE
Proposal title	UPSCALING OF INNOVATIVE PROCESSES FOR THE RECYCLING OF PET FROM COMPLEX WASTES: A CASE STUDY INVOLVING MECHANICAL AND ENZYMATIC SORTING APPLIED TO TYRES, TEXTILES AND HOSES

Note that for technical reasons, the following characters are not accepted in the Proposal Title and will be removed: < > " &

Duration in months	48
Fixed keyword 1	Circular economy
Fixed keyword 2	Waste recycling
Fixed keyword 3	Polymers and plastics
Fixed keyword 4	Textiles
Fixed keyword 5	Consumer products and services
Free keywords	*Enzymatic depolymerization, Multicomponent clothing textile waste, End-of-life tyres, Polyethylene terephthalate PET, Lay-flat hoses, Technical garments, smart sorting, life cycle analysis, FAIR data*

Abstract *

PET is widely used in plastics and textiles leading to over 20 Mt/y of complex waste worldwide for which no closed-loop recycling is viable today. Most complex waste is landfilled or incinerated. There is an urgent need to develop a circular solution to convert complex PET wastes from plastics and textiles back into high-added value products.

WHITECYCLE unites 3 brand-owners, 1 PET converter, 2 waste managers, 1 digital deeptech for smart sorting, the world leading enzymatic recycling SME, 1 LCA company, 3 UNIs/RTOs, 1 cluster and 1 management firm. They aim to demonstrate two new processes combining strong scientific and industrial know-how: (i) innovative identification, sorting and separation technologies that will dramatically increase the PET content of complex waste streams to 80%, and (ii) a disruptive enzymatic recycling process that is expected to yield pure PET monomers sustainably even for impure waste streams. PET monomers obtained will be repolymerised and recycled. Thus, 2 t of waste will be used to demonstrate 3 highly technical PET-containing pilot series: 100 tyres, 1,500 m of lay-flat hoses and 400m2 of multicomponent fabric that will be coated and used to manufacture 4 lines of technical garments. Process design kits, LCA and production cost estimates will be provided to PET manufacturers and waste management companies for rapid deployment and assure social acceptance. First projections show that WHITECYCLE's recycled PET will be competitive with virgin PET.

The project will conduct a full circle loop from real complex waste feedstock to representative product of the 3 use-cases at TRL 5. Then, a strong upscale study will allow the process steps to reach TRL 6 to 8. By 2030, WHITECYCLE will enable the annual recycling of more than 2 Mt of PET, which corresponds to the amount of additional recycled PET needed to meet the EU's 2030 targets. This will reduce emissions by around 2.06 Mt CO2eq and avoid the landfilling of more than 1.8 Mt of PET.

Remaining characters 1

Administrative forms

Proposal ID **101059639**

Acronym **WHITECYCLE**

Has this proposal (or a very similar one) been submitted in the past 2 years in response to a call for proposals under any EU programme, including the current call? (•) Yes ( ) No

Please give the proposal reference or contract number.
101003701

Administrative forms

Proposal ID **101059639**

Acronym **WHITECYCLE**

Declarations

Field(s) marked * are mandatory to fill.

1) We declare to have the explicit consent of all applicants on their participation and on the content of this proposal. * ☒

2) We confirm that the information contained in this proposal is correct and complete and that none of the project activities have started before the proposal was submitted (unless explicitly authorised in the call conditions). ☒

3) We declare:
- to be fully compliant with the eligibility criteria set out in the call
- not to be subject to any exclusion grounds under the EU Financial Regulation 2018/1046
- to have the financial and operational capacity to carry out the proposed project.

☒

4) We acknowledge that all communication will be made through the Funding & Tenders Portal electronic exchange system and that access and use of this system is subject to the Funding & Tenders Portal Terms and Conditions. ☒

5) We have read, understood and accepted the Funding & Tenders Portal Terms & Conditions and Privacy Statement that set out the conditions of use of the Portal and the scope, purposes, retention periods, etc. for the processing of personal data of all data subjects whose data we communicate for the purpose of the application, evaluation, award and subsequent management of our grant, prizes and contracts (including financial transactions and audits). ☒

6) We declare that the proposal complies with ethical principles (including the highest standards of research integrity as set out in the ALLEA European Code of Conduct for Research Integrity, as well as applicable international and national law, including the Charter of Fundamental Rights of the European Union and the European Convention on Human Rights and its Supplementary Protocols. Appropriate procedures, policies and structures are in place to foster responsible research practices, to prevent questionable research practices and research misconduct, and to handle allegations of breaches of the principles and standards in the Code of Conduct. ☒

7) We declare that the proposal has an exclusive focus on civil applications (activities intended to be used in military application or aiming to serve military purposes cannot be funded). If the project involves dual-use items in the sense of Regulation 428/2009, or other items for which authorisation is required, we confirm that we will comply with the applicable regulatory framework (e.g. obtain export/import licences before these items are used). ☒

8) We confirm that the activities proposed do not
- aim at human cloning for reproductive purposes;
- intend to modify the genetic heritage of human beings which could make such changes heritable (with the exception of research relating to cancer treatment of the gonads, which may be financed), or
- intend to create human embryos solely for the purpose of research or for the purpose of stem cell procurement, including by means of somatic cell nuclear transfer.
- lead to the destruction of human embryos (for example, for obtaining stem cells)

These activities are excluded from funding. ☒

9) We confirm that for activities carried out outside the Union, the same activities would have been allowed in at least one EU Member State. ☒

The coordinator is only responsible for the information relating to their own organisation. Each applicant remains responsible for the information declared for their organisation. If the proposal is retained for EU funding, they will all be required to sign a declaration of honour.

False statements or incorrect information may lead to administrative sanctions under the EU Financial Regulation.

2 - Participants

List of participating organisations

#	Participating Organisation Legal Name	Country	Action
1	MANUFACTURE FRANCAISE DES PNEUMATIQUES MICHELIN	FR	
2	MANDALS AS	NO	
3	CARBIOS	FR	
4	KORDSA TEKNIK TEKSTIL ANONIM SIRKETI	TR	
5	ESTATO UMWELTSERVICE GMBH	DE	
6	TLC Synergies	France	
7	DEUTSCHE INSTITUTE FUR TEXTIL- UND FASERFORSCHUNG DENKENDORF	DE	
8	INSTITUT FRANCAIS DU TEXTILE ET DE L'HABILLEMENT	FR	
9	AXELERA - ASSOCIATION CHIMIE-ENVIRONNEMENT LYON ET RHONE-ALPES	FR	
10	CENTRE NATIONAL DE LA RECHERCHE SCIENTIFIQUE CNRS	FR	
11	UNIVERSITE DE POITIERS	FR	
12	ERASME (Centre D'Excellence Jean Monnet sur le Développement Durable)	FR	
13	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	ES	
14	HOGSKULEN PA VESTLANDET	NO	
15	IPOINT-SYSTEMS GMBH	DE	
16	DYNERGIE	FR	
17	INDUSTRIA DE DISENO TEXTIL SA	ES	

Organisation data

PIC	Legal name
971320462	*MANUFACTURE FRANCAISE DES PNEUMATIQUES MICHELIN*

Short name: MICHELIN

Address

Street	PLACE DES CARMES DECHAUX 23
Town	CLERMONT FERRAND
Postcode	63040
Country	France
Webpage	www.michelin.fr

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	20/04/1955	- no
SME self-assessment	unknown	
SME validation	20/04/1955	- no

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant
Controls	MANDALS AS

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Christophe**

Last name* **Le-clerc**

E-Mail* **christophe.le-clerc@michelin.com**

Position in org. Senior Fellow

Department Advanced Research

☐ Same as organisation name

☒ Same as proposing organisation's address

Street PLACE DES CARMES DECHAUX 23

Town CLERMONT FERRAND

Post code 63040

Country France

Website www.michelin.com

Phone +33 6 74 62 92 02

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Jean-Michel	Fauvin	jean-michel.fauvin@michelin.com	+33679392645
Nizar	Didane	nizar.didane@michelin.com	+ 33611037881
Thomas	Simonelli	thomas.simonelli@michelin.com	+ 33 663215102
Lise	Maisonneuve	lise.maisonneuve@michelin.com	+ 33 617558972
Gorka	Gorazureta	ggorazureta@dynergie.eu	+xxx xxxxxxxxx
Mathieu	Raoutier	mraoutier@dynergie.eu	+xxx xxxxxxxxx
Michael	Cogne	michael.cogne@michelin.com	+33 685428423
Francois	Bataille	francois.bataille@michelin.com	+33673131756
Colin-Yann	Jacquin	colin-yann.jacquin@michelin.com	+33634427329

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Christophe	Leclerc	Man	France	christophe.le-clerc@michelin.com	Category B Senior re	Leading		Orcid ID
Dr	Lise	Maisonneuve	Woman	France	lise.maisonneuve@michelin.com	Category C Recognis	Leading		Orcid ID
Dr	Nizar	Didane	Man	France	nizar.didane@michelin.com	Category C Recognis	Leading		Orcid ID
Mr	Thomas	Simonelli	Man	France	thomas.simonelli@michelin.com	Category C Recognis	Team member		Orcid ID
Mr	Florian	Buisson	Man	France	florian.buisson@michelin.com	Category C Recognis	Team member		Orcid ID
Dr	Francois	Bataille	Man	France	francois.bataille@michelin.com	Category B Senior re	Team member		Orcid ID

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☒
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*A bi-dimensional assessment to measure the performance of circular economy: a case study of tires end-of-life management (https://link.springer.com/chapter/10.1007/978-3-319-66981-6_4)* *Haulage truck with 32 tons' average load which usual lifetime is 600000 km.* *Data come from the Research Center of Michelin in Clermont-Ferrand (France) and the EcoInvent 2.2 database. Hereafter is some relevant features to interpret the results. 63.5% of end-of-life tires*
Dataset	*[B] Handbook of properties of textile and technical fibres (AR Bunsell - 2018 - books.google.com)*
Publication	*Cobalt-iron-manganese catalysts for the conversion of end-of-life-tire- derived syngas into light terminal olefins (https://pubmed.ncbi.nlm.nih.gov/29493817/)* *Abstract Co-Fe-Mn/?-Al2O3 Fischer?Tropsch synthesis (FTS) catalysts were synthesized, characterized and tested for CO hydrogenation, mimicking end- of-life-tire (ELT)-derived syngas. JP Falkenhagen, L Maisonneuve? - ? A European Journal, 2018 - Wiley Online Library*
Other achievement	*Chez Michelin, une recherche orient?e par la mobilit? durable* *Annales des Mines-Responsabilite et environnement, 68-71, 2016/3* *Terry Gettys (https://www.cairn.info/revue-responsabilite-et-environnement-2016-3-page-68.htm)*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
BlackCycle Project - H2020-SC5-2019-2	*Creation of a massive circular economy of the ELT in technical applications such as the Tire industry, by producing high-technical raw materials from ELT and pyrolysis by-products.*
BioButterFly Project	*From 2012-in progress: Development of renewable Butadiene avoiding fossil consumption. This project is funded by the French State*
EMPREINTE Project	*Starting September 2020 to 2025. This project aims to reduce environmental footprint of our products by maximizing the ratio of bio-sourced and recycled materials (polymers, oils, fillers, textile, steel). This project is funded by the French State*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Tyres design software	*Creation Software and database for modeling behavior and manufcaturinf tyres*
Testing and characterization raw Materials laborat	*Physical and Chemical Labotories*
Manufacturing and testing tyres facilities	*More than 60 Tyres Manufacturng factories, and 3 tyres testing Shops*
Lehigh Activities	*Converts End-of-Life Tyre into micronized rubber powders, and Retread Tyres. (Michelin Subsidiary)*
Mandals Holes	*Resarch and Manufacturing Holes for Water (Michelin Subsidiary)*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
998763314	*MANDALS AS*

Short name: MANDALS

Address

Street	NORDRE BANEGATE 26
Town	MANDAL
Postcode	4515
Country	Norway
Webpage	-www.mandals.com

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	06/08/2008 - yes
SME self-assessment	unknown
SME validation	06/08/2008 - yes

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant
Is controlled by	MANUFACTURE FRANCAISE DES PNEUMATIQUES MICHELIN

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☒
Public procurer of results	☒
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Nordgård-Hansen, E., Schlanbusch, R., & Sørensen, T. (2017). Modeling of a lay-flat plastic hose extrusion. Modeling, Identification and Control, Vol 31, No. 1, pp 1 -11, ISSN 1890-1328.*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
917994906	*CARBIOS*

Short name: CARBIOS

Address

Street	3 rue Emile Duclaux
Town	SAINT-BEAUZIRE
Postcode	63360
Country	France
Webpage	www.carbios.fr

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	02/09/2019	- yes
SME self-assessment	31/12/2018	- yes
SME validation	unknown	

Departments carrying out the proposed work

Department 1

Department name	Innovation	☐ not applicable
	☐ Same as proposing organisation's address	
Street	8 rue de la groli?re	
Town	Clermont-Ferrand	
Postcode	63100	
Country	France	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title: Dr

Gender: (•) Woman ( ) Man ( ) Non Binary

First name*: **Marion**

Last name*: **Beguerie**

E-Mail*: **marion.beguerie@carbios.com**

Position in org.: Innovation funding manager

Department: Innovation

☐ Same as organisation name

☐ Same as proposing organisation's address

Street: 11 rue patrick depailler

Town: Clermont-Ferrand

Post code: 63000

Country: France

Website: https://www.carbios.com/en/

Phone: +33473865176

Phone 2: *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Frederique	Guillamot	frederique.guillamot@carbios.fr	+33473865176

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Marion	Beguerie	Woman	France	marion.beguerie@carbios.com	Category B Senior le	Leading		
Mrs	Frederique	Guillamot	Woman	France	frederique.guillamot@carbios.com	Category A Top grade	Team member		

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☒
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*An engineered PET depolymerase to break down and recycle plastic bottles* *Nature, 2020 Apr;580(7802):216-219* *Here Carbios describe an improved PET hydrolase that ultimately achieves, over 10 hours, a minimum of 90 per cent PET depolymerization into monomers, with a productivity of 16.7 grams of terephthalate per litre per hour (200 grams per kilogram of PET suspension, with an enzyme concentration of 3 milligrams per gram of PET).*
Other achievement	*Press release 23_04_2021 - Carbios and Michelin take a major step towards developing 100% sustainable tires* *Michelin has successfully validated the use of Carbios? enzymatic recycling technology for PET[1] plastic waste in its tires*
Other achievement	*Press release 24_06_2021 - Global Consumer Brands Unveil World?s First Enzymatically Recycled Bottle* *Each Consortium company has successfully manufactured sample bottles ? based on Carbios? enzymatic PET recycling technology ? for some of their leading products including: Biotherm?, Perrier?, Pepsi Max?** and Orangina?*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
ThanaplastTM	*Identification, optimisation and demonstration at laboratory scale of the activity of the enzymes towards depolymerisation of high-content PET plastic waste: 90% of depolymerisation of PET in 48h in water (2012-2017)*
C.E. PET ADEME	*Extension at laboratory scale of the activity of the enzymes on high-content PET textile waste. Demonstration at pilot scale of depolymerisation and monomer purification of both high-content PET plastic and textile waste. (2018-2021)*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
R&D laboratory	*Carbios' facilities include an R&D laboratory at which depolymerisation at lab scale (TRL 4) are performed (50 g of PET waste in a 500 mL reactor; 1 kg of PET waste in a 10 L reactor). The laboratory is equiped with analytical equipement.*
Pilot unit	*A pilot unit, running since 2018, allows the depolymerization and the development of purification scheme for monomers at TRL 5-6 (100 kg of PET waste in a 1 m3 reactor + filtration A, ultrafiltration, discoloration, precipitation, filtration B, crystallisation and drying process units).*
Pre-industrial unit	*Carbios is building a pre-industrial unit (TRL 7) that will be operational in Q3 2021 comprising equipment for:* *- pretreatment of high content PET waste by extrusion/expansion* *- depolymerization in a 20 m3 reactor (2t/week of PET waste)* *- monomers purification through 18 process units*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
929405113	*KORDSA TEKNIK TEKSTIL ANONIM SIRKETI*

Short name: KORDSA

Address

Street	ALIKAHYA FATIH MAH SANAYICI CAD 90 IZMIT
Town	KOCAELI
Postcode	41310
Country	Turkey
Webpage	www.kordsaglobal.com/

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is yes (small- and medium-sized enterprise) for the call.

SME self-declared status	unknown
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Ms

Gender ◉ Woman ○ Man ○ Non Binary

First name* **Nazan**

Last name* **Coskun**

E-Mail* **nazan.coskun@kordsa.com**

Position in org. Project Specialist

Department Global Technology - Project Management

☐ Same as organisation name

☒ Same as proposing organisation's address

Street ALIKAHYA FATIH MAH SANAYICI CAD 90 IZMIT

Town KOCAELI

Post code 41310

Country Turkey

Website https://kordsa.com/

Phone +90 531 718 5280

Phone 2 +90 262 316 77 28

Other contact persons

First Name	Last Name	E-mail	Phone
Alper	Gul	alper.gul@kordsa.com	+90 5557237288
Cansu	Yalcinkaya	cansu.bozdag@kordsa.com	+xxx xxxxxxxxx
Sule	Askin	sule.askin@kordsa.com	+90 5317311679
Burak	Ilgun	burak.ilgun@kordsa.com	+90 262 3167273
Erhan	Kop	erhan.kop@kordsa.com	+90 532 7721218

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Mr	Alper	Gul	Man	Turkey	alper.gul@kordsa.com	Category A Top grade	Leading		Researcher ID
Mr	Erhan	Kop	Man	Turkey	erhan.kop@kordsa.com	Category A Top grade	Team member		Researcher ID
Dr	Gökçe	Uğur Erinç	Woman	Turkey	gokce.ugur@kordsa.com	Category A Top grade	Team member		Researcher ID
Mrs	Emine	Güven	Woman	Turkey	emine.guven@kordsa.com	Category A Top grade	Team member		Researcher ID
Dr	Yasin	Şen	Man	Turkey	yasin.sen@kordsa.com	Category A Top grade	Team member		Researcher ID
Mrs	Zekiye	Baykara	Woman	Turkey	zekiye.baykara@kordsa.com	Category A Top grade	Team member		Researcher ID
Mr	Bahadır	Kaya	Woman	Turkey	bahadır.kay@kordsa.com	Category A Top grade	Team member		Researcher ID
Mrs	Ekin	Ersoy	Woman	Turkey	ekin.ersoy@kordsa.com	Category A Top grade	Team member		Researcher ID
Ms	Rüya	Keskinkaya	Woman	Turkey	ruya.keskinkaya@kordsa.com	Category A Top grade	Team member		Researcher ID

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☐
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☒
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*New continuous polymerization line and 4 end polyamide 6.6 yarn production line design 2012.*
Publication	*Optimization of thermal stabilizer in polyamide 6.6 condensation polymerization 2012.*
Publication	*Synthesis of aliphatic polyketon polymer and production of high strength yarn by wet spinning method. 2009.*
Publication	*Development of polyamide 6.6 yarn for airbag industry 2013.*
Publication	*Optimized thermal stabilizer amount is used in our current fiber product in Line-1 (Turkey.)*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
PolynSPIRE	*The main objective is to demonstrate a set of innovative, cost-effective and sustainable solutions, aiming at improving the energy and resource efficiency of post-consumer and post-industrial plastic recycling processes, targeting 100% waste streams containing at least80% of plastic materials. To this end, three innovation pillars are addressed at TRL7: A) Chemical recycling assisted by microwaves and smart magnetic catalysts as a path to recover plastic monomers and valuable fillers.*
Cokoon	*Kordsa and Continental have together developed a new sustainable adhesive technology standard for bonding textile reinforcing materials with rubber compounds. Reinforcing materials of this kind are used in the tire industry, as well as in the production of mechanical rubber goods like hoses and conveyor belts. The new technology enables the bonding activation of textile reinforcing materials without the use of resorcinol and formaldehyde.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Differential Scanning Calorimeter (DSC)	
Dynamic Mechanical Analyzer (DMA)	
Optical microscopes	
Fourier Transform Infrared Spectrometer (FTIR)	
Gel Permeation Chromatography (GPC)	
X-Ray Diffractometer (XRD)	
Instron mechanical test equipments	
Rheometers	
Viscometers	

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
899011812	*ESTATO UMWELTSERVICE GMBH*

Short name: ESTATO

Address

Street	DR KILIAN STRASSE 11
Town	WEIDEN
Postcode	92637
Country	Germany
Webpage	www.estato-umweltservice.de

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is yes (small- and medium-sized enterprise) for the call.

SME self-declared status	unknown
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

Department 1

Department name: Recycling

☐ not applicable

☒ Same as proposing organisation's address

Street: DR KILIAN STRASSE 11

Town: WEIDEN

Postcode: 92637

Country: Germany

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Alexander**

Last name* **Prokein**

E-Mail* **alexander.prokein@estato-umweltservice.de**

Position in org. Managing Director

Department Recycling

☐ Same as organisation name

☒ Same as proposing organisation's address

Street DR KILIAN STRASSE 11

Town WEIDEN

Post code 92637

Country Germany

Website www.estato-umweltservice.de

Phone +49 1622701409

Phone 2 +49 9613065227

Other contact persons

First Name	Last Name	E-mail	Phone
Mike	Bubolz	mike.bubolz@estato-umweltservice.de	+49 162 270 5006
Heiko	Arlt	heiko.arlt@estato-umweltservice.de	+49 162 270 5007
Alois	Treml	alois.treml@estato-umweltservice.de	+49 961 306 5940

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Service	*Business Model - Sustainable development* *Waste management, treatment and recovery for nearly all material streams. Experiences in tire collection and selection. Industrial recycling processes for ELT (current capacity 60.000 tons per year). Provision of environmental and energy consulting*
Good	*Production - Output of Tire granulation process* *Different kinds of rubber granulates or powders, Textile fibre fraction from tire recycling and Recycling technology for waste tires or rubber materials*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
Implementing granulation process	*Development, installation and technical modification (process improvements on industrial scale) of three granulation plants since 2001*
Research project on technical use for fibres / tex	*Joint ventures and different research / development activities for an experimental technical solution for textile fibre quality improvement or transformation ? technical and market analysis, potencial industrial design development reports, project plans and business cases, but no final realisation*
BlackCycle H2020	*Participation at this european project to close the loop from ELT to new tires and applications (Grant Agreement Number 869625, ?Horizon 2020 ? the Framework Programme for Research and Innovation (2014-2020)?* *Leader of WP 4: Tyre Selection*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
supply chain / sorting line	*Warehouse to store, collect and select tires including half-automated sorting lines.*
available space	*Install / integrate sorting technologies for textile at the site (prototype)*
feedstock selection	*Existing tire collection and mechanical recycling processes, know-how of selection and raw material quality / supply as well as feedstock capacity*
tire granulation lines / production machines	*Three tire granulation plants (6 fine granulation lines in 24-hours-operation) to produce required textile fibre from ELT and capacity (70.000 tons per year)*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
895942441	*TLC Synergies*

Short name: SYNERGIES TLC

Address

Street	Zone artisanale de Terre Neuve
Town	Gilly sur Isère
Postcode	73276
Country	France
Webpage	

Specific Legal Statuses

Legal person	yes
Public body	unknown
Non-profit	unknown
International organisation	unknown
Secondary or Higher education establishment	unknown
Research organisation	unknown

SME Data

Based on the below details from the Participant Registry the organisation is unknown (small- and medium-sized enterprise) for the call.

SME self-declared status	unknown
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mrs

Gender ◉ Woman ○ Man ○ Non Binary

First name* **Clara**

Last name* **Potton**

E-Mail* **c.potton@synergies-tlc.com**

Position in org. Recycling project manager

Department TLC Synergies

☒ Same as organisation name

☒ Same as proposing organisation's address

Street Zone artisanale de Terre Neuve

Town Gilly sur Isère

Post code 73276

Country France

Website www.synergies-tlc.com

Phone +337 50371495

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Paul	Dumesny	p.dumesny@synergies-tlc.com	+33761889412
Etienne	Wiroth	e.wiroth@synergies-tlc.com	+33620875452

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
REWIND	*Develop a recycled cotton yarn for the sport industry*
FRIVEP	*Organise the collection, sorting and valorisation of professional clothing in France*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
998753711	*DEUTSCHE INSTITUTE FUR TEXTIL- UND FASERFORSCHUNG DENKENDORF*

Short name: DITF

Address

Street	KORSCHTALSTRASSE 26
Town	DENKENDORF
Postcode	73770
Country	Germany
Webpage	www.ditf.de

Specific Legal Statuses

Legal person	yes
Public body	yes
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	10/09/2008	- no
SME self-assessment	unknown	
SME validation	10/09/2008	- no

Departments carrying out the proposed work

Department 1

Department name	Competence Center Polymers & Composites	☐ not applicable
	☒ Same as proposing organisation's address	
Street	KORSCHTALSTRASSE 26	
Town	DENKENDORF	
Postcode	73770	
Country	Germany	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title: Dr

Gender: ○ Woman ◉ Man ○ Non Binary

First name*: **Mark**

Last name*: **Steinmann**

E-Mail*: **mark.steinmann@ditf.de**

Position in org.: Head of Competence Center Polymers & Composites

Department: Competence Center Polymers & Composites

☐ Same as organisation name

☒ Same as proposing organisation's address

Street: KORSCHTALSTRASSE 26

Town: DENKENDORF

Post code: 73770

Country: Germany

Website: www.ditf.de

Phone: +497119340274

Phone 2: *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Hagen	Altmann	hagen.altmann@ditf.de	+497119340396

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Mark	Steinmann	Man	Germany	mark.steinmann@ditf.de	Category A Top grade	Leading		
Mr	Hagen	Altmann	Man	Germany	hagen.altmann@ditf.de	Category A Top grade	Leading		

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Poly(ethylene furanoate) along Its Life-Cycle from a Polycondensation Approach to High-Performance Yarn and Its Recyclate* *doi.org/10.3390/ma13214826* *Development of polyethylene furanoate (PEF) as a potential substitute for PET in fibers for textile and technical applications.*
Publication	*New acyllactam compounds useful to increase the molecular weight of polycondensation products such as polyester or polyamide compounds* *DE102005037418 (B3)* *New acyllactam compounds useful to increase the molecular weight of polycondensation products such as polyester or polyamide compounds*
Publication	*Untersuchungen zur gezielten Schrumpfeinstellung bei der Herstellung von Synthesefasern aus PET (eng: Investigations into the targeted shrinkage adjustment in the production of synthetic fibres from PET)* *Autorenmanuskript, FIZ-Technik, TEMA-Datenbank 2004* *Investigations into the targeted shrinkage adjustment in the production of synthetic fibres from PET)*
Publication	*Einfluss der Rohstoffbasis und des Katalysatarsystems auf das Eigenschaftsprofil von Polyethylenterephthalat (PET) (eng: Influence of the raw material base and the catalyst system on the property profile of polyethylene terephthalate)* *Autorenmanuskript, Copyright FIZ Technik e.V., 264471, TOGA (07.07.2008)* *Influence of the raw material base and the catalyst system on the property profile of polyethylene terephthalate)*
Publication	*Beeinflussung des thermischen und spannungsinduzierten Kristallisationsverhaltens von PET durch Comonomere (eng: Influence of comonomers on the thermal and stress-induced crystallization behavior of PET)* *Stuttgarter Kunststoff-Kolloquium, 17 (2001), 1/V9, 1-5* *Influence of comonomers on the thermal and stress-induced crystallization behavior of PET)*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
AIF IGF 15154	*Diol-modified polyester types: New application possibilities of diol-modified polyester types in the fibre sector*
AIF IGF 14687 BG	*Influence of the raw material on the profile of PET: Influence of the raw material 1,4-benzenedicarboxylic acid dimethyl ester (terephthalic acid dimethyl ester, DMT) or 1,4-benzenedicarboxylic acid (terephthalic acid, purified terephthalic acid, PTA) and the catalyst system on the property profile of polyethylene terephthalic acid*
AIF IGF 12428 N	*Oligomer content: Influence of the production method on the molar mass distribution and the oligomer contents of polyesters*
AIF IGF 12318	*PET modification: modification of PET to increase productivity in fibre production and to improve processability*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Steel pressure reactors (2 L, 5 L, 20 L, 50 L)	*For scale-up of PET polycondensation, steel pressure reactors in the Polymer lab at DITF in size from 1 L up to 50 L will be used to investigate the conditions of the synthesis for high IVs in the project.*

Tumble dryer (50 L)	*Additionally a Tumble dryer (50 L) will be used in the project to carry out SSP of PET after reactor polycondensation.*
Analytics for polymer development	*Namely IR, NMR, element analysis, end group determination, High and low temperature GPC, viscosimetry, color determination.*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
999599454	*INSTITUT FRANCAIS DU TEXTILE ET DE L'HABILLEMENT*

Short name: IFTH

Address

Street	rue des reculettes 14
Town	PARIS
Postcode	75013
Country	France
Webpage	www.ifth.org

Specific Legal Statuses

Legal person	yes
Public body	yes
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is no (small- and medium-sized enterprise) for the call.

SME self-declared status	unknown
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

Department 1

Department name	Research Direction under Scientific Direction	☐ not applicable
	☐ Same as proposing organisation's address	
Street	42 ter Avenue Guy de Collongue	
Town	Ecully	
Postcode	69100	
Country	France	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title: Dr

Gender: ○ Woman ◉ Man ○ Non Binary

First name*: **Pascal**

Last name*: **Rumeau**

E-Mail*: **prumeau@ifth.org**

Position in org.: Deputy-Head of Research

Department: Scientific Direction

☐ Same as organisation name

☒ Same as proposing organisation's address

Street: rue des reculettes 14

Town: PARIS

Post code: 75013

Country: France

Website: www.ifth.org

Phone: +33472861620

Phone 2: *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Olivier	Certhoux	ocerthoux@ifth.org	+33472861623
Christian	Jouzel	direction@ifth.org	+33144081900

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
	Carole	Aubry	Woman	France	caubry@ifth.org				
	Pascal	Rumeau	Man	France	prumeau@ifth.org				
	Philippe	Mailler	Man	France	pmailler@ifth.org				
	Elisabeth	Govart	Woman	France	egovart@ifth.org				
	Philippe	Vancoetsem	Man	France	pvancoetsem@ifth.org				
	Maxime	Gourdon	Man	France	mgourdon@ifth.org				
	Arnaud	Monnier	Man	France	amonnier@ifth.org				
	Alexa	Revouy	Woman	France	arevouy@ifth.org				
	Jacques	Maguin	Man	France	jmaguin@ifth.org				
	Jean-Philipe	Bourg	Man	France	jpbourg@ifth.org				
	Lahcen	Filali	Man	France	lfilali@ifth.org				
	Isabelle	Kuster	Woman	France	ikuster@ifth.org				

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Polypropylene-based SRP composites: a new feature for stamping process - ECCM19 - 19th European Conference on Composite Materials Nantes, France, 22-26th June 2020 - Jean-Charles Fontanier, Sofien Bouzouita, Carole Aubry-Meneveau, Philippe Mailler and Pascal Rumeau*
Publication	*New Process for High Rate Production of Composite Parts based on Thermoplastics Comptes Rendus des JNC 21 – Bordeaux INP – 1-3 juillet 2019 - A. Giroud, P. Jacquet, V. Charreton, J. Maupetit, G. Orange, X. Brochet, J. Loureaux, P. Monnier, P. Rumeau, S. Comas-Cardona, A. Hautefeuille, J. Colmars, Q. Steer*
Publication	*Numerical study of the mechanical behaviour of braided composites - iTechStyle 2nd International Conference of Textile and Clothing (Feb 2018) - Meriem Bounasser, Emmanuelle Vidal Sallé, Philippe Boisse, Philippe Mailler, Pascal Rumeau*
Publication	*Symposium ISF 2014 – Tokyo, Oct. 2014 - Textiles in Building: Development and Progress (Lecture) and Innovation and up-scaling of smart, green and customized fibers for functionalized textiles (Keynote) – Pascal Rumeau*
Publication	*Preparation of multinuclear microparticles using a polymerization in emulsion process - F. Salaün · E. Devaux · S. Bourbigot · P. Rumeau - Journal of Applied Polymer Science 02/2008; 107(4):2444 – 2452*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
CAREFIL	*Development of quality yarns from frayed fibers from recycling*
PARCABAHF	*Low and high frequency acoustic composite walls : Reduction of the environmental impact and improvement of the life cycle of materials by recovery of technical textile waste (glass / PVC yarn base) from the various production stages*
SILENT WALL	*Development of fibrous panels based on natural and recycled plant fibers (flax, hemp) with high sound insulation performance*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Cutting and Fraying of textile wastes	*Cutting of textile wastes : 5 to 150mm* *Fraying of textile wastes : 10 à 50 kg/h*
Defibrating of fluff	*Defibrating of fluff*
Non-woven prototyping	*3 pilot lines allowing various configurations comprising Feeder, Card, Cross-lapper, Needleloom, Airlaid, Hydroentanglement, Spraybonding, IR/Air Bonding/Dryer, Calender, needle-punching, Winder. Available ranges according to the configurations:* *o Length of fibers: 1-75 mm* *o 10-4000 g/m²* *o Wid*
Compounding line	*1 to 30 kg/h*

Meltspinning line	*o 1 to 8 kg/h* *o Up to 450°C* *o 1000 to 4200 m/min* *o Foy and POY yarn* *o 20 to 1 000 dTex*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
993284075	*AXELERA - ASSOCIATION CHIMIE-ENVIRONNEMENT LYON ET RHONE-ALPES*

Short name: AXELERA

Address

Street	ROND POINT DE L ECHANGEUR LES LEVEES
Town	SOLAIZE
Postcode	69360
Country	France
Webpage	www.axelera.org

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is no (small- and medium-sized enterprise) for the call.

SME self-declared status	unknown
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Dr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Sebastien**

Last name* **Beclin**

E-Mail* **sebastien.beclin@axelera.org**

Position in org. Recycling mission coordinator

Department AXELERA - ASSOCIATION CHIMIE-ENVIRONNEMENT LYON ET RHONE-ALPES

☒ Same as organisation name

☒ Same as proposing organisation's address

Street ROND POINT DE L ECHANGEUR LES LEVEES

Town SOLAIZE

Post code 69360

Country France

Website https://www.axelera.org/fr

Phone +33 (0)4 28 27 04 83

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Jennifer	Shaw-Taberlet	jennifer.shaw-taberlet@axelera.org	+33 (0)7 88 45 04 04
Noemie	Vecchio	noemie.vecchio@axelera.org	+xxx xxxxxxxxx
Laure	Hugonet	laure.hugonet@axelera.org	+33 (0)6 84 58 39 20

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Jennifer	Shaw-Taberlet	Woman	France	jennifer.shaw-taberlet@axelera.org	Category C Recognised	Team member		

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☒
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Axelera publishes success stories about the high-impact innovations and business offers of its members. These are available in English at https://www.axelera.org/en/editions/. The cluster also circulates 3 weekly newsletters to its network the disseminates presentations provided at workshops and panel discussions.*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
BlackCycle (H2020)	*This ongoing European project aims to enable circular recycling of the rubber and carbon black in tires. Axelera manages dissemination, communication, outreach and strategic intelligence.*
MineTheGap (H2020)	*This ongoing European project serves to support cleantech and digital SMEs in transferring their technological solutions to SMEs in the European mining sector. Axelera is WP leader responsible for business services provided to SMEs and also contributes to dissemination.*
PyroCO2 (H2020)	*This ongoing European project serves to support the emergence of industrially and economically viable carbon-capture and use (CCU) for the chemical industry. Axelera is in charge of facilitating the emergence of large-scale projects in France and supporting other regions to help similar CCU hubs to emerge.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Spaces for meetings and large-scale workshops	*Axelera has free access to various sized meeting rooms in Lyon, Paris and Brussels via its 2 locations and partners.*
Customer relations tool	*Axelera?s digital infrastructure includes its website and integrated contact management solution (EUDONET).*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
999997930	*CENTRE NATIONAL DE LA RECHERCHE SCIENTIFIQUE CNRS*

Short name: PPRIME

Address

Street	RUE MICHEL ANGE 3
Town	PARIS
Postcode	75794
Country	France
Webpage	www.cnrs.fr

Specific Legal Statuses

Legal person	yes
Public body	yes
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is not an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	18/11/2008	- no
SME self-assessment	unknown	
SME validation	18/11/2008	- no

Departments carrying out the proposed work

Department 1

Department name	Institut PPRIME - UPR CNRS 3346	☐ not applicable
	☐ Same as proposing organisation's address	
Street	4 avenue de la Varsovie	
Town	Angouleme	
Postcode	16021	
Country	France	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Thami**

Last name* **Zeghloul**

E-Mail* **thami.zeghloul@univ-poitiers.fr**

Position in org. Associate professor

Department Institut PPRIME

☐ Same as organisation name

☐ Same as proposing organisation's address

Street 4 avenue de la Varsovie

Town Angouleme

Post code 16021

Country France

Website https://pprime.fr/

Phone *+xxx xxxxxxxxx*

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Ahlem	Benabderrahmane	ahlem.benabderrahmane@univ-poitiers.fr	+33545673245
Coraline	Daniel	coraline.daniel@cnrs.fr	+33549453569
Lucien	Dascalescu	lucien.dascalescu@univ.poitiers.fr	+33545673245

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Thami	Zeghloul	Man	France	thami.zeghloul@univ-poitiers.fr	Category B Senior le	Leading		
Prof	Lucien	Dascalescu	Man	France	lucien.dascalescu@univ-poitiers.fr	Category A Top grade	Team member	0000000345847558	Orcid ID
Dr	Ahlem	Benabderrahmane	Woman	Algeria	ahlem.benabderrahmane@univ-poitiers.fr	Category C Recognis	Team member		

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Electrostatic separation of metals and plastics from waste electrical and electronic equipment* *Dascalescu L., Zeghloul T., Iuga A. WEEE Recycling. Research, Development, and Policies, Elsevier, 2016, pp. 75-106* *Comprehensive review of the state of the art of the electrostatic separation technologies*
Publication	*Experimental study of a tribo-aero-electrostatic separator for finely-grinded matter* *Messal S., Zeghloul T., Mekhalef A, Medles K., Dascalescu L., Journal of Electrostatics, 89 (2017) 59-68* *Validates the feasibility of tribo-aero-electrostatic separation of the fine fractions generated during the recycling process of certain industrial wastes.*
Publication	*Distinct recovery of copper and aluminum from waste electric wires using a roll-type electrostatic separator* *Salama A., Richard G., Medles K., Zeghloul T., Dascalescu L., Waste Management, 76 (2018) 207-216* *Points out the possibility of using electrostatic separation techniques for the recovery of certain conductive materials from granular wastes. The same physical principle will be put to action for the separation of insulating fibers from the more conductive rubber granules*
Publication	*Triboelectric charging of granular polymers previously exposed to dielectric barrier discharges in atmospheric air* *Benabderrahmane A., Zeghloul T., Medles K., Tilmatine A., Dascalescu L., IEEE Trans. Ind. Appl., 56 (2020) 3061 - 3067* *Demonstrates that the exposure to a dielectric barrier discharge may modify the triboelectric charging of certain granular polymers and facilitate their electrostatic separation.*
Other achievement	*Procede et dispositif de separation electrostatique de materiaux granulaires.* *Dascalescu, L., Zeghloul, T., Medles, K., (2019) Brevet d'invention FR 3078638 -2019-09-13* *The pattented method associates electrical and aerodynamic forces for achieving the selective sorting of more or less finely-ginded matter. It is expected that a device making use of the same types of forces be effective for the extraction of fibers from end-of-life tires and similar products*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
Tribo-aero-electrostatic separation process	*Several industrial projects financed between 2008 and 2016 by APR2 company, Bonni?res sur Seine* *A new tribo-aero-electrostatic separation process was pattented, than laboratory validated, and four industrial machines were build for the selective sorting of plastics from waste electric and electronic equipment (output: 600 kg/h)*
R&D of electrostatic separators for the recycling	*Industrial project financed between 2015 and 2018 by CITF company, Saint Cybardeaux* *Based on the results of laboratory studies, a pilot-scale corona-electrostatic separator was designed and engineered, then an industrial unit was built for processing up to 200 kg waste per hour*
Triboelectric charging devices	*Industrial project financed between 2018 and 2022 by CITF company, Saint Cybardeaux* *Several laboratory models of triboelectric charges (fluidized bed-, vibratory-, rotating drum-types were conceived and tested, then an industrial pilot has been manufactured. Tests are in progress in view of using an original multi-cylinder rotating type device for the tribocharging of plastic mixtures in view of electrostatic separation.*

Novel electrostatic separator	*Industrial project financed between 2020 and 2021 by Michelin company, Clermont Ferrand R&D of the laboratory model of a novel electrostatic separator for the recovery of PET fibers from end-of-life tires* *The laboratory models of three modules of a novel electrostatic separator for the recovery of PET fibers from end-of-life tires have been designed, engineered and manufactured. Test are in profess to fine-tune the oprtation of the laboratory equipment*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	**Short description (Max 300 characters)**
Metal-belt-conveyor-type multi-functional electros	*Laboratory equipment designed for being able to perform feasability tests for the main electrostatic separation methods, based on electrostatic induction, corona discharge and triboelectric charging of granular or powdery matter*
Two-disk-type tribo-aero-electro-static separator	*Laboratory equipment designed to perform the tribo-electrostatic separation of finly grinded materials in a fluidized bed generated between two rotating disk electrodes connected to high-voltage supplies of opposite polarities*
Two-metal-belt-conveyor-type tribo-aero-elecytrost	*Patented industrial-pilot equipment designed to perform the tribo-electrostatic separation of granular materials in a fluidized bed generated between two metal-belt-conveyor-type electrodes connected to high-voltage supplies of opposite polarities*
Standard free-fall-type electrostatic equipment	*Laboratory equipment designed to operate with any tribo-charging device (vibratory, rotary, fluidized bed , ?)*
Two standard rotating-roll-type corona-electrostat	*Two laboratory separators currently used for the separation of the constituents of granular mixtures, based on the differences in conductivity and/or shape*
Tribocharging devices	*Vibratory-, fluidized-bed-, and rotary-cylinder- types custom designed laboratory devices for tribocharging experiments*
High-voltage supplies	*Eight high-voltage supplies for the experimental study of electrostatic processes (output voltage ranging from 30 kV to 100 kV, and output currents ranging from 0.3 to 5 mA).*
Electrostatic voltmeters 3 kV, 8 kV, 10 kV and 20	*A complete range of electrostatic voltmeters and electrostatic probes for contact or non-contact measurement of the electrical potential*
Eight electrometers and Faraday pails	*Instruments for the precision measurements of the electric charge of granular or powdery materials*
High-voltage amplifier 30 kV	*Laboratory equipment for generating DC and AC high-voltage for the experimental study of electrostatic processes*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
999859608	*UNIVERSITE DE POITIERS*

Short name: UNIVERSITE DE POITIERS

Address

Street	RUE DE L HOTEL DIEU 15 HOTEL PINET
Town	POITIERS CEDEX
Postcode	86034
Country	France
Webpage	www.univ-poitiers.fr

Specific Legal Statuses

Legal person	yes
Public body	yes
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	yes
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is no (small- and medium-sized enterprise) for the call.

SME self-declared status	unknown
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

Department 1

Department name	Institut PPRIME - UPR CNRS 3346	☐ not applicable
	☐ Same as proposing organisation's address	
Street	4 avenue de la Varsovie	
Town	Angouleme	
Postcode	16021	
Country	France	

Links with other participants

Type of link	Participant

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
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    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
896288634	*ERASME (Centre D'Excellence Jean Monnet sur le Développement Durable)*

Short name: ERASME

Address

Street	POLYTECH BUREAU C104, 2, avenue BLAISE PASC
Town	Aubière
Postcode	63170
Country	France
Webpage	https://erasme.uca.fr/

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	no
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is no (small- and medium-sized enterprise) for the call.

SME self-declared status	unknown
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ◯ Woman ◉ Man ◯ Non Binary

First name* **Arnaud**

Last name* **Diemer**

E-Mail* **diemera@aol.com**

Position in org. Scientific Coordinator

Department ERASME (Centre D'Excellence Jean Monnet sur le Développement Durable)

☒ Same as organisation name

☒ Same as proposing organisation's address

Street POLYTECH BUREAU C104, 2, avenue BLAISE PASCAL

Town Aubière

Post code 63170

Country France

Website https://www.erasmecentre.eu

Phone +33672250475

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Marie	Schellens	marieschellens@gmail.com	+32495618015
Claudiu Eduard	Nedelciu	aduard.nedelciu@gmail.com	+40720597727

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Ms	Nathalie	Spittler	Woman	Austria	nathalie@spittler.at	Category C Recognis	Team member	0000-0003-3493-965X	Orcid ID
Mr	Claudiu Eduard	Nedelciu	Man	Romania	eduard.nedelciu@gmail.com	Category C Recognis	Team member	0000-0003-0884-0656	Orcid ID
Mr	Arnaud	Diemer	Man	France	diemera@aol.com	Category B Senior le	Leading	0000-0002-1996-3400	Orcid ID

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☒
Contributions from the social sciences or/and the humanities	☒
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*DIEMER A., DIERICKX F. (2021), Circular Economy, A New Paradigm for Europe, in DIEMER A., NEDELCIU E., SCHELLENS M., MORALES M., OOSTDJIK M. (2021), Paradigms, Models, Scenarios and Practices for Strong Sustainability, Editions Oeconomia* *The aim of circular economy is to decouple from the linear economic model, shifting from the extract-produce-consume-throw away model to a "circular" economic design. This study proposes contrasts with this narrow and reductive vision of circularity.*
Publication	*TAFFURI, A., SCIULIO, A., DIEMER, A., NEDELCIU, C.E. (2021) Integrating Circular Bioeconomy and Urban Dynamics to define an innovative management of bio-waste: the case study of Turin. Sustainability, 13(11), 6224.* *This study explores the use of systems thinking and system dynamics modelling for the implementation of Circular Bioeconomy in urban areas. It can thus provide insights into Circular Economy deployment and sustainable natural resource management.*
Publication	*DEBLADIS L., DIAKHATE F., GOINEAU M., TAILLANDIER M., DIEMER A (2019), MIchelin, du diagnostic global à l'économie circulaire, Revue Francophone du Développement Durable, hors série n°7, Décembre.* *The article presents the success story of Michelin for the value chain of tires, from global diagnosis to its commitment to the circular economy and the economy of functionality.*
Publication	*DIEMER, A., NEDELCIU C.E., SCHELLENS, M., GISLADOTTIR, J. (2018). Challenges for sustainability in criticl raw material assessments. International Journal of Management and Sustainability, 7(3), 156-179.* *This study uses systems thinking and qualitative system dynamics modelling to understand social and economic drivers for critical raw materials. This methodology is relevant for the economists who work on Economy of natural resources or Economy of Environment.*
Publication	*DIEMER A., DIERICK F., GLADKYKH G., MORALES M., PARRIQUE T., TORRES J. (2016), European Union and Sustainable Development, Challenges and Prospects, Editions Oeconomia.* *The first part is concerned by the different dimensions (economic, social, cultural…) of sustainable development. The second part is focus on the thematic scope such as biodiversity, energy transition and climate change, industrial ecology, urban metabolism, food which are issues for the policy and the strategy of EU.*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
RESCUE SDGs *2021 - 2023* *Germany*	*This research project is commissioned by German Environmental Agency (UBA) and GWS, it aims to answer the questions (1) what synergies or trade-offs exist between the SDGs of the 2030 Agenda and with regard to the envisaged transformations towards integrated climate and resource protection, and (2) what are pathways to ambitious climate and resource protection that at the same time also lead to the best possible SDG outcomes?*
IATB LOOP *2021-2023* *AUF (Francophonia Agency)*	*IATB (Integrated Assessment Tools Box for Living on One Planet) commissioned by Francophonia Agency, proposes to develop three tools (Circles of sustainability, toile maker and System Dynamics Model) to map flows and stocks and designed circular Economy Framework at the cities Level. 5 countries have been involved (France, Norway, Italy, Lituania and Burkina Faso*

JEAN MONNET EXCELLENCE CENTER ON SUSTAINABILITY 2	*The Jean Monnet Center was financed by European Commission and its objective is to use expertise on sustainability at UCA and from different partners (25 european partners) to match climate, biodiversity, circular economy, education issues. The Center organized symposium, summer schools and Webinar on Circular Economy.*
ADAPTECON II MSCA-ITN-2015-EJD 2015 - 2019 MARIE	*The objective was to produce a new economic model (project ADAPTECON) including system dynamics approach and planet boundaries. 12 PhD have been recruited, two ESR had to design circular economy for Steel Industry and climate change models. Three partners : University of Clermont Auvergne, University of Iceland and University of Stockholm.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	**Short description (Max 300 characters)**

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
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    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
998993786	*IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA*

Short name: IRIS

Address

Street	CALLE VELAZQUEZ, NO 94 PRIMERA PLANTA
Town	MADRID
Postcode	28006
Country	Spain
Webpage	www.iris.cat

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	unknown
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	31/12/2019	- yes
SME self-assessment	31/12/2019	- yes
SME validation	24/02/2009	- yes

Departments carrying out the proposed work

Department 1

Department name	IRIS Innovation Lab	☐ not applicable
	☐ Same as proposing organisation's address	
Street	39-41 Carretera d'Esplugues	
Town	Cornella de Llobregat (Barcelona)	
Postcode	08940	
Country	Spain	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title: Dr

Gender: (•) Woman ( ) Man ( ) Non Binary

First name*: **Ana Maria**

Last name*: **Lopez Sabiron**

E-Mail*: **alopez@iris-eng.com**

Position in org.: PROJECT AND INNOVATION MANAGER

Department: PMO

☐ Same as organisation name

☐ Same as proposing organisation's address

Street: 39-41 Carretera d'Esplugues

Town: Cornella de Llobregat (Barcelona)

Post code: 08940

Country: Spain

Website: https://www.iris-eng.com/

Phone: +349 35542500

Phone 2: *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Albert	Torres	albert.torres@iris-eng.com	+34935542500
Alexandra	Poch	apoch@iris-eng.com	+34935542500
Colm	Digby	cdigby@iris.cat	+34935542500
Joan	Puig	jpuig@iris-eng.com	+34935542500

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Dr	Ana Maria	Lopez Sabiron	Woman	Spain	alopez@iris-eng.com	Category B Senior le	leading	https://orcid.org/0000-0002-8509-7109	Orcid ID
Dr	Laura	Rodriguez Turienzo	Woman	Spain	lrodriguez@iris-eng.com	Category A Top grad	leading	https://orcid.org/0000-0003-4656-1674	Orcid ID
Dr	Laurent	Philippet	Man	France	lphilippet@iris-eng.com	Category B Senior le	leading	https://orcid.org/0000-0001-6296-6669	Orcid ID
Dr	Victor	Olmos	Man	Spain	volmos@iris-eng.com	Category B Senior le	leading	https://orcid.org/0000-0003-0132-6849	Orcid ID

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☒
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Review on the photonic techniques suitable for automatic monitoring of the composition of multi-materials wastes in view of their posterior recycling* *https://doi.org/10.1177/0734242X21997908* *This review focuses on the materials identification step (taking place after waste collection and before sorting and reprocessing) but in outlook, further insights on sorting are given as well as future prospects which can contribute to increasing the circularity of the plastic composites? value chains.*
Publication	*Sensing Technology* *https://doi.org/10.1002/9783527804153.ch6* *This chapter provides key metrological concepts, which are fundamental in the context of resource efficiency by rationalizing production processes. It describes relevant sensor technologies from as basic as energy metering, to the latest advancements in spectroscopy, process analytical technologies (PAT), soft-sensors, microelectromechanical system (MEMS)-based sensors and smart sensors.*
Good	*VISUM InLine* *https://www.iris-eng.com/device/visum-in-line/* *Industrial NIR analyzer for real-time and continuous chemical composition monitoring adapted to several types of geometries and unit operations (conveyor belts, pipes, tubes, etc.) and unit process machinery (e.g. blenders). It is designed for: (1) Continuous monitoring of Quality Attributes (CQAs); (2) Real-time and simultaneous measurement of Critical Control Parameters (CCPs); (3) Anomaly Detection.*
Other achievement	*DEVICE FOR DETECTING FOREIGN MATTER IN AN EDIBLE SUBSTANCE OR IN A CONTAINER CONTAINING THE EDIBLE SUBSTANCE AND METHOD EMPLOYING THE DEVICE* *Number: EP19382101*
Other achievement	*AN INTEGRATED OPTICAL SYSTEM FOR THE EXAMINATION OF SAMPLE MATERIALS* *https://worldwide.espacenet.com/patent/search/family/060270416/publication/EP3467478A1?q=pn%3DES2817438T3*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
MULTICYCLE	*MULTICYCLE, H2020 SPIRE, GA 820695, Advanced and sustainable recycling processes and value chains for plastic-based multi-materials*
PRESERVE	*PRESERVE, H2020-NMBP-BIOTEC, GA 952983, High performance sustainable bio-based packaging with tailored end of life and upcycled secondary use*
MERLIN	*MERLIN, H2020-SC5-2020-2, GA 101003883, Increasing the quality and rate of MultilayER packaging recycLINg waste*
MULTIPLE	*MULTIPLE, H2020 ICT, GA 871783, Multimodal spectral sensors and orchestrated deep models for integrated process optimisation*
CIRCULAR FOODPACK	*CIRCULAR FOODPACK, H2020-SC5-2020-2, GA 101003806, Circular Packaging for Direct Food Contact Applications*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Spectroscopic Laboratory	*Equipped with several instruments, namely, a FTIR-ATR (Interspectrum), 2 UV-VIS (PerkinElmer and Ocean Optics), an Extended-SWIR (StellarNet) and a NIR (Mightex) spectrophotometers as well as a SWIR Hyperspectral imager (Specim-Xenics).*

Electronics, mechanical and optical workshops	*Equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others*
Food and wet chemistry lab	*Equipped with a DSC, a Surface Energy analyser, an HPLC, a Cobb tester, microbiological equipments, microspcope and microtome for structural analysis, migration cells, colourimeter (for colour and opacity measurement), Ford cup viscometers, water vapour permeation cells, turbidimeter, etc.*
Optical photonics-based technologies and PAT syste	*HSI and NIR based cameras suitable for the surface and composition analysis of different materials and for the optimization and monitoring of numerous kinds of processes.*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? (•) Yes ( ) No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
916768923	*HOGSKULEN PA VESTLANDET*

Short name: HVL

Address

Street	INNDALSVEIEN 28
Town	BERGEN
Postcode	5020
Country	Norway
Webpage	www.hvl.no/

Specific Legal Statuses

Legal person	yes
Public body	yes
Non-profit	yes
International organisation	no
Secondary or Higher education establishment	yes
Research organisation	yes

SME Data

Based on the below details from the Participant Registry the organisation is no (small- and medium-sized enterprise) for the call.

SME self-declared status	unknown
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

Department 1

Department name	Department of Environmental Sciences	☐ not applicable
	☐ Same as proposing organisation's address	
Street	6 Royrgata	
Town	Sogndal	
Postcode	6856	
Country	Norway	

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Prof.

Gender ◉ Woman ○ Man ○ Non Binary

First name* **Valeria Jana**

Last name* **Schwanitz**

E-Mail* **valeria.jana.schwanitz@hvl.no**

Position in org. Professor in Energy Economics and Climate Change

Department Department of Environmental Sciences

☐ Same as organisation name

☐ Same as proposing organisation's address

Street 6 Royrgata

Town Sogndal

Post code 6856

Country Norway

Website www.hvl.no

Phone +4795764411

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
August	Hubert Wierling	august.hubert.wierling@hvl.no	+4795764411
Terje	Gravdal	terje.gravdal@hvl.no	+4755587054
Guillaume	Lapeyre	guillaume.lapeyre@hvl.no	+xxx xxxxxxxxx

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Prof	Valeria Jana	Schwanitz	Woman	Germany	valerias@hvl.no	Category A Top grade	Leading	https://orcid.org/0000-0002-8752-2754	Orcid ID
Prof	August	Wierling	Man	Germany	augustw@hvl.no	Category A Top grade	Team member	https://orcid.org/0000-0002-7443-7593	Orcid ID

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☒
Co-definition of research and market needs	☒
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☐
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☒
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
COMETS project	*H2020. COMETS (Collective Action Models for the Energy Transition and Social Innovation) is filling the knowledge gaps around citizen engagement in the energy transition to renewable sources by investigating and quantifying the aggregate contribution of Collective Action Initiatives (CAIs) in the energy sector at both national and European levels. Role of HVL: building inventory and estimation/scenario development.*
EERAdata project	*H2020 (lead). The goal of EERAdata is to develop, explore, and test a FAIR and open data ecosystem for low-carbon energy research data, where FAIR stands for F-indable, A-ccessible, I-nteroperable, R-e-usable.*
DRES2Markets project	*H2020. Main aim of the DRES2Market project is to develop comprehensive and affordable approach to facilitate effective participation of distributed generation based on renewable energies in the electricity markets, and enable to provide balancing and reserve services according to market criteria. Role of HVL: contribution to social, environmental, and economic impact analysis.*
AMPERE project	*FP7. The AMPERE project analysed climate mitigation pathways and associated mitigation costs under technology and policy limitations. In a series of multi-model comparisons, it evaluated model differences and the relationship between model results and historical trends. Role of applicant: lead of model validation, model intercomparison.*
RELEASE project	*Norwegian Research Council. RELEASE develops knowledge that assists them in such decision-making, while simultaneously providing new theoretical and empirical contributions to real options theory, social theory, restoration ecology, and local sustainable development.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
966592488	*IPOINT-SYSTEMS GMBH*

Short name: IPOINT

Address

Street	Ludwig-Erhard-Strasse 58
Town	Reutlingen
Postcode	72760
Country	Germany
Webpage	www.ipoint-systems.de

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	21/12/2000	- yes
SME self-assessment	unknown	
SME validation	21/12/2000	- yes

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mrs

Gender ◉ Woman ○ Man ○ Non Binary

First name* **Maria**

Last name* **Dos Santos**

E-Mail* **maria.dossantos@ipoint-austria.at**

Position in org. Team leader R&D

Department Product office

☐ Same as organisation name

☐ Same as proposing organisation's address

Street 58 Ludwig-Erhard-Strasse

Town Reutlingen

Post code 72760

Country Germany

Website https://www.ipoint-systems.com

Phone +431272037012

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Anna	Bicker	anna.bicker@ipoint-systems.de	+xxx xxxxxxxxx

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier
Mr	Jan	Hedemann	Man	Austria	jan.hedemann@ipoint-systems.de	Category A Top grade	Leading		
Mr	David	Nerjes	Man	Austria	andreas.beck@ipoint-systems.de	Category A Top grade	Team member		
Mrs	Anja	Kratz	Woman	Austria	david.nerjes@ipoint-systems.de	Category A Top grade	Team member		
Mr	Andreas	Beck	Man	Austria	anja.kratz@ipont-systems.de	Category A Top grade	Team member		

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☒
Technology developer	☒
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)
Publication	*Schiffleitner, A; Prox, M; Wahl, A (2020): Proceedings of Electronic Goes Green Conference 2020 Reaching Carbon-neutrality with Role-based Access to LCA Information of Materials, Parts and Components*
Publication	*Koffler, C., Amor, B., Carbajales-Dale, M., M. Proc et al. (2020). Int J Life Cycle Assess 25, 478 482. On the reporting and review requirements of ISO 14044*
Publication	*Vieira, M., Cenian, K., De Schreyver, A., Genest, A. et al. SETAC (2018), Europe 28th Annual Meeting Rome* *The end of an era: is data and model exchange across LCA software tools finally possible?*
Publication	*Klein, M, Stock, M(2016), Uwf 24, 25-28* *?kobilanzierer auf Datensuche*
Publication	*Schiffleitner, A; Hedemann, J; Prox, M; Wahl, A (2018), Proceedings of Care Innovation Conference Vienna Nov 2018* *Live LCA - New Level of Possibilities based on automated LCA - a User Requirements Point of View*

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)
Live LCA: (2017-2020)	*Development of a software solution that collects and links the relevant data from the supplier network, internal production, use phase and recycling and reuse. Automation and integration of data collection and processing. User-specific profiles to support individual data investigation. Creation of scenar ios, identification of risk components and materials and selection of alternatives. Assessment and control of the environmental impact of complex products, using LCA and other*
MultiMaK I+II: (2013-2014, 2015-2018, BMBF)	*During MultiMaK, an initial col lection of design rules has been prepared, which provides the construction depart ment with numerous suggestions for the development of components in multi material design. Special focus was placed on providing specific knowledge for the design and production of components made of fibre-plastic composites. Further more, the foundations for the methodical development of components in multi material construction were laid.*
LCT (2019-2021)	*Life Cycle Technologies for hybrid structures in the frame of ?Forschung scampus Open Hybrid Lab Factory?. The project?s goal was to design multi-ma terial components in a life cycle-oriented way. LCT primarily investigated the use and end-of-life options with the aid of targeted condition monitoring. An im portant component is a holistic analysis and evaluation considering technical, eco nomic, and ecological criteria.*
MAFIMA: (2019 - 2022)	*Comprehensive research on most-effective use of Material Flow Cost Accounting (MFCA). Creation of Scenarios with IT-supported algo rithms, suggestion of measures for savings, resource efficiency, research on Data Management in versatile Companies. The results was a simulated environment in which algorithms for scenario building and the expert system can be tested proto typically.*
EcoSteel: (2017 - 2020, BMBF)	*Material flow model to improve the sustainability and resource efficiency of electric steel production in the electric arc furnace. Be sides ifu Hamburg, the Institute for Industrial Furnace Engineering and Thermal Engineering (IOB) of RWTH Aachen University contributes its expertise. Coop eration with several steel mills will ensure the practical relevance of the results.*

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)
Umberto	*Architecture concept building and programming expertise*
e!Sankey	*Visualisation of material and energy flows as Sankey diagrams*

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
952411961	*DYNERGIE*

Short name: DYNERGIE

Address

Street	1-3 PLACE GIOVANNI DA VERRAZZANO
Town	LYON
Postcode	69009
Country	France
Webpage	www.dynergie.eu

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is an SME (small- and medium-sized enterprise) for the call.

SME self-declared status	31/12/2015	- yes
SME self-assessment	31/12/2015	- yes
SME validation	unknown	

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title Mr

Gender ○ Woman ◉ Man ○ Non Binary

First name* **Brice**

Last name* **Lorho**

E-Mail* **blorho@dynergie.eu**

Position in org. Manager

Department DYNERGIE

☒ Same as organisation name

☒ Same as proposing organisation's address

Street 1-3 PLACE GIOVANNI DA VERRAZZANO

Town LYON

Post code 69009

Country France

Website https://www.dynergie.eu/

Phone +15183202166

Phone 2 *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Cedric	Raoul	craoul@dynergie.eu	+337 82 67 29 17

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier

Role of participating organisation in the project

Project management	☒
Communication, dissemination and engagement	☒
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☐
Prototyping and demonstration	☐
IPR management incl. technology transfer	☐
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

PIC	Legal name
894827911	*INDUSTRIA DE DISENO TEXTIL SA*

Short name: INDITEX

Address

Street	AVENIDA DE LA DIPUTA EDIFICIO INDITEX
Town	A CORUNA
Postcode	15143
Country	Spain
Webpage	www.inditex.com

Specific Legal Statuses

Legal person	yes
Public body	no
Non-profit	no
International organisation	no
Secondary or Higher education establishment	no
Research organisation	no

SME Data

Based on the below details from the Participant Registry the organisation is yes (small- and medium-sized enterprise) for the call.

SME self-declared status	unknown
SME self-assessment	unknown
SME validation	unknown

Departments carrying out the proposed work

No department involved

Department name *Name of the department/institute carrying out the work.*

☒ not applicable

☐ Same as proposing organisation's address

Street *Please enter street name and number.*

Town *Please enter the name of the town.*

Postcode *Area code.*

Country *Please select a country*

Links with other participants

Type of link	Participant

Main contact person

This will be the person the EU services will contact concerning this proposal (e.g. for additional information, invitation to hearings, sending of evaluation results, convocation to start grant preparation). The data in blue is read-only. Details (name, first name and e-mail) of Main Contact persons should be edited in the step "Participants" of the submission wizard.

Title: Mrs

Gender: (•) Woman ( ) Man ( ) Non Binary

First name*: **Sham**

Last name*: **Sharif Sokker**

E-Mail*: **shamsh@cn.inditex.com**

Position in org.: Corporate Environmental Sustainability - Innovation Lead

Department: NDITEX Environmental Sustainability Department

☐ Same as organisation name

☒ Same as proposing organisation's address

Street: AVENIDA DE LA DIPUTA EDIFICIO INDITEX

Town: A CORUNA

Post code: 15143

Country: Spain

Website: https://www.inditex.com/es/home

Phone: + 0034 648408690

Phone 2: *+xxx xxxxxxxxx*

Other contact persons

First Name	Last Name	E-mail	Phone
Adolfo	Neira Gomez	adolfong@ext.inditex.com	+34 679 248 055

Researchers involved in the proposal

Title	First Name	Last Name	Gender	Nationality	E-mail	Career Stage	Role of researcher (in the project)	Reference Identifier	Type of identifier

Role of participating organisation in the project

Project management	☐
Communication, dissemination and engagement	☐
Provision of research and technology infrastructure	☐
Co-definition of research and market needs	☐
Civil society representative	☐
Policy maker or regulator, incl. standardisation body	☐
Research performer	☐
Technology developer	☐
Testing/validation of approaches and ideas	☒
Prototyping and demonstration	☒
IPR management incl. technology transfer	☒
Public procurer of results	☐
Private buyer of results	☐
Finance provider (public or private)	☐
Education and training	☐
Contributions from the social sciences or/and the humanities	☐
Other If yes, please specify: (Maximum number of characters allowed: 50)	☐

List of up to 5 publications, widely-used datasets, software, goods, services, or any other achievements relevant to the call content.

Type of achievement	Short description (Max 500 characters)

List of up to 5 most relevant previous projects or activities, connected to the subject of this proposal.

Name of Project or Activity	Short description (Max 500 characters)

Description of any significant infrastructure and/or any major items of technical equipment, relevant to the proposed work.

Name of infrastructure of equipment	Short description (Max 300 characters)

Gender Equality Plan

Does the organization have a Gender Equality Plan (GEP) covering the elements listed below? ○ Yes ◉ No

Minimum process-related requirements (building blocks) for a GEP

- **Publication:** formal document published on the institution's website and signed by the top management
- **Dedicated resources:** commitment of human resources and gender expertise to implement it.
- **Data collection and monitoring:** sex/gender disaggregated data on personnel (and students for establishments concerned) and annual reporting based on indicators.
- **Training:** Awareness raising/trainings on gender equality and unconscious gender biases for staff and decision-makers.
- **Content-wise, recommended areas** to be **covered** and addressed via concrete measures and targets are:
    - work-life balance and organisational culture;
    - gender balance in leadership and decision-making;
    - gender equality in recruitment and career progression;
    - integration of the gender dimension into research and teaching content;
    - measures against gender-based violence including sexual harassment.

Proposal ID **101059639**

Acronym **WHITECYCLE**

3 - Budget

?

No.	Name of beneficiary	Country	Role	Personnel costs/€	Subcontracting costs/€	Purchase costs - Travel and substistence /€	Purchase costs - Equipment/€	Purchase costs - Other goods, works and services/€	Internally invoiced goods and services/€ (Unit costs-usual accounting practices)	Indirect costs/€	Total eligible costs	Funding rate	Maximum EU contribution to eligible costs	Requested EU contribution to eligible costs/€	Max grant amount	Income generated by the action	Financial contributions	Own resources	Total estimated income
1	Manufacture Francaise Des Pneumatiques Michelin	FR	Coordinator	1,138,192	791,800	61,300		290,500		372498.00	2654290.00	70	1858003.00	1,858,003	1858003.00		0	0	1858003.00
2	Mandals As	NO	Affiliated	121,000				34,000		38750.00	193750.00	70	135625.00	135,625	135625.00				135625.00
3	Carbios	FR	Partner	347,150	108,200	14,000	149,960	46,000		139277.50	804587.50	70	563211.00	563,211	563211.00				563211.00
4	Kordsa Teknik Tekstil Anonim Sirketi	TR	Partner	195,500	81,500	58,500	105,000	189,000		137000.00	766500.00	70	536550.00	536,550	536550.00				536550.00
5	Estato Umweltservice Gmbh	DE	Partner	200,000		16,000		19,000		58750.00	293750.00	70	205625.00	205,625	205625.00				205625.00
6	Tlc Synergies	FR	Partner	277,400	200,000	45,000	84,000	5,000		102850.00	714250.00	100	714250.00	499,975	499975.00				499975.00
7	Deutsche Institute Fur Textil- Und Faserforschung Denkendorf	DE	Partner	206,120		22,000		35,000		65780.00	328900.00	100	328900.00	328,900	328900.00				328900.00
8	Institut Francais Du Textile Et De L'habillement	FR	Partner	326,589	6,000	20,000	101,200	20,000		116947.25	590736.25	100	590736.00	590,736	590736.00				590736.00
	Axelera - Association										298537.50		298538.00		298538.00				298538.00

9	Chimie-environnement Lyon Et Rhone-alpes	FR	Partner	134,550		12,000		92,280		59707.50		100		298,538					
10	Centre National De La Recherche Scientifique Cnrs	FR	Partner	110,016	60,000	28,500		1,600		35029.00	235145.00	100	235145.00	235,145	235145.00				235145.00
11	Universite De Poitiers	FR	Affiliated	117,634						29408.50	147042.50	100	147043.00	147,043	147043.00				147043.00
12	Erasme (Centre D'excellence Jean Monnet Sur Le Développement Durable)	FR	Partner	235,800	90,470	10,000		15,600		65350.00	417220.00	100	417220.00	417,220	417220.00				417220.00
13	Iris Technology Solutions, Sociedad Limitada	ES	Partner	294,000		40,500		270,600		151275.00	756375.00	70	529463.00	529,463	529463.00				529463.00
14	Hogskulen Pa Vestlandet	NO	Partner	284,716		9,600		5,000		74829.00	374145.00	100	374145.00	374,145	374145.00				374145.00
15	Ipoint-systems Gmbh	DE	Partner	257,540	0	48,000	0	14,400	0	79985.00	399925.00	70	279948.00	279,948	279948.00				279948.00
16	Dynergie	FR	Partner	121,316		24,000		5,000		37579.00	187895.00	70	131527.00	131,526	131526.00				131526.00
17	Industria De Diseno Textil Sa	ES	Partner	314,170		25,000		20,000		89792.50	448962.50	70	314274.00	0	0.00				0.00
			TOTAL	4,681,693	1,337,970	434,400	440,160	1,062,980	0	1654808.25	9612011.25		7660203.00	7,131,653	7131653.00	0	0	0	7131653.00

4 - Ethics & security

Ethics Issues Table

1. Human Embryonic Stem Cells and Human Embryos			Page
Does this activity involve Human Embryonic Stem Cells (hESCs)?	○ Yes	◉ No	
Does this activity involve the use of human embryos?	○ Yes	◉ No	

2. Humans			Page
Does this activity involve human participants?	○ Yes	◉ No	
Does this activity involve interventions (physical also including imaging technology, behavioural treatments, etc.) on the study participants?	○ Yes	◉ No	
Does this activity involve conducting a clinical study as defined by the Clinical Trial Regulation (EU 536/2014)? (using pharmaceuticals, biologicals, radiopharmaceuticals, or advanced therapy medicinal products)	○ Yes	◉ No	

3. Human Cells / Tissues (not covered by section 1)			Page
Does this activity involve the use of human cells or tissues?	○ Yes	◉ No	

4. Personal Data			Page
Does this activity involve processing of personal data?	◉ Yes	○ No	41
Does it involve the processing of special categories of personal data (e.g.: genetic, biometric and health data, sexual lifestyle, ethnicity, political opinion, religious or philosophical beliefs)?	○ Yes	◉ No	
Does it involve profiling, systematic monitoring of individuals, or processing of large scale of special categories of data or intrusive methods of data processing (such as, surveillance, geolocation tracking etc.)?	○ Yes	◉ No	
Does this activity involve further processing of previously collected personal data (including use of preexisting data sets or sources, merging existing data sets)?	◉ Yes	○ No	41
Is it planned to export personal data from the EU to non-EU countries? Specify the type of personal data and countries involved	○ Yes	◉ No	
Is it planned to import personal data from non-EU countries into the EU or from a non-EU country to another non-EU country? Specify the type of personal data and countries involved	○ Yes	◉ No	
Does this activity involve the processing of personal data related to criminal convictions or offences?	○ Yes	◉ No	

5. Animals			Page
Does this activity involve animals?	○ Yes	◉ No	

6. Non-EU Countries			Page
Will some of the activities be carried out in non-EU countries?	◉ Yes	○ No	37

KORDSA is a large industrial company based in Turkey. KORDSA will be in charge of transforming the recycled PET monomers through different industrial processes in order to obtain cords for the production of tyres and yarns for the production of lay-flat hoses. These materials are not hazardous and non toxic. They are not critical materials.

In case non-UE countries are involved, do the activities undertaken in these countries raise potential ethics issues?	○ Yes	◉ No	

It is planned to use local resources (e.g. animal and/or human tissue samples, genetic material, live animals, human remains, materials of historical value, endangered fauna or flora samples, etc.)?	○ Yes	● No	
Is it planned to import any material (other than data) from non-EU countries into the EU or from a non-EU country to another non-EU country? For data imports, see section 4.	● Yes	○ No	37

During the pilot-scale loop, KORDSA (TURKEY) will send 25kg of cords to MANDALS (NORWAY) and 50kg cords to MICHELIN (FRANCE).
During the pre-industrial loop, KORDSA (TURKEY) will send 250kg to MANDALS (NORWAY) and 700kg of cords to MICHELIN (FRANCE).

Specific attention will be paid to the materials being sent cross-border and proper documentations will be provided to the relevant authorities in order to fully respect the Turkish and the EU laws regarding transit of merchandises.
These materials are not hazardous and non toxic. They are not critical materials.

Is it planned to export any material (other than data) from the EU to non-EU countries? For data exports, see section 4.	● Yes	○ No	37

During the pilot-scale loop, 90kg of yarns will be sent from IFTH (FRANCE) to KORDSA (TURKEY).
During the pre-industrial loop, 1300kg of polymer will be sent from TECHNIP ZIMMER - subcontractor (GERMANY) to KORDSA (TURKEY).

Specific attention will be paid to the materials being sent cross-border and proper documentations will be provided to the the relevant authorities in order to fully respect the Turkish and the EU laws regarding transit of merchandises.
These materials are not hazardous and non toxic. They are not critical materials.

Does this activity involve low and/or lower middle income countries, (if yes, detail the benefit-sharing actions planned in the self-assessment)	○ Yes	● No	
Could the situation in the country put the individuals taking part in the activity at risk?	○ Yes	● No	

7. Environment, Health and Safety			Page
Does this activity involve the use of substances or processes that may cause harm to the environment, to animals or plants.(during the implementation of the activity or further to the use of the results, as a possible impact) ?	○ Yes	● No	
Does this activity deal with endangered fauna and/or flora / protected areas?	○ Yes	● No	
Does this activity involve the use of substances or processes that may cause harm to humans, including those performing the activity.(during the implementation of the activity or further to the use of the results, as a possible impact) ?	○ Yes	● No	

8. Artificial Intelligence			Page
Does this activity involve the development, deployment and/or use of Artificial Intelligence? (if yes, detail in the self-assessment whether that could raise ethical concerns related to human rights and values and detail how this will be addressed).	● Yes	○ No	19

9. Other Ethics Issues			Page
Are there any other ethics issues that should be taken into consideration?	○ Yes	● No	

I confirm that I have taken into account all ethics issues above and that, if any ethics issues apply, I will complete the ethics self-assessment as described in the guidelines How to Complete your Ethics Self-Assessment ☒

Ethics Self-Assessment

Ethical dimension of the objectives, methodology and likely impact

Our project involves the import/export of materials with Turkey, i.e., a non-EU country.
However these imports and exports are technical materials that are being transformed in Turkey by KORDSA in order to manufacture end-products. These materials are quite similar to currently commercialized products and, not in any way, possibly problematic materials such as animal and/or human tissue samples, genetic material, live animals, human remains, materials of historical value, endangered fauna or flora samples, fossils, etc.
These exchanges are intended to reinforce the links between stakeholders of the value chain and accelerate the dissemination of circular economy principles.
Consequently, we do not consider there is any issue for the import/export of such materials with Turkey.
On top of this, these merchandises exchanges will be fully respecting the Turkish and the EU laws regarding custom border controls etc.
The project serve to solve a persistent problem in Europe with the highest ecological and social standards.

Remaining characters 3968

Compliance with ethical principles and relevant legislations

MICHELIN as project coordinator will lead and assess the execution of the ethical requirements within the project in accordance with applicable national, European and international legislation. Ethical standards and guidelines will be rigorously applied, irrespective of the country in which the research is carried out.

The project will ensure that all beneficiaries/partners will follow EU and national regulations regarding data protection and will obtain approval from local/national authority in charge of data protection, if applicable. Indeed, all ethical issues that this project may be exposed will be handled by the affected partners and in accordance with the applicable legislation. The work of the project will be subject to the following ethical-related directives and regulations:

• The Charter of Fundamental Rights of the EU (2000/c 364/01)

• Directive 95/46/EC of 24 October 1995 on the protection of individuals with regards to the processing of personal data and the free movement of such data

• REGULATION (EU) 2016/679 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation)

KORDSA (TURKEY) is a key supplier for MICHELIN and thus has been evaluated by EcoVadis, the world's most trusted business sustainability ratings. KORDSA has been ranked in the first quarter of the rankings of the companies evaluated by EcoVadis with an overall score of 67/100 whereas the average industry score is 46/100. Specific attention will be paid to the materials being sent cross-border and proper documentations will be provided to the the relevant authorities in order to fully respect the Turkish and the EU laws regarding merchandises in transit.

Partners will be compliant with the General Data Protection Regulation (GDPR) during their collection of personal data during

communication, dissemination and network-building activities. Partners have appointed Data Protection Officers (DPO), and the contact details of the DPOs of Michelin and Axelera will be made available to the Commission and all data subjects involved in the project. Partners confirm that all of the data we intend to process is relevant and limited to the purposes of the research project (in

accordance with the 'data minimisation principle'). The purpose is to ensure the dissemination and broad uptake of the results of the project. The personal data will be limited to:

• Name (mandatory)

• Organisation (mandatory)

• Title (mandatory)

• Email address (mandatory)

• Telephone number (optional)

A description of the technical and organisational measures that will be implemented to safeguard the rights and freedoms of the data subjects will be submitted as a deliverable by Month 2 of the project (D7.1 – GDPR compliance report). For further processing of previously collected personal data, an explicit confirmation that Axelera and other partners have lawful basis for the data processing and that the appropriate technical and organisational measures are in place to safeguard the rights of the data subjects will be included in this deliverable. This will include the above-listed contact data provided to us by project partners for potential inclusion in the mailing list for dissemination activities. Detailed information on the informed consent procedures in regard to data processing will be kept
on file by Axelera.

Regarding Artificial Intelligence, the developed technologies will not replace existing or future jobs, because it enables sorting on mixtures that humans cannot manually sort. The AI sorting technology will not harm and will improve Human agency and oversight (decision-support tools enabling circular waste management), Technical robustness and safety (new processes enabled), Privacy and data governance (data will be generated which it was not previously possible to have), Transparency (the tool will enable stakeholders to know what percentage of materials they have and what percentage of materials is recycled), Diversity, non-discrimination and fairness (the tool will help increase fairness between actors dealing with secondary raw materials (currently "waste") and actors dealing with primary raw materials, Societal and environmental wellbeing (job creation, less GHG emissions, less landfilling, and Accountability (enabling polluter pays principles to be further applied for companies that produce consumer products and allowing accountability of secondary raw materials suppliers to confirm the amount of recycled material content).

Remaining characters 280

Administrative forms

Proposal ID **101059639**

Acronym **WHITECYCLE**

Security issues table

1. EU Classified Information (EUCI)[2]			Page
Does this activity involve information and/or materials requiring protection against unauthorised disclosure (EUCI)?	◯ Yes	◉ No	
Does this activity involve non-EU countries?	◯ Yes	◉ No	

2. Misuse			Page
Does this activity have the potential for misuse of results?	◯ Yes	◉ No	

3. Other Security Issues			Page
Does this activity involve information and/or materials subject to national security restrictions? If yes, please specify: (Maximum number of characters allowed: 1000)	◯ Yes	◉ No	
Are there any other security issues that should be taken into consideration? If yes, please specify: (Maximum number of characters allowed: 1000)	◯ Yes	◉ No	

[2]According to the Commission Decision (EU, Euratom) 2015/444 of 13 March 2015 on the security rules for protecting EU classified information, "European Union classified information (EUCI) means any information or material designated by an EU security classification, the unauthorised disclosure of which could cause varying degrees of prejudice to the interests of the European Union or of one or more of the Member States".

[3]Classified background information is information that is already classified by a country and/or international organisation and/or the EU and is going to be used by the project. In this case, the project must have in advance the authorisation from the originator of the classified information, which is the entity (EU institution, EU Member State, third state or international organisation) under whose authority the classified information has been generated.

[4]EU classified foreground information is information (documents/deliverables/materials) planned to be generated by the project and that needs to be protected from unauthorised disclosure. The originator of the EUCI generated by the project is the European Commission.

5 - Other questions

Two-stage calls

The full stage-2 proposal must be consistent with the short outline proposal submitted the to stage 1 - in particular with respect to the proposal characteristics addressing the concepts of excellence and impact

Are there substantial differences compared to the stage-1 proposal? ◯ Yes ◯ No

Essential information to be provided for proposals including clinical Trials / studies / investigations

Clinical study means, for the purpose of this document, any systematic prospective or retrospective collection and analysis of health data obtained from individual patients or healthy persons in order to address scientific questions related to the understanding, prevention, diagnosis, monitoring or treatment of a disease, mental illness, or physical condition. It includes but it is not limited to clinical studies as defined by Regulation 536/2014 (on medicinal products), clinical investigation and clinical evaluation as defined by Regulation 2017/745 (on medical devices), performance study and performance evaluation as defined by Regulation 2017/746 (on in vitro diagnostic medical devices).

Are clinical studies / trials / investigations included in the work plan of this project? ◯ Yes ◉ No

LARGE-SCALE CIRCULARITY OF PET VIA SMART SORTING, ENZYMATIC RECYCLING OF COMPLEX WASTE - CIRCULAR PRODUCTION OF 3 HIGH-SPECIFICATION PET-CONTAINING PRODUCTS: TYRES, MULTI-MATERIAL CLOTHING AND HOSES

-

WHITECYCLE - WHITE TECHNOLOGY TEXTILE CYCLE

LIST OF PARTICIPANTS

Part. No.	Participant organisation name	Short name	Country
1(Coord.) *2*	MANUFACTURE FRANÇAISE DES PNEUMATIQUES MICHELIN *(WITH LINKED THIRD PARTY: FENNER MANDALS AS)*	MICHELIN *MANDALS*	FR *NO*
3	CARBIOS SA	CARBIOS	FR
4	KORDSA TEKNIK TEKSTIL ANONIM SIRKETI	KORDSA	TR
5	ESTATO UMWELTSERVICE GMBH	ESTATO	DE
6	TLC SYNERGIES	SYNERGIES	FR
7	DEUTSCHE INSTITUTE FÜR TEXTIL – UND FASERFORSCHUNG DENKENDORF	DITF	DE
8	INSTITUT FRANÇAIS DU TEXTILE ET DE L'HABILLEMENT	IFTH	FR
9	AXELERA	AXELERA	FR
10 *11*	CENTRE NATIONAL DE LA RECHERCHE SCIENTIFIQUE CNRS INSTITUT PPRIME *(WITH LINKED THIRD PARTY: UNIVERSITE DE POITIERS)*	PPRIME	FR *FR*
12	ERASME (CENTRE D'EXCELLENCE JEAN MONNET SUR LE DEVELOPPEMENT DURABLE)	ERASME	FR
13	IRIS TECHNOLOGY SOLUTIONS SOCIEDAD LIMITADA	IRIS	SP
14	HOGSKULEN PA VESTLANDET	HVL	NO
15	IPOINT	IPOINT	DE
16	DYNERGIE	DYNERGIE	FR
17	INDUSTRIA DE DISENO TEXTIL SA	INDITEX	SP

CALL & PROJECT INFORMATION

Type of action: Innovation action (IA)
Duration of the project: 48 Months
Deadline: 06 October 2021

COORDINATING PERSON DETAILS

Name	Michael Cogne
Organisation	MANUFACTURE FRANÇAISE DES PNEUMATIQUES MICHELIN (MICHELIN)
Email	michael.cogne@michelin.com

TABLE OF CONTENT

1 EXCELLENCE 3
1.1 OBJECTIVES AND AMBITION 3
1.2 METHODOLOGY 8
2 IMPACT 17
2.1 PROJECT'S PATHWAYS TOWARDS IMPACT 18
2.2 MEASURES TO MAXIMISE IMPACT – DISSEMINATION, EXPLOITATION, AND COMMUNICATION 20
2.3 SUMMARY 27
3 QUALITY AND EFFICIENCY OF THE IMPLEMENTATION 28
3.1 WORK PLAN AND RESOURCES 28
3.2 CAPACITY OF PARTICIPANTS AND CONSORTIUM AS A WHOLE 29

GLOSSARY

AI	Artificial Intelligence	**LCA**	Life-Cycle Assessment
ELH	End-of-life hose	**LCC**	Life-Cycle Cost
ELT	End-of-life tyre	**LIBS**	Laser Induced Breakdown Spectroscopy
EOL	End-of-life	**Mt / kt / t**	Million tonnes / kilo tonnes / tonne
EC	European Commission	**NIRS**	Near Infrared Spectroscopy
EU	European Union	**PET**	Polyethylene terephthalate
FAIR	Findable Accessible Interoperable Reusable	**POC**	Proof of concept
GHG	Greenhouse gas	**REF**	Reference
HMLS	High-Modulus Low-Shrinkage	**SSP**	Solid state polycondensation
HSI	Hyperspectral imaging	**XRF**	X-ray Fluorescence
IV	Intrinsic viscosity	**r-PET**	Recycled Polyethylene terephthalate
ICP-MS	Inductively coupled plasma mass spectrometry	**v-PET**	Virgin polyethylene terephthalate
		3T	Tension, Time and Temperature

EXECUTIVE SUMMARY

PET is widely used in plastics and textiles leading to **over 20 Mt/y of complex waste worldwide for which no closed-loop recycling is viable today**. Most complex waste is landfilled or incinerated. There is an urgent need to develop a circular solution to convert complex PET wastes from plastics and textiles back into high-added value products.

WHITECYCLE unites 3 brand-owners, 1 PET converter, 2 waste managers, 1 digital deeptech for smart sorting, the world leading enzymatic recycling SME, 1 LCA company, 3 UNIs/RTOs, 1 cluster and 1 management firm. They aim to demonstrate two new processes combining strong scientific and industrial know-how: (i) innovative identification, sorting and separation technologies that will dramatically increase the PET content of complex waste streams to 80%, and (ii) a disruptive enzymatic recycling process that is expected to yield pure PET monomers sustainably even for impure waste streams. PET monomers obtained will be repolymerised and recycled. Thus, **2 t of waste will be used to demonstrate 3 highly technical PET-containing pilot series: 100 tyres, 1,500 m of lay-flat hoses and 400m^2 of multicomponent fabric that will be coated and used to manufacture 4 lines of technical garments**. Process design kits, LCA and production cost estimates will be provided to PET manufacturers and waste management companies for rapid deployment and assure social acceptance. First projections show that WHITECYCLE's recycled PET will be competitive with virgin PET.

The project will conduct a full circle loop from real complex waste feedstock to representative product of the 3 use-cases at TRL 5. Then, a strong upscale study will allow the process steps to reach TRL 6 to 8. By 2030, WHITECYCLE will enable the annual recycling of more than 2 Mt of PET, which corresponds to the amount of additional recycled PET needed to meet the EU's 2030 targets. This will reduce emissions by around 2.06 Mt CO2eq and avoid the landfilling of more than 1.8 Mt of PET.

1 Excellence

1.1 Objectives and ambition

1.1.1 General presentation of WHITECYCLE

The European Commission's (EC) Circular Economy Action Plan adopted in March 2020 has identified plastics as among the 7 key product value chains as a priority, given their environmental impacts and circularity potentials. Indeed, as the consumption of plastics is expected to double in the coming 20 years, the EC has taken targeted measures to address the sustainability challenges posed by this ubiquitous material and will continue to promote a concerted approach to tackle plastics pollution at global level. To increase uptake of recycled plastics and contribute to the more sustainable use of plastics, the EC has proposed mandatory requirements for recycled content and waste reduction measures for key products such as packaging, construction materials and vehicles, also considering the activities of the Circular Plastics Alliance. **The European strategy is ambitious, envisioning the redefinition of the plastic value chain to make it 100% circular by 2050. It requires ambitious industrial actions coordinated at the European level. That's why WHITECYCLE aims at the industrial-scale implementation 3 very different circular value chains by 2030, within 4 years after the project.**

WHITECYCLE particularly addresses the use case of polyethylene Terephthalate (PET), **third most used polymer in the world** with more than **82 Mt produced in 2019 worldwide**[1] **and** used in various applications (Figure 1).

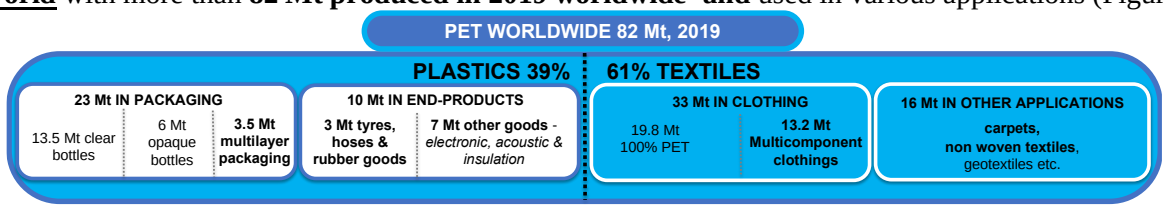

Figure 1 : PET amounts in the two targeted applications i.e., plastics and textile in 2019

If PET recycling processes are commonly implemented **for easy-to-collect and pure mono-material products**, (7 Mt of recycled PET (r-PET) produced in 2017 worldwide), **the situation is very different for multi-material products and massive volumes of PET waste currently end up landfilled or incinerated without any recycling possibility**. Indeed, there are no available solutions at the industrial scale, for the recycling or upcycling of these complex wastes due to current technical limitations related to the sorting, separation, and purification processes of the retrieved mono-materials. This complex mixed waste stream includes multilayer packaging, multicomponent textiles, and composites plastics (i.e., tyres, hoses etc.). In a recent study, Singh et al.[2] estimated 44 Mt of PET is used today in a such complex compositions that it cannot be recycled with classical technologies. In our project, with conservative figures, **a first estimation of eligible waste feedstock exceeds 20 Mt per year** (1.5 Mt end-of-life tyres (ELT) and end-of-life hoses (ELH), 13.2 Mt multicomponent clothing, and 3.5 Mt multilayer packaging, 2 Mt of other goods and industrial applications). Such large amounts of waste generate soil and water contamination that can adversely affect soil and water biodiversity, significant greenhouse gas (GHG) emissions and many other negative impacts on human health. Rethinking the functioning of such a complex value chain requires efforts by all its key players (i.e., plastics and textiles producers, end-users, recyclers, retailers, consumers) to drive investment in the right direction.

The objective of WHITECYCLE is to overcome the barrier posed by feedstock complexity via a set of versatile, disruptive, and fully scalable technological processes (mechanical, physico-chemical, and enzymatic) **capable of infinitely retrieving and recycling PET, even from complex mixed waste** to yield pure, industrially relevant monomer at any scale. WHITECYCLE includes a clear plan for replication of the technologies to be implemented at the industrial scale (i.e., > TRL8) beyond the project, across a wide range of sectors implementing or manufacturing PET-containing products. Information recorded in detail and in standardized ways (using FAIR data principles) will enable interoperability among the components, transparency of the processes involved, and the reuse in similar contexts as well as new, innovative ones (data as an asset).

The main project outcomes will be:

- 3 industrially viable circular value chains from bin to highly technical products namely tyres, lay-flat hoses, and multicomponent textiles at TRL 6 or above for the different steps (recognition, sorting, enzymatic depolymerization, purification, monomer reuse).
- Comprehensive LCA, LCC, social acceptability studies and FAIR data for each of the 3 use-cases
- Agreements for upscaling and implementation from 12 brand-owners, 3 PET manufacturers and 3 waste managers / recycling equipment manufacturers

[1] IHS Markit 2020
[2] Singh et al., Techno-economic, life cycle, & socioeconomic impact analysis of enzymatic recycling of PET, Joule (2021), doi.org/10.1016/j.joule.2021.06.015

1.1.2 USE-CASE #1: PET IN TYRES AND REINFORCED RUBBER GOODS

Tyre manufacturers must find solutions at the EU scale to produce sustainable tyres, since natural resources are decreasing, while waste is increasing. ELT are non-biodegradable wastes, generated in large amounts all over the world[3]. **Worldwide, 1.6 billion new passenger car tyres are sold each year[4] representing 11 Mt of ELT.**

Tyres are the only parts of the car that touch the road. They must strike a balance between safety, traction, durability, energy efficiency and manufacturing cost. Due to these competing needs, tyres are made of very complex composites that perform in harsh conditions. Figure 2 depicts the overall structure of a passenger car tyre. Tyre is made up of elastomers (natural & synthetic rubber), filler (carbon black & silica), reinforcement material (metals & PET textiles based on highly technical fibre), plasticizers and other chemicals. Some fractions are currently very well retrieved such as rubber and metal, but the embedded textile part composed of the **carcass plies (**thin textile PET fibre cords bonded into the rubber) and the cap plies made of PET, polyamide or hybrid polyamide/aramid is not. In addition, the textile acts as a pollutant for compounds and metals valorisation.

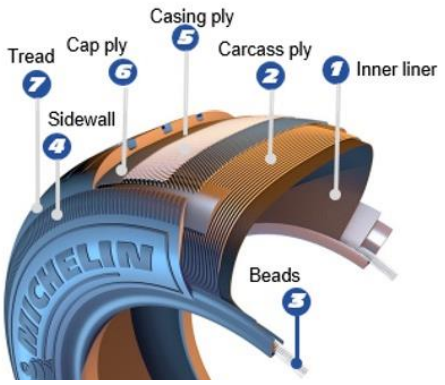

Figure 2 : Main elements of a tyre

PET for tyre cords represents 7 % of ELT mass or 800 kt worldwide per year (greatest application for industrial polyester fibres (3 Mt).

In Europe about 3.4 Mt/year of ELT are treated to recover materials or to produce energy[5]. Of these, 47 % of collected ELT are shredded. This shredding process outputs a stream of dirty fibrous textile (about 20 % of the total volume of shredded ELT, i.e., 320 kt/year). The recovered dirty fibrous textile represents a special waste[6] (see Figure 3), containing PET, that must be disposed of and mostly used in cement-plant for energy recovery. The dirty fibrous textile contains approximately 50 % of rubber particles, 37 % PET, 13 % other fibres. In Europe, it represents a feedstock of 250 kt of waste PET if all ELT were shredded to produce secondary raw material.

Figure 3 : Dirty fibrous content from ELT

With an expected yield of 80 %, WHITECYCLE aims to recover 200 kt/y of PET in EU and 640 kt worldwide. Tyre manufacturers have been able to use r-PET from mechanical recycling, but it needs very selected feedstock of clear bottles which waste volumes will be inadequate for large scale use. Sorting and processing PET from ELT remain an ambitious goal due to the complexity of tyre structures and to the lack of available technologies to address PET composite recycling. The technologies developed within WHITECYCLE for tyre recycling could be applied to many other specific rubber goods associating polymeric matrix and PET fibres like in belts, and seals.

Given their strict technical specifications, complexity, and massive volume of PET from tyre waste, tyres represent a very impactful use-case for <u>PET enzymatic recycling with already available waste</u>. The goal is to recycle PET from tyres into new tyres in an infinite circular loop.

1.1.3 USE-CASE #2: PET IN MULTICOMPONENT CLOTHING TEXTILES

The circular economy has become a major preoccupation for the textile industry during the last years. Consumer expectations about transparency, material origin and production conditions are increasing, pushing the industry to operate rapid, drastic, and sustainable changes particularly regarding end-of-life (EOL) clothing. **Clothing textiles are the fourth highest-pressure category for the use of primary raw materials and water, after food, housing, and transport, and fifth for GHG emissions**[7]. By 2025, the separate collection of textile waste will be compulsory in Europe[8]. There is thus a need for technologies to enable the circular reuse of this resource.

Recycling technologies are being developed, offering great perspectives for the industry (for instance, sorting technologies, mostly based on Near InfraRed Spectroscopy NIRS). However, they often rely on a mono-material, mono-colour, mono-layer feedstock which is not representative of collected clothing feedstock.

[3] Marco Marconi et al. / Procedia CIRP 69 (2018) 944 – 949
[4] https://www.statista.com/statistics/625275/global-tyre-market-volume/
[5] EASME. Recycling rubber to reduce noise, 2015, https://ec.europa.eu/easme/en/news/recycling-rubber-reduce-noise.
[6] European Waste Catalogue – EWC code 19.12.08
[7] EEA Briefing report Nov 2019.
[8] https://ec.europa.eu/info/news/new-waste-rules-will-make-eu-global-front-runner-waste-management-and-recycling-2018-apr-18_en

Indeed, **most clothing textiles are multi-material**, and multicolour, making their identification quite complex (Figure 4). **Such feedstock is mostly landfilled,** and their sorting remain an enormous issue. Multilayer textile clothing can be classified into different categories:

- Layers assembled with a seam or lasers including:
  - Suits, dresses, or cloth with a lining: once the garment is cut into pieces of 10x10cm most of the pieces are separated and can be considered monolayer
  - Winter jackets and cold protection including padding. Different layers assembled with multiple seams or lasers. Even when being cut most of the pieces will remain assembled together with at least 3 layers
- Layers that have been pasted on the whole surface. Only a chemical reaction can separate these layers
  - Coated textile
  - Textile laminated with a membrane

Figure 4 : Clothing shreds

The proportion of multilayer textile clothing among the multicomponent textile feedstock is precisely unknown at this stage as no recent study of characterisation of the specific feedstock have been run. Indeed, given the complexity of these piece, they are not recyclable at this stage and are mostly landfilled. An estimate can however by done, based on the type of clothing. Jacket, outdoor and sport clothing, rain and cold protection clothing and a large part of professional clothing are multilayer, which could represent 40 % of the textile feedstock, i.e., 13.2 Mt of PET containing multilayer textile clothing.

Based on production figures, it is reasonable to assume that on average, the chemical composition clothing textile waste is made of 50 % of PET **which corresponds to a feedstock of approximately 33 Mt.**

In the EU, a study published in 2019 showed that an average of 26 kg of textile was purchased per inhabitant and 11kg was discarded [9]. In 2030, with a considered waste textile feedstock based on 26kg and a population of 447 million, a 50 % content in PET and a 40 % multicomponent fraction, **it gives 2.3 Mt usable feedstock in the EU.**

WHITECYCLE targets the production of 1.85 Mt of r-PET coming from today's landfilled multicomponent textile waste. The future tools to recognize and sort could also be used in other fields such as multilayer packaging recycling.

1.1.4 USE-CASE #3: PET IN LAY-FLAT HOSES

The circular economy is still not a focus area in lay-flat hose production and use. The competition is focused on price and quality, not environmental impacts, like CO_2 footprint. Hoses are treated as disposables with few eco-considerations imposed by regulations or customers. WHITECYCLE will take an important role in changing this, and MANDALS wishes to be the first lay-flat hose producer in the world offering hoses made from recycled materials (see Figure 5).

Figure 5 : A lay-flat hose

A lay-flat hose consists in a PET-based weave that is totally embedded into thermoplastics or rubber material. The weave determines the strength of the hose, the outer layer protects the weave from abrasion, cuts, chemicals and UV, and the inside layer provides a leak-tight barrier for the fluid transported. MANDALS co-invented, produces, and sells a significant volume of the circular looms needed to produce lay-flat hoses. We estimate 1,500 circular looms worldwide. Based on this, we believe about **120 kt/year of hose is produced where 25 w% is PET, and that a similar weight is transformed to waste each year**.

For these 3 streams, WHITECYCLE targets to transform 2.56 Mt of EU mostly landfilled PET waste into 2.06 Mt of r-PET: 200 kt PET coming from waste tyres, 1.85 Mt kt from PET multicomponent clothing waste and 10 kt from hoses with a yield of 80 %. It will demonstrate the viability of the innovative technologies on currently non-valorised and available complex feedstocks.

1.1.5 Overall and specific objectives along with associated KPIs

The overall objectives of the WHITECYCLE project are to **develop a full, flexible, infinitely circular PET value chain based on** a white technology, namely, enzymatic depolymerization where the enzyme is a biological catalyst with high substrate specificity operating in environmentally friendly conditions. The project **aims at demonstrating**

the relevance of r-PET in the high-specification applications with positive effects on environment. The project will provide:

Original processes:

- New recognition and sorting technologies applied to two feedstocks: multilayer clothing and complex plastics (ELT, ELH) to maximize the PET fraction entering the enzymatic recycling process.
- Amorphization technology of semi-crystalline PET from complex waste which is key for any enzymatic process.
- Enzymatic depolymerization technology which enables to efficiently process mixed materials, unlike any existing conventional recycling processes limited by the necessary high purity level of their waste feedstocks.
- Purification of PET monomers coming from complex waste to manage high technical grade PET production.

During the project, the targeted yields for these first steps are respectively 90 %, 95 % and 80 %. After full industrialization the whole recovery targeted yield is 80 % regarded the targeted feedstock.

High-specifications application use cases:

- **Tyres (TRL8):** Pre-industrial scale manufacturing and testing of r-PET containing tyres (pilot series of 100).
- **Hoses (TRL6):** Pilot-scale manufacturing and testing of r-PET containing lay-flat hoses (1,500 m).
- **Clothing (TRL6):** Pre-industrial scale manufacturing of r-PET containing multicomponent garments (4 lines of clothing items)

Systemic circular economic approach, dissemination, communication, replication and social acceptance:

- Optimise circularity through the limitation of pollution and consumption of water, energy in the circular value-chain.
- Evaluate the life-cycle economic and environmental KPIs of the complete circular solution.
- Demonstrate the viability of project technologies with at least 4 additional brand-owners beyond the consortium during the project to enable the rapid adoption in additional sectors.
- Prepare the future commercialisation of developed technologies for broad-scale replication.
- Document processes and procedures along the entire supply chain with the FAIR data implementation protocol.
- Anchor concepts for sustainable industrial transformation, which foster social inclusion and transparency along with economic sustainability, while minimizing adverse impacts on the environment and society.

OBJECTIVE 1: To sort and separate PET from multilayer waste feedstocks - *MICHELIN, ESTATO, SYNERGIES, PPRIME, IRIS*	
PET sorting methods will be adapted to the complex waste feedstocks. Different processing technologies will be implemented towards ELT and clothing waste (i.e., combination of recognition, mechanical, aerodynamic, and electrostatic methods) to reach 80 % PET content from fibres/rubber mixes and multilayer textile clothing. Regarding ELT and ELH waste stream, a lab-scale unit will be optimised to run by PPRIME to produce **1.5 t of PET fibres** Then, a pre-industrial unit will be built, installed, and tested at ESTATO's facilities with **sorting capabilities of up to 5 t of PET fibres a day.** Regarding clothing waste stream, an optical and photonic based pilot unit system for polymer recognition will be developed and built by IRIS and used by SYNERGIES to manually sort multilayer clothing items according to their composition to allow the constitution of a usable feedstock of 80 % of PET content. Then, an automatic inline pre-industrial optical and photonic polymers recognition pilot unit will be developed and built by IRIS, adapted, and tested on the existing industrial monolayer textile waste sorting line at SYNERGIES. **The sorting capability of this pre-industrial unit will be up to 20 kt of textile waste a year**	
KPI	• **KPI #1.1**: Automated demonstrator units for the efficient separation of PET fibres from composites (ELT, ELH) and multilayer (clothing) with more than 80% of PET and no metallic parts. o Regarding ELT: 150 kg sorted ELT at M13, 1.5 t at M27 (TRL6). o Regarding clothing: 50 kg sorted multilayer clothing textile at M17, 0.5 t at M27 (TRL6). • **KPI #1.2**: Yield of 90% of PET recovered compared to total PET in ELT and complex clothing with significant PET content.

OBJECTIVE 2: To pre-treat separated PET waste - *MICHELIN, IFTH*	
Enzymatic depolymerization requires a low crystallinity level (< 15%) and high specific area to exhibit excellent monomer recovery yield. WP2 focuses on implementing first, a pilot-scale and second, a pre-industrial pre-treatment process of PET complex waste to reduce the crystallinity of concentrated PET feedstocks enabling enzymatic depolymerization Various physico-chemical principles for heating, and cooling will be explored to determine the best combination of processes (e.g., extrusion, radiation, quenching, activated gas, micronisation etc.) to provide WP3 with pre-treated PET waste.	
KPI	• **KPI #2.1**: Pre-treated PET waste < 15 % crystallinity: analytical 100kg (M12), 1 t (M28, TRL6). • **KPI #2.2**: Yield of PET recovered compared to total PET >95%.

OBJECTIVE 3: To recover purified monomers from complex PET at pilot scale and to recover purified monomers from high PET content waste at pre-industrial scale – *CARBIOS, MICHELIN*	
To perform the enzymatic depolymerization of PET contained in composite textile waste and complex textile clothing waste at pilot scale and develop a purification process suitable to produce high purity monomers from the low PET content waste feedstocks containing diverse impurities (obtain the same quality as virgin monomers). To produce 2 t of purified monomers from high PET content waste at preindustrial scale (2t/week).	
KPI	• **KPI #3.1**: 80% PET depolymerization in 24h at lab scale (50 g, TRL4, from ELT M12, clothing M15) • **KPI #3.2**: To produce recycled monomers with a purity comparable to those of petrochemical monomers 100kg in M19 for ELT feedstock and 100kg in M22 from clothing waste feedstock (TRL 6). • **KPI #3.3:** To assess best end-of life options for impurities filtered out of the reaction mixture at demo-scale (MICHELIN)(TRL6) • **KPI #3.4**: To produce 1-2 t of purified monomers from high PET content waste M22 (TRL 7)
OBJECTIVE 4: To produce high technical PET fibrous materials - *IFTH, KORDSA, DITF*	
To perform polycondensation and solid-state polymerization (SSP) on recycled monomer to obtain low and high molecular weight r-PET polymer with similar properties as virgin ones. To perform high speed spinning to produce textile yarn and high modulus low shrinkage yarn (HMLS). To perform twisting and dipping on high performance yarn to prepare tyre cords.	
KPI	• **KPI #4.1**: Polycondensation and SSP of r-PET (160kg M26, 1800kg M27) • **KPI #4.2**: Production of technical yarn (130kg of yarn M28, 800kg of HMLS yarn M30) • **KPI #4.3**: Production of tyre cord (50kg single end cord M28, 600kg dipped fabrics M34)
OBJECTIVE 5: To produce at pre-industrial level r-PET containing tyres, garments, and lay-flat hoses - *MICHELIN, MANDALS, INDITEX*	
Demonstrating in relevant industrial conditions the recyclability of PET from complex waste to challenging PET application, as tyres, technical hoses, and clothing with quality equivalent to those made with virgin PET (v-PET). The demonstration will prove, in a first phase targeting TRL5 on prototype processes, that r-PET is a relevant product for a high technical application: tyre (25 pilot tyres produced to check safety and performances), lay flat hoses (production and testing of 25 m hose where at least 40% or v-PET is replaced by r-PET) and suitable for commodity application such as garment field (20 m² multicomponent fabric produced to check feasibility, and performances). In the second phase, scale-up feasibility on industrial representative facilities will be evaluated (100 tyres, TRL8, 1,500 meters hoses, TRL6, 400 m^2 coated fabric and 4 lines of garments, TRL6)	
KPI	• **KPI #5.1**: Recycled PET tyre fabrication: 100 r-PET car passenger pilot tyres and tests – M45 • **KPI #5.2:** Recycled PET hose fabrication: 1,500 meter lay flat hose and tests – M45 • **KPI #5.3:** Recycled PET clothing fabrication: 400 m^2 multicomponent coated fabrics + 4 lines of garments and tests – M45
OBJECTIVE 6: To quantify the impact for sustainability dimensions for the use cases and implement the FAIR data protocol & prototyping for broader reuse - *ERASME, HVL, IPOINT, all.*	
The impact of use cases regarding social, economic, and environmental dimensions is quantified and qualified with assessment methods that evaluate the entire supply chain (social LCA). All project data is handled by a FAIR data protocol. A prototype is made available for reuse in a similar industrial context. The protocol supports communication, dissemination, and exploitation of project results within and beyond WHITECYCLE.	
KPI	• **KPI #6.1**: 1 LCA + LCC per use case (with a focus on quantifying environmental impacts) • **KPI #6.2**: 1 Social impact quantification/qualification per use case (incl. social acceptance analysis)
OBJECTIVE 7 : To transfer project results and collaborate with at least 18 companies beyond the consortium who will replicate circular recycling and production in many applications - *AXELERA, all.*	
Material transfer agreements and specific agreements will be signed with at least 12 brand owners beyond the consortium who will use r-PET from the project to replicate circular production use case, using budgets beyond the project. Similarly, the full process design package will be prepared for at least 3 PET manufacturers for enzymatic recycling. Likewise, agreements will be signed with at least 3 waste producers and/or recycling equipment producers to test their feedstocks and to adopt WHITECYCLE sorting and separation technologies. For more details, see section 2. The FAIR data reference implementation file will be developed into a blueprint for FAIRification of data in related industries (to the level of machine-actionability).	
KPI	• **KPI #7.1**: 12 brand-owners, 3 PET manufacturers, and 3 large-scale waste managers sign an agreement to test, upscale and/or deploy WHITECYCLE solutions • **KPI #7.2:** 1 use case with a prototype for an industrial FAIR data protocol is uploaded for reuse to the European Open Science Cloud.

1.1.6 List of key technologies and their progresses in terms of TRL during the project

At the project, the TRLs of the breakthrough processes proposed will be advanced, and these new ecotechnologies will be de-risked in the eyes of companies who will be eager to adopt and upscale them, as presented below.

Key techno.	Justifications
Sorting system for PET in ELT and ELH TRL3 → TRL6	A set of vibratory, aerodynamic, and electrostatic separator devices already individually available will be combined at lab scale to assess the sorting of the PET textile fraction from ELT shreds. Based on the results, modifications and enhancements to the whole process will be carried out to settle a new pre-industrial-scale process. **Pre-industrial unit will be built to target a 500 t/y separation of PET containing fabric from post-consumer ELT shred waste streams.**
Sorting system for PET contained in clothing shreds TRL3 → TRL6	IRIS will design and develop a novel recognition monitoring system based on different technologies used successively: characterization, feedstock processing, optical/photonic and automatic sorting technologies, with the aim of separating among the multilayered waste according to their composition, to constitute a stock containing more than 80% of PET content. After validation on off-line sorting (0.5t produced), the automatization of the processes and use of smart algorithms will guarantee a high speed of the recognition sorting system. **It will be upscaled and adapted to be integrated into SYNERGIES automated textile waste sorting centre with the aim of reaching an output of 20 kt/y of clothing shreds containing more than an 80% of PET, ready to be further pre-treated before PET recycling.**
Pre-treatment (amorphization + micronisation) TRL3 → TRL6	Experimental studies based on analytical mixes are done with thermal processes (heating/cooling through radiation, extrusion, quenching) and via the activated gas route and micronization. These processes are tested at 1 kg scale allowing the identification of optimised parameters. **A first batch at lab scale will produce 1 t of pre-treated waste with less than 15 % of crystallinity. A pre-industrial unit is then designed and built with a targeted through put of 200 t/y.**
PET enzymatic depolymerization TRL4 → TRL7	CARBIOS will perform depolymerization at laboratory scale (TRL4: 50 g waste in a 500 mL reactor). After validation of pre-treatment, **scale-up of depolymerization** will be performed at pilot scale (TRL 6: 100 kg of waste in a 1m^3 reactor). The purification process developed for high PET content waste will be adapted to low PET content waste and 100 kg of purified monomer will be produced from each of both feedstocks. CARBIOS will produce 1-2 t of monomers at the preindustrial unit (TRL 7: 2t/week) from high PET content waste.
Tyres use-case TRL5 → TRL8	At first, a first close loop on complex waste feedstock will validate the circularity concept. Due to limited quantity, pre-industrial/proto equipment will be used to make 25 tyres (spinning, single end cord dipping, tyre building). **The second loop based on high PET content waste will allow the use of representative industrial apparatus (spinning, dipping, calendaring lines) to build 100 prototype tyres to meet TRL8 requirement level**.
Hoses/Garment use-cases TRL5 → TRL6	**25 m lay flat hoses building trial will meet TRL6 requirement level**. Representative garments will be produced at first with 40kg of yarn produced on pre-industrial/proto facilities. The second loop will allow the use of full preindustrial condition (TRL6). This experiment will demonstrate relevancy of r-PET versus hose (1,500 m hoses) and garment (400 m^2 fabric for technical garments + 4 lines of garments) performance requirements.

1.2 Methodology

1.2.1 Project concept and main activities

The WHITECYCLE project's methodology derives from the ambitious idea of turning the whole PET value chain into a full circular loop. **This implies (i) to demonstrate the overall concept through exchanging materials from one process to the next at TRL5, and (ii) to at least pre-industrially upscale all the innovative processes throughout a circular chain while making them energy and economically efficient**. To serve this purpose, 2 circular sets of activities are carried out in the project (Figure 6).

(i) The first set is represented by the 8 disk sections of the inner part, i.e., the 8 pilot-scale processes (TRL3-5). At this stage, actual materials will be processed along the circular value chain (PET retrieval from 10 t of ELT and 2 t of clothing waste + revalorization it into 25 prototype tyres, 20 m^2 textile and 25 hoses). (ii) The second sets of activities are on the outskirt of the disk and are split in two: the upstream processes will be upscaled as opposed to the downstream processes that will be implemented to carry out PET retrieval and revalorization from 2 t of PET bottles into 100 tyres, 1,500 m hoses and 400 m^2 of fabric for technical garments that will be used to manufacture 4

lines of garments. On top of this approach, a comprehensive sustainability analysis of the loop is implemented consisting of characterizing and revalorizing the by-products all along the value chain.

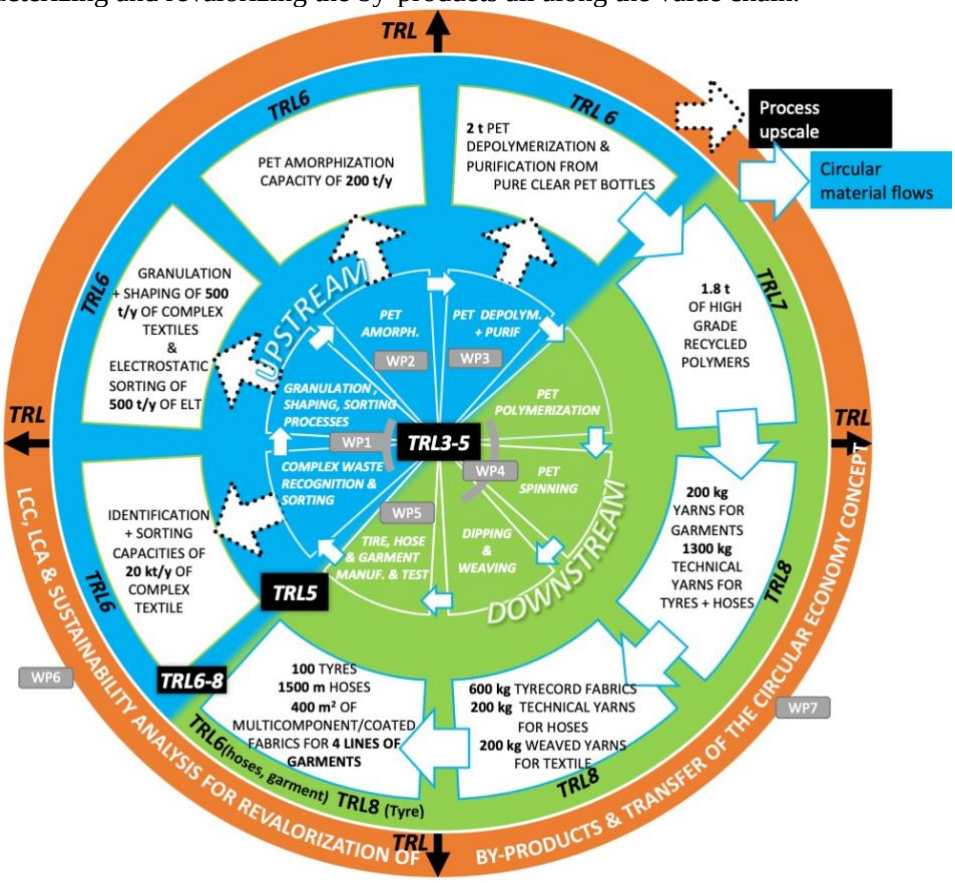

Figure 6 : WHITECYCLE is built around two sets of activities

At project's end, the technologies will be sufficiently de-risked in the eyes of industry that the corresponding value chains will become circular within 4 years after the project. The methodology is presented in Figure 7.

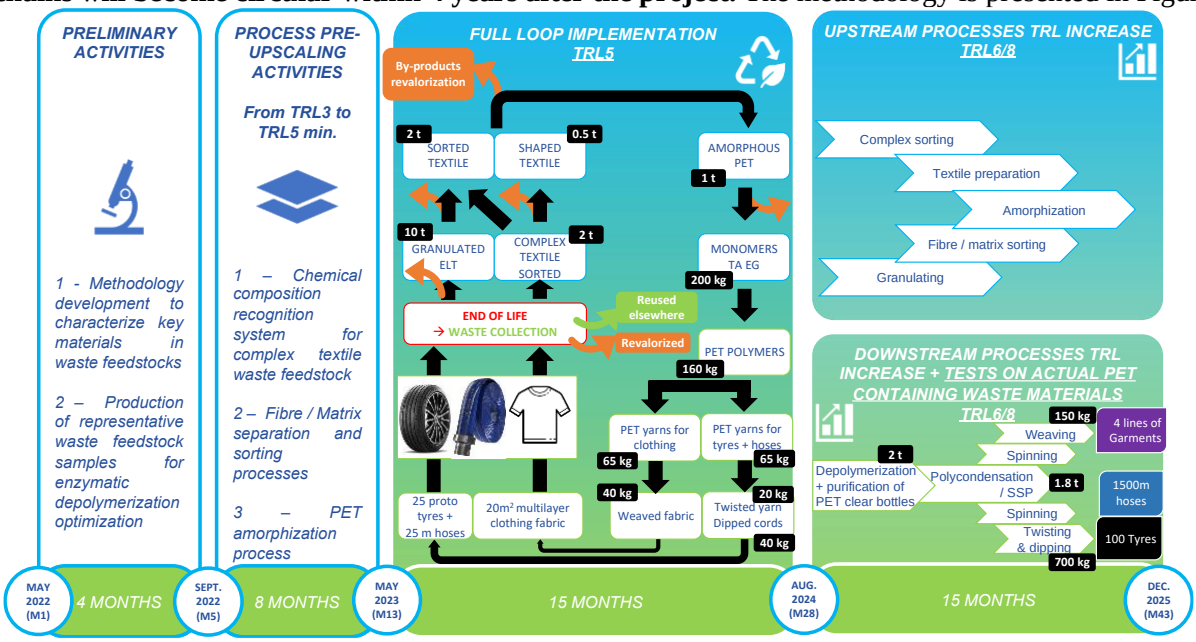

Figure 7 : Overall concept of the WHITECYCLE project

1 - 2 - First, IRIS and MICHELIN will define the most appropriate sets of optical technologies (NIIR, LIBS, etc.) to be used for the recognition of the chemical composition (i.e., the PET content) of the textile waste. In the meantime, SYNERGIES, ESTATO, ITFH will prepare representative waste samples that will be provided to CARBIOS to validate the enzymatic depolymerization process and to develop the monomer purification.

3 – 4 - 5 - Second, a process pre-upscaling phase will be taking place dealing with (i) developing a PET recognition demonstrator device (IRIS) that will be exploited by SYNERGIES for the sorting of multilayer textile wastes, (ii) developing the fibre/matrix electrostatic separation demonstrator (PPRIME & MICHELIN) to retrieve the textile

fraction from ELT shreds and (iii) developing the PET amorphization process (IFTH). At this stage, these processes will reach TRL5 and be ready for the full loop activities.

6 – Third, demonstration activities will be carried out at TRL5 involving all the partners of the WHITECYCLE value chain. 2 t of textiles waste and 10 t of granulated ELT and ELH will be sorted, shaped, and prepared to ultimately increase the PET concentration of both waste streams, reaching up to 80%. Both streams will be amorphized, micronized (IFTH and MICHELIN) and depolymerized (CARBIOS) leading to the retrieval of up to 200 kg of monomers. Then, DITF will oversee the production of 160 kg of polymers:

1. The high intrinsic viscosity IV (0.95) r-PET polymers will be further processed through spinning (IFTH), dipping, twisting (KORDSA) to ultimately be implemented in the production of 25 tyres (MICHELIN). Yarns will be also processed by MANDALS to make 25m of hoses
2. The Low IV (0.65) r-PET will be further processed through spinning and weaving (IFTH) to manufacture 20 m^2 of multicomponent clothing fabric (INDITEX).

7 – The activities will be decoupled as follow:

- ON THE UPSTREAM PROCESSES: Upstream processes TRL increase to reach 6 to 8 without exchange of materials from one process to the next. Upscaled processes validation.
- ON THE DOWNSTREAM PROCESSES: Downstream processes TRL increase to reach 6 to 8 with an exchange of material from one process to the next, starting from the enzymatic depolymerization. Indeed, CARBIOS will depolymerize 2 t of clear PET bottles and the resulting monomers will be further processed by the partners (IFTH, DITF, KORDSA) to retrieve 450 kg of cords for garments and lay-flat hoses and 700 kg of dipped fabrics for tyres. MICHELIN will manufacture and test 100 tyres, MANDALS will manufacture 1,500 m lay-flat hoses, and INDITEX will weave 400 m^2 coated/multicomponent fabric and manufacture 4 lines of technical garments.

1.2.2 Main challenges related to the chosen methodology

1.2.2.1 ELT sorting and separation of textile fibres

Main issues: End of Life Tyre (ELT) shredding generates large masses of a dirty textile mixture of rubber and fibrous materials. Today it is mainly incinerated for energy recovery, sometimes with some added value for cement plants. The use as a secondary raw material is very limited. **The main issues are: (i) the complexity of this feedstock containing non-magnetic flying fibres and rubber dust, and (ii) the lack of efficient sorting systems dedicated to this composite.**

State-of-the-Art: Dirty textile is composed of a mix of equal parts of rubber and fibrous materials. Clean textile is composed of 75 % of PET fibres. Shredding tyres generate different types of products.

- Shredded parts, over 20 mm and bigger chips which include textile[10] and sometimes tyre steel parts.
- Rubber granulates or chips, from less than 1 mm to 20 mm which are sorted from dirty textile fibres[11].

In the field of ELT shredding, through continuous granulating process (grinding or knife cutting), Eddy current is used to separate metallic parts from polymeric parts, sieving is used to separate the different rubber parts regarding their size. Finally, airflow/cyclone effect is used to remove the flying particles (rubber dust and fibrous elements). Several downcycling strategies have been studied: expansive soils, geopolymer concrete, but without traction. Within REFIBRE European LIFE project a process to clean textile fibres up to 99% was mentioned, coupled with a compacting operation, toward bituminous conglomerates and plastic compounds applications. The WHITECYCLE project will position itself as logical successor of REFIBRE but focused on technology allowing circular economy *ad infinitum*. The first stage is the constitution of high-rate PET fibres feedstock which is a challenge due to their low difference in mass and density (1.2 for rubber vs 1.35 for PET), no magnetic effect, very small pieces (< 20 µm fibres) and dark colour due to rubber. No solutions are available.

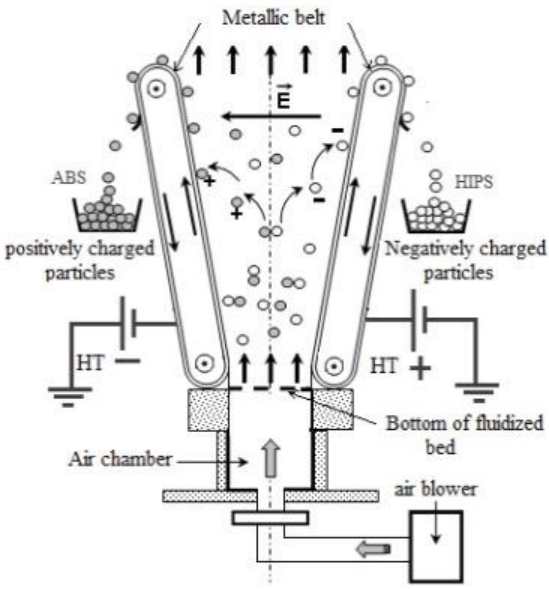

Figure 8 : System developed by PPRIME to sort ABS/HIPS mixes from WEEE

Background: PPRIME is specialized in applied electrostatics research and has conducted extensive work on electrostatic separation of granular or micronized plastics. Dascalescu reported ability to separate granular plastic using tribo-aero-electrostatic separator in the context of waste of electric and electronic equipment (WEEE)[12] as seen on Figure

[10] *Aliapur powergom technical sheet, Aliapur geogom technical sheet*
[11] *Aliapur technigom technical sheet*
[12] Tribo-aero-electrostatic Separator for Coarse Granular Insulating Materials M. Miloudi, L. Dascalescu, et al, 2013

8. **PPRIME also reports combination of shredding and tribo-electrostatic separation**[13]. PPRIME has 6 experimental installations to study tribo-aero-electrostatic phenomena. This valuable knowledge will be applied to purify the textile fraction of ELT shreds. **MICHELIN and PPRIME carried out encouraging pre-studies leading to feasibility at 100 g scale of sorting fibrous content from rubber dust.**

Beyond State-of-the-Art: The project will bring a totally new combination of technologies (shredding, sieving, tribo-aero-electrostatic process) to sort PET fibrous part from shred. Hence, the project will reach lab scale and consequently pilot-scale tests.

1.2.2.2 Post-consumer clothing PET recognition, sorting and separation

Main issues: Nowadays, a relevant part of used and new textiles ends up in landfills, although partial recycling of the materials would be possible. To do so, **it is necessary to know the exact composition of the materials and sort them according to their nature**. Composition recognition based on the cloth label tag is not reliable and take too much time to be cost efficient. The sorting needs to be automated. **If surface detection technologies are promising for monolayer textile recognition, they are not suitable for multilayer** as they can only identify one side of the textile.

State-of-the-Art: There are already some methods available for material identification that bring accurate results (differential scanning calorimetry (DSC), gas or liquid chromatography, inductively coupled plasma mass spectrometry (ICP-MS), X-ray fluorescence (XRF) etc. Nevertheless, they are offline lab methods and not suitable for automated industrial inline identification. Currently, manual methods are applied. However, its application is limited, because staff cannot sort different materials with the required velocity and accuracy, and the exact composition of textiles cannot be identified manually given the large rage of existing blends.

In this sense, Near InfraRed spectroscopy (NIRS) is widely used in waste industry. During past years technologies providers identified the textile as a potential market for their application and started to adapt them. Valvan balding system developed the Fibresort technology[14] to sort monolayer complete cloths. Recently Tomra Sorting system announced the launch of new industrial textile sorting plant in collaboration with Stadler[15]. Here again only mono layer can be identified. These technologies bring some limitations for multi-layered, black, or complex textile. Consequently, it is necessary to bring additional techniques to be used in combination.

Background: **IRIS has a large experience** in developing fast, customized, online cutting-edge photonic-based solutions **for different inline and real-time processing monitoring focused on Vis-NIR and SWIR spectroscopy and other techniques (Hyperspectral imaging (HSI), Laser Induced Breakdown Spectroscopy (LIBS), etc.), coupled with advanced chemometrics analysis and Industrial Artificial Intelligence Solutions (e.g., machine learning).** IRIS add high values for driving improved process quality and efficiency in the identification of the substances and the processability of the information. These tools have been proven to be accurate in identifying the composition of materials in mono and multi-component flows. Specifically, IRIS' results in the Multicycle[16] project for the plastic waste sector showed the viability of using NIR-HSI to identify the different layers in laminated plastic packaging films, including PET, and showed the feasibility of solutions like LIBS for black plastic and complex materials to overcome the limitations of Vis-NIR and SWIR-HSI[17]. In textiles, IRIS has already applied HSI for the determination of concentration of a medical active compound on wound textiles or CN-NL nanoparticles in novel textile[18].

MICHELIN has implemented Terahertz technology to determine nature and thickness of multimix compounds.

SYNERGIES is setting up an innovative industrial plant for dismantling and sorting monolayer textiles. To do so, Synergies is involved in several research projects aiming to design this innovative project by working with the equipment provided on the modifications needed on their existing equipment's to answers textile needs.

Beyond State-of-the-Art: WHITECYCLE will envisage to develop a recognition system for PET contained in multi-layered or coated clothing shreds based on the material composition identification to define the respective recycling route and separate them from non-recyclable parts. The final recognition prototype will encompass Vis-NIR and SWIR-HSI and LIBS technology with Artificial Intelligence (AI) Solutions (e.g., machine learning) to define fast and accurate models to elaborate data and provide real-time identification of multi-layer textile clothing. Both soft and hardware will be optimized, and their application extended to operate at inline mode in SYNERGIES' waste clothing textile facility and be in communication with existing sorting equipment providing the compositional information. The AI solution will be plugged on the RECYCLE platform that is developed by Refashion.

[13] Shredding as simultaneous size-reduction and tribo-charging operation, Particulate Science and Technology, 2019, Dascalescu et al.
[14] http://www.valvan.com/products/equipment-for-used-clothing-wipers/sorting-equipment/fibresort/
[15] https://recycling.tomra.com/blog/tomra-and-stadler-deliver-the-worlds-first-fully-automated-textile-sorting-plant
[16] *H2020 MULTICYCLE Advanced & sustainable recycling processes and valuechains for plastic-based multi-materials (SPIRE 820695).*
[17] *Araujo-Andrade C, Bugnicourt E, Philippet L, et al. Review on the photonic techniques suitable for automatic monitoring of the composition of multi-materials wastes Waste Management & Research. 2021;39(5):631-651. doi:10.1177/0734242X21997908*
[18] https://www.mdpi.com/2079-4983/11/2/32

1.2.2.3 Sorted PET waste pre-treatment

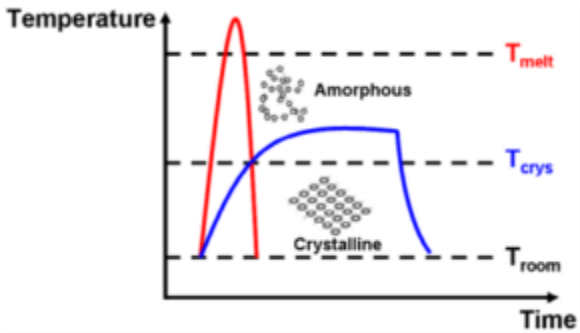

Figure 9 : crystallization & amorphization process

Main issues: The depolymerization process needs a high content of PET:

1. with an **amorphous microstructure** to allow the active site of the enzyme to fully shape polymer molecules and catalyse the hydrolysis of ester bonds
2. available for the enzyme (not coated by extra-materials) with **high specific area** and wettability

Composite materials and especially technical ones present the highest challenge regarding those two points. In hoses, tyres and some technical garments, the PET used are highly crystalline (crystalline rate > 40%) and very high intrinsic viscosity (IV>0.8dL/g). The multilayers and composites in the use-cases targeted long-life products so adhesion need to be strong between fibres and matrix (Figure 9).

State-of-the-Art: The introduction of white chemistry, and specifically enzymatic catalysis increases the need for amorphous organic microstructure. It is reported in other fields such as polysaccharide pre-treatment to allow enzymatic modification (Dome[19]). As described by Singh et al.[20] one main issue is to define an efficient pre-treatment amorphization process. **The process of amorphization of an organic structure is not very developed**. Two major ways can be chosen to reduce the crystallinity of a structure. The first is a **thermal route** (Figure 8) with **heating** above the melting temperature (through extrusion or oven) followed by a **rapid cooling** or "quenching" preventing the polymer from crystallizing. For all thermal routes, the extra-material could highly interfere through melting, degrading. A second route is radiation or **activated gases of low or high energy**. The morphologic changes induced in macromolecular materials resulting from scission, recombination may lead to various amorphization phenomena. The increase of surface area is more known with micronization approach or generation of high aspect ratio objects (fibres, foams).

Background: CARBIOS has developed a pre-treatment process for high PET content plastic waste to TRL 6 to decrease its crystallinity and increase its specific surface area[21]. It consists in an extrusion of PET flakes, followed by a quick cooling and micronization or extrusion expansion in one step.[22] This process is inadequate for PET waste containing a high level of extra material generate heavy fumes. MICHELIN has developed different treatments to modify the surface energy of tyre cords, and among these, plasma treatment was proven to amorphize textile surface[23].

IFTH possesses important lab scale range of processes to shape fibrous material into non-woven product, conduct treatment on polymer to modify microstructure (thermal stabilization, extrusion) or enhance wettability (gas, plasma)

Beyond State-of-the-Art: **Different technical paths will be developed by MICHELIN and IFTH with thermal or activated gas routes to reach amorphous PET with high specific surface, in presence of remaining impurities**. The best way to conduct polymer pre-treatment will be implemented to preindustrial scale of 20 kg/h use (TRL 6) and evaluated through LCA.

1.2.2.4 Enzymatic PET depolymerization and monomer purification

Main issues: **The recycling of complex waste such as composite plastic waste or complex clothing waste is limited by:**

- **Quantity of impurities** (colorants, pigments, rubber, elastane, cotton, coating…).
- **Low PET content high intrinsic viscosity (IV)** of 0.95 dL/g of PET in the composite (ELT, ELH).

The recycling of such waste requires a process able to:

- Depolymerize very selectively PET into monomers without reacting with impurities
- Efficiently purify the monomers from the mixture to obtain purity equivalent to oil-based monomers.

State-of-the-Art: Today, mechanical recycling that consists in melting and mechanical filtration of the PET waste, is the only solution developed in Europe at industrial scale to recycle PET. This process does not allow the removal of impurities from the waste feedstocks inducing major limitations: coloured waste can only be recycled for darker applications, only extra pure inlet waste can be treated to target demanding (high IV) applications, limited number of recycling loops.

[19] Changes in the crystallinity degree of starch having different types of crystal structure after mechanical pre-treatment, K. Dome &all, Polymers, 2020,12,641
[20] Singh et al., Techno-economic, life-cycle, & socioeconomic impact analysis of enzymatic recycling of PET, Joule (2021), doi.org/10.1016/j.joule.2021.06.015
[21] *WO2017198786A1*
[22] *WO 2021/123299, WO 2021/123301*
[23] FR2991344, FR2996806, FR3013625, 3013622

Alternative technologies are under development, but not industrial yet in Europe. Chemical depolymerization of PET allows to go back to monomers that can be purified and repolymerized to obtain high value PET. Chemical depolymerization occurs generally at high temperature resulting in degradation of impurities or fewer selectivity of the catalysts towards PET. Indeed, most chemical processes cannot treat multicomponent or complex feedstocks.
A more selective process is biological depolymerization using enzymes. Enzymatic depolymerization of PET has never been achieved before in satisfying yield and today existing methods are limited to laboratory experiments and display low yield (50% with seven 3-day rounds of enzyme addition[24]). European project P4SB has focused on other biological tools for PET degradation, limited at lab scale without communicated efficiency.

Background: From 2015 to 2017, in the framework of the Thanaplast™ project financed by Bpifrance, CARBIOS has identified and optimized an efficient enzyme able to selectively depolymerize 90 % of PET waste in only 10 h at moderate temperature (60-75 °C) in water, without any organic solvent and at atmospheric pressure (TRL5 at the end of the project). Furthermore, **the enzyme is highly selective for PET (Figure 10) and is thus very tolerant to multimaterial feedstocks** (no need of drastic sorting for high purity of PET in waste, no inhibition by other polymers, dyes, or pigments). **The enzymatic recycling technology developed by CARBIOS has been acknowledged as a breakthrough by the scientific community with a publication in the prestigious Nature.[25]**

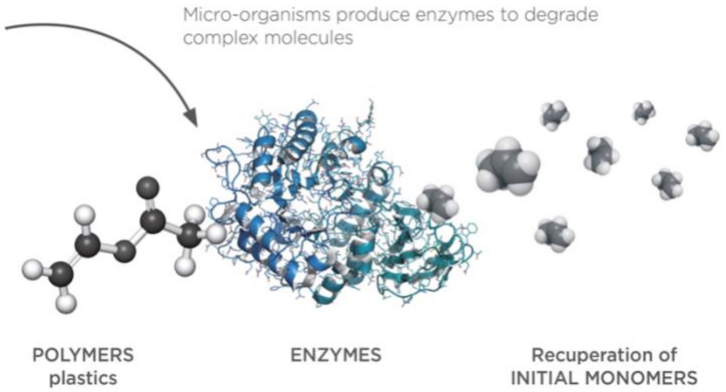

Figure 10 : Enzymatic depolymerization of polymer

Scale-up of the enzymatic recycling process has already been performed from laboratory scale (500 mL reactor treating batches of 50 g of PET waste - **TRL4**) to pilot scale (1 m^3 reactor treating batches of 100 kg of PET waste - **TRL6**) without any loss of performance. Clear food contact bottles have been produced by L'Oréal, Nestlé Waters, PepsiCo and Suntory Beverage & Food Europe from enzymatic r-PET obtained from high PET content colored plastic waste[26]. This result clearly demonstrates the performance of enzymatic recycling to perform circular economy without loss of quality. Various high PET content wastes (coloured and opaque bottles, white textile production scraps and post-consumer-coloured clothes) have been treated at pilot scale. Importantly, the monomers resulting from the CARBIOS process coming from colored or opaque plastic waste have, once repolymerized, made it possible to obtain high tenacity fibre meeting MICHELIN's requirement for tyres application.[27]
CARBIOS has inaugurated a preindustrial unit (2 t of PET waste treated per week in a 20 m^3 reactor, **TRL7**) in September 2021. After commissioning and first trials, first t of purified TA and purified MEG from high content PET waste will be produced by the end of 2022. The preindustrial unit will be able to treat 100 t / year of PET starting from 2023. Operations of this preindustrial unit will be supported by a European LIFE project from October 2021 to December 2024. **CARBIOS will thus be able to produce and purify 1-2 t of monomers from high PET content waste at the preindustrial unit (TRL7) for this project.**
CARBIOS is planning to begin the building of the first industrial plant of 40 kt in 2023 reaching TRL8 by 2025.
In parallel of this scale-up, CARBIOS is starting to extend the process to complex waste and in collaboration with MICHELIN, has performed first depolymerization trials at lab scale of analytical mixtures of rubber/PET with 70 to 90% PET content that exhibits correct yield.
Beyond State-of-the-Art: Leveraging the selectivity of enzymatic depolymerization offers the opportunity to expand the eligible range of waste feedstocks for recycling. CARBIOS will validate that a high yield can be achieved with the pre-treated complex waste at TRL 4. Then, CARBIOS will optimize the conditions of the whole process for ELT, ELH, and complex clothing waste at TRL 6 **showing that enzymatic recycling can be adapted to complex feedstocks not recycled today**.

1.2.2.5 Polymerization (PET polycondensation and SSP) & PET filament – spinning twisting and dipping,

Main issues: During the enzymatic depolymerization of PET, purified terephthalic acid (TA) and mono ethylene glycol (MEG) are generated. However, due to different sources of PET waste, some other chemicals acting as impurities can also be present after the purification. While TA and MEG are good for making up a new r-PET polymer, other impurity chemicals can have detrimental effect on the polycondensation of r-PET quality. For

[24] Kaabel, bioRxiv, 2020, Clean Enzymatic depolymerization of highly crystalline polyethylene terephthalate in moist-solid reaction mixtures
[25] Tournier et al.,Nature volume 580, pages216–219 (2020) An engineered PET depolymerase to break down and recycle plastic bottles
[26] CARBIOS' press release: 24/06/2021
[27] CARBIOS' press release: 23/04/2021

example, monofunctional phthalic acids can act as chain terminating agents, resulting in low molecular weight r-PET polymers. As such, achieving high intrinsic viscosities (IV) becomes difficult if the number of impurities is high.
The amount of crystallinity is very important in terms of achieving the High-Modulus Low-Shrinkage (HMLS) property. As such, if the polycondensation does not afford high molecular weight (i.e., high IV values), it is very difficult to achieve the desired technical yarn, and further adequate twisted, dipped cord.
The main issues regarding the PET multifilament spinning are:

- Spinnability of the r-PET due to the presence of residual impurities or the polymer chains organization
- Spinning and drawing behaviours of r-PET filaments to reach required specifications for tyre and hose
- Scaling up to produce PET multifilament yarns in large quantity (700 kg) for end-users' partners.
- Low amount of r-PET high IV chips which will not be significant enough to feed the preindustrial processes.

During twisting, and weaving quality problems may arise due to fluffy cord - broken filaments, which in turn reduce the mechanical properties of the final tyre-cord and hose-weave fabric During dipping, quality problems such as uneven coating of the adhesive layer may occur and cause inefficient adhesion between fibres and rubber compound.
State-of-the-Art: **The different processes involved are well defined and industrialized for over 20 years**:

- the direct polycondensation process has been developed only once the purity of the Purified Terephthalic Acid PTA reach an acceptable level, and with a specific range of catalysts
- the solid-state polymerization SSP at temperatures between the glass transition and the onset of melting to reach correct intrinsic viscosity IV (0.6 for garment application, 1 for technical application)
- the yarn spinning and drawing at high speed
- the twisting of cord on direct twisting machine
- weaving, knitting, braiding and consequent dipping or coating for the various application

The introduction of perturbations in this stream has a significative effect especially on optimized products.
Fibres can be broadly divided into two segments depending on the PET polymers characteristics and the spinning process involved in the production of the yarn:

- Apparel textile: LOY (Low Oriented Yarn), POY (Partially Oriented Yarn), FOY (Fully Oriented Yarn) or FDY (Fully Drawn Yarn) multifilament yarn with mechanical properties until 4 cN/dTex in tenacity, 60% in elongation at break, 8 % in shrinkage in general with molecular weight of 20,000 g/mol, which refers to 0.55 < IV< 0.67 dL/g
- Technical textile: HT (High Tenacity), HMLS (High Modulus Low Shrinkage), LS (Low Shrinkage), SLS (Super Low Shrinkage) having a minimum of 6,5 cN/dTex in tenacity with high molecular weights (IV>0.95 dL/g).

The macromolecular chains organization in the polymer filament is a key parameter to reach a HMLS yarn properties. The preferably orientation of the polymer chains in the drawing direction during spinning process allows controlling the amorphous and crystalline structures (Figure 11) to obtain high tenacity and low shrinkage filaments.

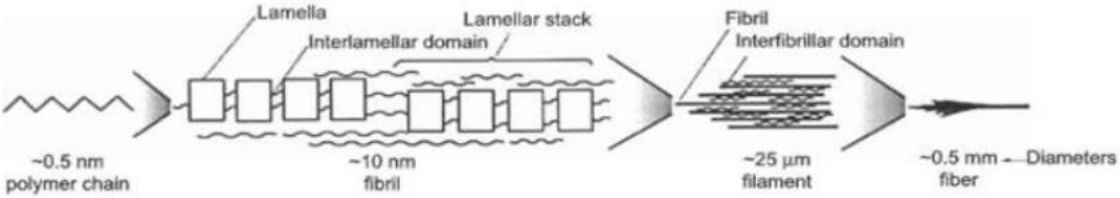

Figure 11: Polymer chains orientation in a spin-drawn filament

Flakes of PET bottles are first transformed into granulates by compounding process and then processed by melt spinning to obtain multifilament yarns or staple fibres. For instance, Indorama Ecorama® fibres are staple fibres produced from r-PET Ecorama® resin based on bottle. The IV of these resins are comprised between 0.6 and 0.7 d/dL and used for textile applications. r-PET industrial yarns are also available. Morssinkhof Sustainable Products produces the MOPET-r Yarn 4235T from used bottles and other post-consumer and post-industrial packaging for hoses application (tenacity = 7 cN/dTex and shrinkage=3%). **Until now, no r-PET yarns coming from tyre or complex textiles are commercially available for industrial and textile applications.**
PET yarn is turned around itself to form a twisted cord through different methods such as, ring or direct twisting. There is no weaving process in single end cord (SEC) production in demonstration-scale production. In preindustrial scale studies twisted yarns are woven together held together with weft fibres. The last step called “dipping” is coating cords with a heat curable adhesive layer. The fabric is passed through a pool of water based adhesive formulations and then further cured. The main parameters are glue recipe, process conditions (time, temp., tension during glue curing) to reach required mechanical and adhesive performance.
Background: DITF has long experience in novel raw material and catalyst systems, process aids and co-monomers especially on polyester on their dedicated facilities (20L) as noticeable through numerous AiF projects 14687, 12428 and 12318. French Textile Technical Center, IFTH has strong know-how in spinning processes and yarn applications.

For both clothing and technical textile, PET multifilament yarns have already been prototyped by IFTH by the mean of a melt spinning semi-industrial line and the use of commercial PET granules with an IV until 1.1 g/dL.
KORDSA have high knowledge on **e**sterification and SSP processes through close contact to polyester chips producers. KORDSA has also long experience in spinning processes of polyester multifilament tyre grade yarns at industrial scale and is industrially producing PET dipped cords adapted. Their huge experience on both ring and direct twisting on mono-component and hybrid cords and on glue formulations for hard applications will be of interest to deal with r-PET yarns (1 out of every 3-passenger car tyre is reinforced by KORDSA). Among other activities, INDITEX spins garment-grade yarns. MANDALS twist yarn on ring twisters to desired turns per meter and dtex and produce circular weaves on circular looms that has been developed, patented, and produced at the site.
Beyond State-of-the-Art: **Novel recipes of polycondensation and SSP** will be developed to eliminate or tolerate the putative residual impurities to obtain the r-PET comparable with commercial PET clothing and technical grade. **The spinnability of the r-PET polymer** from complex waste will be studied by the mean of the IFTH semi-industrial melt spinning line. The pre-industrial scale studies of the r-PET polymer will be studied by the mean of the KORDSA and INDITEX pre-industrial scale lines. **Optimization studies will be conducted by KORDSA, INDITEX and MANDALS on the bases of varying levels of twisting of r-PET, weaving and dipping processes**.

1.2.2.6 Tyre, hose, and clothing evaluation

Main issues: **Tyre and hose uses are high demanding for HMLS r-PET: new r-PET will require evaluation through extreme solicitations.** The reinforcement will face complex thermomechanical solicitations during tyre and hose manufacturing process and usage (during twisting, thermal treatment, curing, weaving…). These solicitations lead to abrasion and fatigue mechanisms which induce cord damage that needs to be evaluated in a representative environment.
State of The Art: Impact prediction of cord properties, specifically r-PET on long lasting performance of tyres remains a field of research. It has been the subject of numerous publications[28][29][30]. However, none of these tests correctly predicts tyre damage[31] such as fatigue and thermo-mechanical degradations. Scientific research studies both in the perimeter of filaments degradations[32], and in the perimeter of mesoscopic filament geometry[33]. r-PET coming from bottle recycling find some outputs in textile. Recently, CONTINENTAL announced the usage of recycled PET in tyre production[34][35]. This option, integration of r-PET coming from chemical or mechanical recycling ways into tyre, is also studied by MICHELIN. However, the main limit of this alternative is the un-accessibility of mixed/impure materials (e.g., PET coming from tyre, hoses, or garments wastes) as they are non-adapted for mechanical or chemical recycling.
Background: The critical point of r-PET is feedstock management and a probable severe competition with the packaging sector for raw materials distribution. More globally, MICHELIN actively researches materials to meet 100% Sustainable Material Rate by 2050 using Renewable and Recycling technologies.
MANDALS has been a forerunner in lay-flat hose development in the world for one hundred years. To our knowledge, there is no competitors that has plans for developing eco designed hoses. Hoses with r-PET will therefore likely be disruptive in this market.
To evaluate new industrial supplies of PET, MICHELIN requires fabric weaving, dipping, and rubberizing. It usually demands minimum 3 t of textile products per evaluation. Alternative to demonstrate tyre performances with small quantities is to use Single End Cord produced by textile industry at laboratory scale and to reconstitute a small surface of rubberized fabric processing laboratory drum. Simulation by physical characterizations and calculations of thermomechanical behaviour of textile cords, prediction of tyre industrial scale performances by laboratory properties will be complementary elements for industrial performance prevision.
For MANDALS, current methodology to evaluate new PET yarn requires laboratory tests of some 100 g of yarns, for laboratory and pilot scale tests and then pilot scale production of hoses (some kg of yarn). Finally, full-scale tests require some 100 kg of yarns to finished product for small diameter hoses.
Beyond State-of-the-Art: **The demonstration of close loop for so complex product as multicomponent clothing, hoses and tyres regarding production and performance will be a breakthrough.** Tyre, hose, and clothing eco design will be driven by LCA. New methodologies will be required to assess interaction level at each step of PET

[28] [Wilson, 1951] Tyre-Cord Compression Fatigue, Wilson M.W, Textile Research Journal 44-47, 1951
[29] [Hearle, 1967] Fatigue in fibres and plastics (a review) Hearle, J.W.S. Journal of Materials Science Volume 2, Issue 5 Pages 474-488, 1967
[30] [Canaveri, 1979] Laboratory Machine to Evaluate the Resistance of Tyre Cords (Textile or Steel), ASTM Special Technical Publication, pp. 110-121 1979
[31] [Le-Clerc, 2006] Mécanismes microstructuraux impliqués dans la fatigue des fibres thermoplastiques Le Clerc C., Ecole Doct. Ecole des Mines de Paris, 2006
[32] [Bunsell, Le-Clerc 2018] Handbook of Properties of Textile & Technical Fibres, Textile Institute Books, P.595-618, 2018
[33] [Sibellas 2019] Experimental study of the fibre orientations in single and multi-ply continuous filament yarns, Journal of the Textile Institute, September 2019.
[34] https://www.tyretechnologyinternational.com/news/materials/continental-to-use-recycled-pet-in-tyre-production.html consulted in August 2021
[35] https://www.european-rubber-journal.com/news/conti-upcycle-pet-bottles-tyre-reinforcement-2022 consulted in August 2021.

recycling such as, cleaning, pre-treatment, depolymerization, monomer purification, polycondensation, spinning. **Tyre, hoses, and garment performances will be demonstrated with moderate r-PET quantity** through most updated lab-test protocols and low cord consumption.

1.2.3 Research and innovation activities or projects whose results will feed into the WHITECYCLE project

The know-how generated in the previous relevant projects is presented below and will be translated into WHITECYCLE to generate globally unique and impact-focused foreground.

EU related projects	Main objective of the project	Main results linked to WHITECYCLE project
BLACKCYCLE *2020-2023* H2020	It aims to create, develop, and optimize a full circular value chain for the recycling of rubber fraction of ELT to secondary raw materials for tyres (r-carbon black, r-plasticizers) with no waste of resources in any part of the chain and specific attention to the environmental impact (*MICHELIN, ESTATO, AXELERA)*	SRMs will be used to manufacture new ranges of recycled car and truck tyres. MICHELIN will bring knowledge in ELT industrial-scale recycling processes and in tyre eco-design.
RESYNTEX 2015-2019 H2020	It aims to develop circular economy for textiles. This project made improvements in detection of different mono-component materials through hyperspectral technology and on enzymatic depolymerization of natural fibre.	WHITECYCLE will pursue these developments for sorting multilayers and coated products and for efficient depolymerization of PET fibres.
POLYNSPIRE *2018-2022* H2020	It aims to demonstrate a set of innovative and sustainable solutions for efficient plastic recycling, with a focus on PA and PU. It addresses 3 innovation pillars which are implemented in operational environments reaching TRL 7: chemical, mechanical recycling and valorisation (*KORDSA)*	Industrial knowledge about the recycling of polymers.
REFIBRE *2015-2018*	It aims to overcome two existing barriers limiting ELT fibre recycling: (i) absence of economically affordable applications for fibre reuse, and (ii) lack of proper treatment technologies for recovery of clean fibres. The goal is 100% valorisation of the ELT fibre and transformation to useful secondary raw material.	Specific knowledge regarding recycling processes of ELT fibres to be used as secondary raw materials will be brought by MICHELIN to the project. However, this project leads to downcycling.
FIBRESORT *2016-2020* INTERREG	Reduction of the impact of virgin textile materials, and the development of business models and open markets for the growing amounts of recyclable textiles in North-West Europe (NWE). It aims to realise the implementation of FIBRESORT technology as the new industry standard to enable high value textile-to-textile recycling.	SIPTEX and FIBRESORT technologies are one of the technologies that could be integrated into SYNERGIES mono layer sorting plant but is unable to sort multilayers. Technology developed by IRIS in the project could come in complement to this technology
SIPTEX *2016-2018* Swedish program	SIPTEX is a project that evaluates automated textile sorting by building a pilot. The recognition and sorting equipment are based on near-infrared (NIR) technology.	
MERLIN *2021-2024* H2020	It offers innovative solutions for all the processes required to increase the quality and rate of recycled plastic materials coming from multi-layer packaging waste: (i) sorting (optical sensors, AI and robotics), (ii) delamination (depolymerization and solvent-based processes), (iii) recycling and (iv) validation (developing rigid and flexible packaging solutions).	Specific knowledge regarding the sorting of multi-layer packaging waste combining optical sensors, AI and robotics and the depolymerization through solvent-based techniques.
DECOAT *2019-2023* H2020	Its goal is to enable circular use of textiles and plastic parts with 'coatings' (performance coatings, paints and adhesion layers). A triggerable polymer material system based on smart additives and corresponding recycling processes (activating the coating formulation) will be developed.	Specific knowledge regarding matrix/coating separation and polymers recycling processes
DEMETO *2014-2020* H2020	DEMETO proposes sustainable industrial application of chemical treatment for reuse of PET waste streams using microwave depolymerization.	The technologies will be compared regarding environmental impact and robustness
EERAdata *2020-2023* H2020	Develops and tests a FAIR and open data ecosystem in the low-carbon energy field, through broad community involvement and application in four use cases; implements an open platform for	Development of a FAIR data implementation protocol in WHITECYLE

	seamless access to energy data. The goal is to facilitate a mental shift towards FAIR and open data practices. *(HVL)*	
THANAPLAST *2015-2017* Bpifrance	Identification and optimization of an efficient enzyme able to selectively depolymerize PET.	TRL5 at the end of the project has been reached. WHITECYCLE will be the next step of the THANAPLAST project

1.2.4 The inter-disciplinary aspect of the project

WHITECYCLE's multidisciplinary approach combines enzymatic biotech, various other ecotech and processes, materials science, analytical chemistry, digital and AI technologies, risk assessment, business modeling and product design, systems thinking, legal and behavioural studies, as required by the nature of the circularity problems. The consortium includes partners with all these backgrounds, many of whom are accustomed to working in interdisciplinary teams. Although several of these partners are specialized in their fields, two partners, AXELERA and DYNERGIE, have a broad general knowledge base of scientific and industrial issues related to ecotech and the circular economy. These two partners and WP leaders also have experience in bridging any disciplinary or linguistic divides between project partners. In addition, very regular communication between partners will help them to learn to speak each others' "languages".

1.2.4.1 Gender dimension

The consortium has not identified any sex/gender-related issues related to the activities to be carried out, the resulting technology and its potential applications. The consortium recognizes the importance of gender equality. The project will follow European regulations/recommendations on the principle of equality between men and women. During WHITECYCLE, the consortium will follow the Directive 2002/73/EC on the implementation of equal treatment for men and women as guideline, e.g., for access to employment, vocational training, and promotion, working conditions. Coincidentally, in fields like manufacturing, recycling, disposal, waste collection, the male presence is relatively large, while the presence of females is relatively strong in design and business management. The consortium is conscious of the strategic importance to identify and avoid gender bias in science and solutions. The scientific coordinator and the Exploitation Manager will pay special attention to avoid that the designing of methodologies and interpretation of data produced during the project could be affected by gender.

1.2.4.2 Open science and FAIR data management

FAIR data are a prerequisite for enabling increased transparency along the product life cycle, which is a plus for social acceptance, offering the basis for access to comprehensive information about the entire product supply chain. It supports seamless tracking and reporting (even of issues that might only come up later). Implementing a FAIR data protocol for all project and use case data from the start allows faster identification of additional exploitation options through ensuring machine-actionability (e.g., finding unknown patterns by linking to other business cases, material databases, or market data at a large scale). In this way, the range of applicability and transferability of the solutions found in the use cases will be enlarged. As of today, the EU is heading the global development of defining and operationalizing FAIR data principles for humans and machines alike. Application to industrial use cases is yet in its infancy. Very few use cases exist, even fewer reached the level of implementation (i.e., having available coded algorithms to handle data exchange in accordance with the FAIR principles, see Schwanitz et al. 2021 (under review)). WHITECYCLE is an excellent use case for demonstrating the FAIRification cycle because it involves a variety of types of data (experimental, qualitative from stakeholder surveys, text documents that handle regulations, market data; as well as private/restricted data and open data). As part of the FAIRification process, the proper handling of IPR issues is also ensured. For example, all data in the project are assigned with license tags that clarify conditions for accesses and open data; the prominence of data is also tracked in the metadata. FAIR principles foster seamless interoperability with any other FAIR data sources (so-called FAIR data implementation points), which is a prerequisite to enable industry 4.0 technologies (including automated material discovery, digital reporting & compliance) and to realizing the value of data as a business asset.
Regarding **open science**, WHITECYCLE offers a prototype for an industrial FAIR data protocol that can be reused, replicated, and modified. More specifically, WHITECYCLE, through HVL leadership:

- Foresees open access to scientific publications under the conditions required by the grant agreement (GA);
- Manages research data responsibly in line with the FAIR principles as outlined above. This includes the incorporation of open vocabularies in the linking of data and metadata. Notably, machine-actionability is given prominence in the data management plan. Open access to research data is envisaged under the principle 'as open as possible, as closed as necessary'. The conditions required by the GA apply;

- Ensure digital access to all information about the research outputs/tools/instruments needed to validate the conclusions of scientific publications or to validate/re-use research data; the consortium will make use of the possibility to publish early results as preprints. Exceptions from this rule are handled in the GA;
- Confirms that in cases of public emergency, if requested by the granting authority, immediate open access is given to all research outputs under open licenses or, if exceptions apply, access under fair and reasonable conditions to legal entities that need the research outputs to address the public emergency;
- Fosters the implementation of open science principles, namely use of open source software (as far as possible, e.g., for the assessments carried out in WP6), open evaluation (through peer-reviewed, open source publications and timely presentation of results), open engagement of social actors (through incorporating stakeholder opinions in the impacts assessment, WP6); open educational resources (through integrating knowledge from the use cases in curricula and course material of academic partners, see above).

The consortium is experienced in the implementation of FAIR and open data principles and the development of a FAIR and open data ecosystem (e.g., EERAdata project).

2 Impact

2.1 Project's pathways towards impact

2.1.1 Increased deployment and market uptake of new technological solutions to waste management and recycling, and the measurement of recycled content

Deployment and market uptake require viable costs on one hand and buy-in across the value chain on the other hand.

Viable costs: Since the supply of recycled PET is limited compared to the demand, the price of r-PET from mechanical recycling is high and is completely decorrelated from virgin PET. CARBIOS and MICHELIN's first projections show that the enzymatically r-PET will be less expensive that mechanically r-PET and competitive with virgin PET for textile applications. Projection on the production cost will vary with parameters determined during the WHITECYCLE project (pre-treatment process, depolymerization yield, purification scheme, valorisation of coproducts…). **Our strategic positioning comes from the ability to treat various currently non-valorised complex waste feedstocks and to yield high quality r-PET** (Figure 12). The price of the resulting r-PET is not expected to be a limiting factor for market introduction in the next few years.

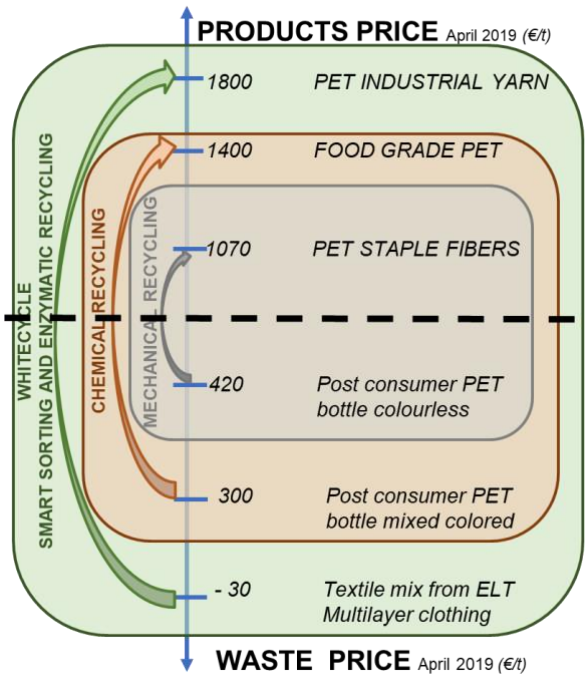

Figure 12 : WHITECYCLE recycling performances versus other comparable solutions

Market uptake across the Value Chain: WHITECYCLE aims to convince at least 12 brand-owners, 3 PET manufacturers, and 3 large-scale waste managers to adopt project results by 2030. These are the key deciders in the value chain. The **strategic importance of brand owners is that they provide the specifications** to PET and textile manufacturers (e.g., for percentage of recycled PET). Brand-owners whose products use PET cover a vast array of sectors including Automotive, Electrical, Electronics and Consumer Goods. **3 brand owners are partners of WHITECYCLE: MANDALS, MICHELIN and INDITEX**, and CARBIOS created a consortium in 2019 that gathers European leaders in packaging namely L'Oréal, Nestlé Waters, Pepsico and Suntory Beverage & Food Europe. A major milestone was recently reached with the production by the consortium members of branded bottles using CARBIOS r-PET. This step successfully validates the quality of the recycled PET resulting from enzymatic recycling (CARBIOS' press release 24/06/2021). Beverage brand owners have committed to incorporate up to about 600 kt of food-grade r-PET by 2025 (e.g., Pepsico 25% recycled content by 2023[36]). The next step is to recycle higher specification products from more complex waste. CARBIOS is currently negotiating with a consortium of textile brand-owners. Even though mandatory textile sorting, and recycling won't apply until 2025[37], brand owners in the textile industry such as **INDITEX**, Gap, H&M, Adidas, Nike, Chanel, Hermes, Kering, Carrefour, Galeries Lafayette, and La Redoute signed a Fashion Pact and are eager to incorporate recycled content in their products.

Regarding PET producers, there is one PET manufacturer in the consortium, but it is important that project technologies be replicated broadly. Therefore, CARBIOS has already set in motion a pragmatic plan. The SME invested and, in September 2021, CARBIOS launched its own industrial demonstration plant to produce r-PET. By the end of 2022, the operation of the plant will enable the complete engineering documents for a first process (Process

[36] https://www.transparencymarketresearch.com/recycled-pet-market.html
[37] WFD, Article 12b DIRECTIVE (EU) 2018/851

Design Package) to be drawn up for the building and operation of a 30-50 kt/year capacity reference unit. During WHITECYCLE, this demo plant will enable designs for processes for more complex waste to be designed. These future industrial-scale units will be operated by PET producers under license agreements with CARBIOS. CARBIOS received expressions of interest from 3 industrial-scale PET producers in Europe requesting to host the first planned reference unit. The site for the installation of the reference unit will be determined in late 2021 along with the signing of a first agreement between CARBIOS and a PET producer. Further discussions with other European PET producers for the licensing of the technology will follow. In this way, CARBIOS is committed to enabling the operation of one industrial-scale r-PET plant by 2027 and 10 by 2031.

The quantities of r-PET produced today are too small (3% of total capacity) **to meet brand-owners' demand**. The shortage is expected to amount to 1.5 Mt by 2025 and **2 Mt by 2030**[38]. According to WHITECYCLE's exploitation plan, **project technologies should produce 2.06 Mt r-PET in 2030, which is well aligned to cover this projected shortage.** This r-PET can be used for every application, including food packaging and applications with stringent mechanical and optical specifications, thus meeting the large market demand expected for r-PET.

Given the shortage of r-PET, **waste managers** are another key actor in the value chain. WHITECYCLE partners, **ESTATO** and **SYNERGIES** are waste management companies with numerous regional, national, and international customers and a commitment to circularity. Implementing WHITECYCLE's new sorting technologies will open new markets for them and increase their margins by enabling them to provide higher value materials to new customers (80% PET content circular raw material for PET manufacturers thanks to the full combination of AI, aeraulic, tribo-charging and electrostatic processes). They plan to upscale and broadly implement tyre and clothing separation and sorting by 2030. In addition, **ALIAPUR** is a tyre collection company that is partly owned by **MICHELIN** and committed to the circularity of tyres. **CARBIOS** in in discussions with major European waste collectors as well as European waste collection organisations (Citéo, ReFashion, Texaid, Dutch Circular Textile Valley…) to feed its pre-industrial unit (100t of PET/year for which a supply contract has already been signed) and the future reference unit (30-50 kt/year). **IRIS, ESTATO and SYNERGIES will prepare process design kits that enable waste management and recycling equipment companies to build and operate optical and electrostatic sorting plants.**

IRIS's AI based solution to identify PET content in complex waste feedstocks coupled to automated sorting and dedicated mechanical preparation steps will for the first time allow waste managers to select, prepare and enrich to at least 80% PET content and has already attracted the interest of SYNERGIES and several other waste management companies. This technology will not replace existing jobs, because it will enable first-ever sorting of materials never before sorted, manually or otherwise. WHITECYCLE will create an IT decision-support platform for textile recycling companies to help them adopt WHITECYCLE technologies. Other smart waste management solutions have already reduced OPEX costs for waste managers while creating value for strategic decision-making[48]. Thanks to data management from feedstock sorting to monomer re-creation, the r-PET can be certified, and the downstream actors (polymer manufacturers, compounders, and end-users…) will be able to prove how much recycled PET they use. This also creates added value that will enable waste management and recycling companies to further invest and install new capacity. This will be aligned with EU regulations requiring the use of recycled raw materials, such as the binding target that recycled content in PET bottles should reach 25% by 2025 and 30% by 2030 (Art. 6 DIR (2019)/904).

WHITECYCLE's AI and sorting processes are valuable for every recycling process (mechanical, chemical, enzymatic), because they increase the quality of the feedstock and the yield of any further recycling steps. Moreover, the technical solutions elaborated for sorting within the framework of this project will be easily adaptable to the treatment of other types of textiles, plastic, and composite containing wastes. Several WHITECYCLE partners are already pushing WHITECYCLE technologies to waste managers, and this project will help de-risk these disruptive technologies in the eyes of these companies. **IRIS, IPOINT, CARBIOS, AXELERA and MICHELIN will continue to engage in targeted techno-push activities to waste managers within and beyond the consortium.**

Europe has a strong and thriving PET manufacturing and processing industry. For example, WHITECYCLE partner, KORDSA, is the leading manufacturer of polyester yarn. The major PET manufacturers in Europe are SABIC, Indorama, BASF, and Pollisan. These companies have expressed their commitment to producing circular PET in their factories and recognize the need to go beyond mechanical recycling. WHITECYCLE partners are in discussions regarding plastic circularity with SABIC and BASF (**AXELERA**).

FAIR data will also be used to prepare arguments for market uptake and replication. For example, data could be shared with stakeholders to justify process KPIs and show that they are reproducible.

[38] HIS Market 2019

The above summary shows the consortium's commitments to increased deployment and market uptake. Partnerships cover the full value chain so that all actors move forward in parallel. **Please see also the exploitation plans and financial projections in section 2.2.1.**

2.1.2 Increased upcycling and recycling rates for the targeted material streams

Since 2006, the amount of recycled plastic waste has doubled, and WHITECYCLE aims to continue this progress by overcoming today's main hurdles: the need to make high quality r-plastics from low-value waste streams.[39] Today, the more complex PET feedstocks targeted in this project are 100% landfilled or incinerated. Furthermore, for textiles, only 0.1-1% is recycled to remake textiles. Enzymatic recycling developed by CARBIOS is the only one to date to have upcycled textile waste into clear bottles. WHITECYCLE will show that even the most complex textile wastes can be efficiently recycled in a closed loop for new textiles or in added-value bottles.
A huge improvement over this current situation, the total yield of PET sorting expected at the end of WHITECYCLE will be at least 80% for tyres and for PET-containing textiles. In conclusion**, through the combination of new recognition, sorting and enzymatic depolymerization, the yield on complex feedstock will go from 0 to 80%.**
Our technology developments pave the way for circularity of high-quality PET; the quality of the sorting (mechanical and enzymatic) will give access to full high-quality recycling of **20 Mt fully deployed at global scale by 2050.**
Within 4 years after the project (2030), WHITECYCLE is expected to enable the circular recycling of a total of 2.06 Mt of PET: 1.85 Mt of PET multicomponent textiles, plus 200 kt of PET from tyres, plus 10 kt of hosing.
The calculation is based on a middle ramp-up with the following assumptions:

- Assuming today's production volumes and that all r-PET produced is used as raw materials for circular production in the same applications meeting the same standards.
- Textile fibres waste represent about 7 % by weight of ELT or 320 kt/y per year of dirty fibres in Europe with a potential volume of 680 kt/y if all available ELT are shredded
- 50 % by mass of ELT dirty fibres is polymer of which 75% is PET: it represents 120 kt/y of PET with a potential of 250 kt/y
- 25% by mass of ELH content is PET
- Due to new applications of secondary raw material, 100 % of collected tyres in Europe will be granulated.
- 11.6 Mt of projected waste clothing in Europe in 2030, with 50 % of PET and 40 % of this feedstock is composed of multicomponent textile clothing with significant PET content (40-95 %), which yields 2.3 Mt/y of extractable PET thanks to the WHITECYCLE project
- Assuming 80% efficiency in the full collection, sorting and recycling process.

The mandatory common EU targets are listed below. By 2030, Europe will require double the current PET reprocessing capacity to process the tonnage of PET expected to be collected if these EU targets are met[40] e.g., around 2 Mt of additional r-PET. WHITECYCLE will enable these targets to be met.

- $\geq$55% of municipal waste must be recycled by 2025; 60% by 2030 and 65% by 2035.
- beverage bottle collection should reach 77% by 2025 and 90% by 2029 (Art. 9 DIR (2019)/904)
- $\geq$50% recycling of all plastic packaging by 2025, and $\geq$55% by 2030.
- $\leq$ 10% of municipal waste can be deposited in a landfill by 2035.
- a separate collection of textile waste will be compulsory in Europe by 2025 with no incineration or landfilling.
- recycled content in PET bottles should reach 25% by 2025 and 30% by 2030 (Art. 6 DIR (2019)/904)

2.1.3 Increased resource efficiency along and across value chains, causing a measurable reduction in GHG emissions

The 2.06 Mt/y by 2030 of r-PET targeted described above will replace both PET incineration and primary production from fossil fuels, thereby increasing resource efficiency and dramatically reducing GHG emissions. On one hand, regarding the end of life, by preventing incineration of end-of-life PET, the project is expected to reduce GHG emissions by **2.78 Mt CO2eq per year by 2030.** This first basic calculation on the perimeter of incineration step relies on the following assumptions: about 50% of complex PET-containing waste is currently incinerated in the EU[40] (the rest being landfilled), and 2.7 t of CO_2 are released for every ton of incinerated plastic on average[41].
On the other hand, based on life cycle assessment methodology, climate impacts are directly linked to the production of PET. On average today, **4.6 kgCO_2eq is emitted per kg PET produced[43].** LCA studies exist on classical mechanical, chemical recycling of PET vs bio based or virgin PET production[42]. A first LCA study for enzymatic

[39] https://www.plasticseurope.org/application/files/1115/7236/4388/FINAL_web_version_Plastics_the_facts2019_14102019.pdf
[40] Plastics Recyclers Europe, 2020
[41] Industrial transformation 2050 — Pathways to net-zero emissions from EU heavy industry, Material Economics, Stockholm (https://materialeconomics.com/publications/industrial-transformation-2050).
[42] *ACS Sustainable Chem. Eng.* 2018, 6, 8, 9725–9733, Journal of Polymers and the Environment volume 27, pages533–548(2019), Journal of Environmental Management, Volume 260, 15 April 2020, 110062

recycling was proposed by Singh et al.[43] including the full recycling loop of clear and colored bottles, amorphization, enzymatic depolymerization and purification. It emphasizes that enzymatic recycling **reduces the total supply chain energy used by 69 to 83 % to produce terephthalic acid (PET monomer precursor) and by 35 to 42 % to produce PET. This results in GHG emissions reductions between 8 to 21% to produce PET (0.5kg to 1kgCO_2eq per kg PET produced) compared with the fossil-derived PET production baseline.** Independently, preliminary LCA studies using "Product Environmental Footprint" (PEF) methodology, conducted at MICHELIN with CARBIOS data (functional unit: To provide 1kg of bottle grade PET granulate at the production plant gate & to manage the corresponding kg of PET waste at end of life used) reproduce this order of magnitude. In the WHITECYCLE proposal and with conservative assumptions we estimate that a total of **2.06 MtCO_2eq** will be saved per year in 2030 (**1 MtCO_2eq saved per Mt PET produced x 2.06 Mt r-PET produced**).

2.1.4 A significantly higher level of involvement of all actors (manufacturers, retailers, consumers, CSOs, public administration etc.) in circular practices

A circular economy will require commitment and organization by public authorities, manufacturers, citizens, and consumers. For them to accept the extra responsibility and burden, WHITECYCLE will consider how the direct benefits of the circular economy can be shared with them, and these benefits must be clear, transparent, and attractive for them to adopt. For example, **AXELERA will forge a link with EU training programmes** (WP6), so that WHITECYCLE partners join trainers and students in preparing the work force for the new jobs the circular economy will create. This will enable European students to get interesting and high-impact jobs. **HVL will explore how the project can create value for citizens, consumers, companies, and public authorities** and then prepare convincing communications to be transmitted broadly by consortium partners and the networks in which they are present. Partners' policy divisions will inform policymakers to ensure that policy requirements are more ambitious, given the new technical possibilities enabled by WHITECYCLE. FAIR data can provide an objective basis for policy- and decision-making by stakeholders. **ERASME will study and outline synergies between partners and companies, local symbiosis exploitation** and improvement in circular economy scheme and value chains to deliver indicators and roadmaps for the partners and industrial companies of the sector to improve replicability of the best practices.

2.1.5 Strengthened competitiveness and job retention and creation potential of circular value chains under different economic and social conditions

The 9 companies in the WHITECYCLE consortium will bring the technologies to the market and have prepared **financial and job projections (see sec. 2.2.2)**. In summary, these **9 companies plan to create 130 direct jobs in Europe and 35 direct jobs in Turkey by 2030** linked to the first stages of exploitation of WHITECYCLE results. Recycling creates an average of ten times more jobs than trash. Many more waste management jobs and companies are needed in Europe to meet the challenges of a circular economy. In nature, almost 20% of the species are dedicated to the degradation of "waste" (end-of-life) biomass[44]. Between 2012 and 2018, the number of jobs linked to the circular economy in the EU grew by 5 % to reach around 4 million. The circular economy has the potential to create more than 700 000 new jobs in Europe by 2030[45], while improving the quality of jobs. Therefore, further indirect jobs will be created in the below-described large and growing markets and through the replication of the project concept in other contexts.

A net exporter (trade balance of 15 billion euros in 2018), the **European plastics industry directly employs more than 1.6 million people in Europe with 60,000 companies and turnover of €360bn[39]** for plastic manufacturing and processing. The **PET market has an annual growth over 4% with European consumption of 15.9 Mt/year** (5.8 Mt plastics and 10.1 Mt fibers) with similar amounts of PET becoming waste each year (70 Mt annual global production). The **r-PET market is forecast to double by 2029, reaching $4.7bn[46]**, notably due to European regulations that impose an increased content of recycled PET in packaging bottles.
In Europe, the **textile industry employs nearly 6% of manufacturing workers[47]**. A Combined Annual Growth Rate (CAGR) of 5% is expected through 2025. Europe is home to hundreds of leading textile retailers and brands, internationally acclaimed designers, thousands of talented emerging designers, and forward-thinking entrepreneurs, researchers, and educators. European high-quality textiles are in growing demand across the world.

[43] Singh et al., Techno-economic, life-cycle, & socioeconomic impact analysis of enzymatic recycling of PET, Joule (2021), doi.org/10.1016/j.joule.2021.06.015
[44] https://www.seuil.com/ouvrage/homo-detritus-baptiste-monsaingeon/9782021352603
[45] https://www.ellenmacarthurfoundation.org/assets/downloads/publications/EllenMacArthurFoundation_Growth-Within_July15.pdf
[46] https://www.environmentalleader.com/2020/03/recycled-pet-market-2029/
[47] https://www.mordorintelligence.com/industry-reports/europe-textile-industry

The **smart waste management market was valued at USD 1.77 billion in 2020** and is expected to reach USD 6.52 billion by 2026, registering a CAGR over 25%, from 2021[48]. 3 of the 5 major worldwide players are European companies with hubs in France and Germany (Suez, Veolia, Pepperl+Fuchs).
In conclusion, WHITECYCLE's circular solutions will **create local jobs** for a fair and inclusive Europe.

2.1.6 Other expected outcomes addressed by the WHITECYCLE project

Up to 80 % of the environmental impacts of products are determined during the design phase. Using recycled products reduces **energy, water, and virgin resource consumption** over its lifecycle of transportation, production, consumption, and disposal. Let's consider water impacts. 79 billion m^3 of water was used by the global textile industry in 2015, using plastic waste instead to make clothing greatly reduces water use. Water scarcity is an increasingly frequent phenomenon affecting **at least 11% of the EU population and 17% of its territory**.
Regarding landfilling, about 50% of complex PET-containing waste is landfilled in the EU (the rest being incinerated) so **WHITECYCLE will prevent landfilling of about 1.03 Mt of PET per year by 2030** (50% x 2.06Mt r-PET produced in a circular loop rather than in the traditional way).
Regarding pollution, PET single-use beverage bottles are among the most common marine litter items found on most beaches in the EU (DIR(2019)904). PET degrades very slowly in water and contributes to growing plastic continents in oceans, dispersion of microplastics in soil and water, with harmful effects on biodiversity. More than 700 species of marine animals have been found with traces of plastics in their digestive tracts.
Moreover, the digitalization of the recognition prototype will enable the collection of data relating to the chemical compositions of textile waste, which is still very unknown to this day. The data will be shared through the FAIR implementation protocol to different shareholders in the perspective to enhance the creation of similar projects, to address the issue of recycling other textile components than PET (e.g., nylon, cotton etc.). This will ultimately facilitate circular economy. The development of digital solutions will allow to draw software modeling practices to help implement a model for data management of the monitoring system which could be easily deployed for other value chains containing textile PET materials as packaging or construction.
During the COVID-19 crisis, an increased volume of personal protection equipment PPE with single use or short use generated a huge complex waste feedstock[49]. Among these products, complex waste streams of multicomponent materials that can be found in masks have to be collected, separated, treated. WHITECYCLE technology will be able to help and improved these complex waste management issues.
In addition, our project will contribute to **a greener industrial Europe** as recommended in **European Green Deal, Industrial Strategy for Europe COM(2020)102**). The **Green Deal** also aims at helping companies become world leaders in clean products and technologies. The 9 companies developing technologies respect the principles of green processes at high yields in mild conditions, without organic solvents, and with valorisation of co-products.

2.1.7 Barriers

The question of when wastes become raw materials: When a waste becomes a raw material, this can create more value for the next steps of the value chain. MICHELIN and other project partners are involved in working groups to solve this problem of inequity of value creation, which slows the activation of the part of the value chain that deals with low-value or negative value "waste".
Regulations: Through its Standards & Regulations departments, partners will determine the status of materials, whether they are considered as waste or secondary raw materials in respect of different regulations to enable their transport, which is limited for "waste" by regulations, including the Waste Framework Directive (WFD), directive 2008/98/EC. The constraints of the REACH regulation (regulation N°1907/2006) will be carefully respected.
Economic competitiveness: The overall cost of r-PET will depend on fluctuating PET waste price and on the process itself. Emerging regulations, listed above, will help ensure competitiveness. Also, the low cost of poorly recycled, complex waste and the low-energy use and cost of the processes will help ensure economic viability.
Reliable feedstock sourcing: A sufficient supply of waste is needed to feed the recycling units. This is highly mitigated by process applicability to quite diverse, impure, low-value waste streams as opposed to the other recycling solutions. Strong and mutually beneficial relationships will be built with several waste management companies, two of which are consortium partners.

2.2 Measures to maximise impact – Dissemination, exploitation, and communication

Protection, exploitation, and dissemination activities will be managed by appointed managers (IPR manager; Exploitation and innovation manager; Communication and Dissemination manager) with clearly defined roles within

[48] https://www.mordorintelligence.com/industry-reports/smart-waste-management-market
[49] Face masks and the environment: Preventing the next plastic problem, 03/2021 (Face masks and the environment: Preventing the next plastic problem -- ScienceDaily)

the WP7 "Dissemination Communication, Strategic Intelligence, Outreach and Replication". The leader of dissemination and outreach will be AXELERA, a public-private network of 370 members with partners across Europe and the world. This network will serve as a solid basis for reaching relevant stakeholders including key partnerships to link innovation projects to early adopter companies.

- The first key partnership is with its 240 member companies of which 120 are small and 100 of which are large multi-national companies located across Europe and/or the world (waste management companies, materials manufacturers, eco-technology manufacturers, engineering firms, etc.).
- Secondly, AXELERA has close partnerships with 15 European clusters, public-private partnerships, and industry groupings in the field of WHITECYCLE who facilitate connections with their member companies (PETCore, P4Planet, EIT Raw Materials, SUSCHEM, CO2Value Europe, FundingBox, Eyde Cluster (NO), Brightlands (NL), Clean (DK), Greenwin [BE], Chemie Cluster Bayern [DE], CleanTech Alpes [CH], Environment Park [IT], IAR [FR], Association Chimie du Végétal [FR]).
- Thirdly, AXELERA maintains strong relationships with 9 large research organizations beyond the WHITECYCLE consortium that collaborate with innovative companies in the field of WHITECYCLE (AIMPLAS, BGR, CEA, Fraunhofer, ICAMCYL, IFPEN, SINTEF, Tecnalia and VTT).
- Fourthly, local development agencies and chambers of commerce across Europe and the world (notably in China and Canada) are key partners on the international scene. AXELERA has a flexible and "smart" contact database platform based on EUDONET software that facilitates communication and collaboration with this large network in line with the General Data Protection Regulation.

Partners will push info via the WHITECYCLE website, social media, newsletters, B2B meetings, workshops, trade fairs, process design kits, scientific publications, FAIR data, video, press releases, leaflets, etc. (WP7).

2.2.1 Dissemination activities

Objectives per stakeholder	Dissemination activities / channels
To convince at least 12 brand-owners, 3 PET manufacturers, and 3 large-scale wastes to be early adopters of project results	Key data will be provided via exploitable metrics to technology scouts from relevant industries so that WHITECYCLE circular recycling processes can be rapidly transferred to at least 18 companies by the end of the project for upscaling and industrial implementation across Europe. Notably, CARBIOS will prepare a process design kit for enzymatic recycling and will transfer it to at least one major PET manufacturer (by 2026) who agrees to produce r-PET at an industrial scale using project technologies. Similarly, IRIS, ESTATO and SYNERGIES will prepare process design kits that enable companies to build and operate optical and electrostatic sorting plants. ESTATO and SYNERGIES aim to build and operate such sorting lines. IRIS plans to exploit the recognition system developed considering SYNERGIES as a preferential customer, before widening the scope to other potential waste managers. The 3 brand-owners in the consortium will convey to their PET suppliers that they would like to purchase large volumes of r-PET produced via WHITECYCLE enzymatic recycling. The consortium will also reach out to additional brand-owners (see sec 2.1), to waste managers and to PET producers to provide additional sales arguments, since brand owners provide the specifications for PET. In addition to already targeted prospects, additional end-user companies will be identified by all partners and by AXELERA during strategic intelligence work. MICHELIN will openly disseminate its knowledge gained on tyre eco-design. WHITECYCLE will create an IT platform enabling the automatic assessment of a waste feedstock's environmental impact and its compatibility with WHITECYCLE's recycling solutions. This tool will be disseminated to recycling companies, especially for textile recycling, to help them to plan their transition to WHITECYCLE technologies. The consortium will organise 2 workshops in conjunction with international events attracting large numbers of potential end-users to spread awareness of the main results of WHITECYCLE. The workshops will include 40 to 60 direct participants, with a focus on presenting the results of demonstrations to potential early adopters of project technologies. The workshops will serve to obtain companies' needs and feedback and to strengthen relationships. Workshop may include virtual or real site visits. In addition, B2B meetings will be a core part of WHITECYCLE dissemination and will involve all partners developing technologies plus AXELERA. The Advisory Board will be composed of ALIAPUR, WESTERN SYDNEY UNIVERSITY, UNIVERSIDAD DE LEON and REFASHION. They will play an important role in the adoption of the project results by external stakeholders. Companies will reach out to specialist and industrial websites and publications such as PETCore. Outreach to clusters will be ensured by AXELERA, as these clusters facilitate access to many early adopters.
To disseminate the project idea, approach, and	Target audiences will include researchers in universities and RTOs in the fields of enzymology, biotech, green chemistry, smart waste sorting, AI, textiles, polymer recycling, LCA and the circular economy from

results to the public research and training communities	different perspectives including materials, engineering, digital, SSH, environmental, toxicology and business research. To disseminate the results of the projects, WHITECYCLE partners plan to submit 10 major scientific publications to peer-reviewed journals and to present results at conferences. AXELERA will prepare easy-to-understand summaries of 3 of the major publications to emphasize its key messages and to increase the impacts for the targeted stakeholders. Partners will present and network at appropriate meetings, such as the Suschem annual meeting, the PETCore Annual Conference, IEEE Industry Applications Society Annual Meeting, European Tyre Recycling Conference, etc. AXELERA will strengthen links between the consortium and EU training networks in the Circular Economy, such as C-PLANET and CAPTURE, whose training schedules coincide well with WHITECYCLE's. WHITECYCLE will provide case studies and data so that these training networks can provide relevant training content to their students, as well as partners' needs concerning technicians, engineers, and scientists. This outreach is expected to lead to internships and/or hires at partner companies. Two biographies of "heroic" people in the consortium making great progress toward the circular economy will be shared, and one of these biographies will cover a woman, such as Frédérique Guillamot, Innovation Manager at CARBIOS, to encourage young women to pursue careers in science and industry, especially in the circular economy.
To communicate clearly with the general public	Via the public website, social media, newsletters, press releases, video, and summaries of publications. Dissemination to the International Network on Public Communication of Science and Technology Partners will participate in broadly disseminated interviews in popular science blogs, online channels, magazines, newspapers, radio, or television. The SSH partners ERASME will host webinars, workshops, and debates for the general public on circular economy topics.

All dissemination and communication will be written in a gender-sensitive way, including submissions designed to encourage women in engineering and scientific careers.

2.2.2 WHITECYCLE exploitation plan

2.2.2.1 Key exploitable results and ownership

An exploitation plan is found below followed by individual exploitation plans as well as financial and job projections.

Key Exploitable Results	Target Exploitation Route	Exploiting Partner	IP protection / dissemination	Target Users/Clients
Knowledge and new processes related to sorting and separation systems for composites and multi-material textiles	Create the first polyvalent automated sorting plant in France at 20 kt scale. Sales of sorted textiles as secondary raw material Licensing of processes to companies. Possible creation of JV or spin-off.	IRIS, SYNERGIES PPRIME MICHELIN	Patents + Process Design Kit. Further R&D to reach other sectors and improve KPIs	Waste managers & recycling equipment manufacturers. Companies that need to improve composite sorting. Companies in PET recycling and production. Companies focusing on automatizing new processes.
Optimized optical system for the chemical identification of PET and other elements in textile-based products	Licensing of processes to companies. Possible creation of joint-venture or spin-off company.	IRIS		
New processes for PET amorphization	Licensing of processes to companies	IFTH MICHELIN		
Optimized process for enzymatic PET depolymerisation from complex composite and multi-material waste including monomer purification	Licensing of processes to companies One operational plant by 2026 & 10 by 2031	CARBIOS	Patents + Process Design Kit. Further R&D to reach other sectors and improve KPIs	Companies in PET recycling and production + Brand owners for PET-containing products + companies in PTA production
Enzyme adapted to complex composite and multi-material textiles	Selling enzymes to licensee (3 PET producers)	CARBIOS	Patents and possibility to extend to other application sectors	Companies in PET recycling and production or companies in PTA production
Knowledge and optimized process related to the polymerization of technical grade r-PET	Production of r-PET, licensing to other companies	DITF / KORDSA	Patent	Companies specialized in technical PET polycondensation
Knowledge related to the manufacturing of r-PET-based hoses	Production and sales of r-PET containing hoses	MANDALS	Patent	Hosing market actors
Knowledge related to the manufacturing of r-PET-based clothing	Production and sales of r-PET containing garments	INDITEX	Patent	Clothing market actors

Knowledge related to the manufacturing of r-PET yarns and cords	Production and sales of r-PET yarns & cords for tyres, hoses and building	KORDSA	Patent	Tyre, hose and building material manufacturers
Knowledge regarding the eco-impacts of tyre components on the circular value chain	Production and sales of eco-designed, r-PET containing tyres	MICHELIN	Open dissemination of eco-design best practices	Plastic & composite industries, SMEs, Public and private organizations
Knowledge related to the manufacturing of r-PET-based tyres	Production and sales of eco-designed, r-PET containing tyres	MICHELIN	Patent	Tyre market actors
IT platform enabling an automatic calculation of a feedstock's environmental impact assessment and its compatibility with WHITECYCLE's recycling process	Access fee to platform	MICHELIN SYNERGIES IRIS	Patent	Recyclers, textile sector

2.2.2.2 Preliminary High-Level Exploitation plan

Overall exploitation framework. The exploitation of project results will remain open to partners who contributed to each result. This will be ensured by the Grant Agreement and Consortium Agreement. Partners will negotiate the terms of their future relationships as partners, suppliers, customers, advisors, prescribers, etc. The exploiting partner(s) will select all deliverables and outcomes that (i) can be transformed into exploitable results, (ii) confer a competitive edge to the exploitable result, (iii) are in accordance with the Intellectual Property policy.
Key Areas of Application. The disruptive eco-KPIs of WHITECYCLE products and process increases their value versus competitors. Currently, the prices of r-PET are not competitive with virgin PET (v-PET), but the situation is expected to evolve with WHITECYCLE. Its novel technologies will make r-PET relevant for higher added value composite and multilayer end-product markets, notably in Europe, North America, Japan, and China.
Competitive Advantages. WHITECYCLE will give value to the complex PET fraction abundant in waste feedstocks. The expected price of r-PET is competitive with v-PET.
Market. The PET market size was valued at **USD 27 billion in 2020** and is projected to reach USD **40 billion by 2028**, growing at a **CAGR of 3.69% from 2021 to 2028**[50].
Sales and Marketing Strategy. Hoses, tyres, and clothing will be distributed through direct sales. Products will have high added value and competitive price. A digital marketing approach will be used to develop communication, sales support tools, and expertise sharing. The key industrial brand-owners, MICHELIN, MANDALS and INDITEX have a very good market position and sales force. As for the process technologies, these will be licensed or jointly exploited through the creation of joint ventures. See also financial project tables below.
advantage. WHITECYCLE therefore, has a great economical potential as this is not the only output of the project.

2.2.2.3 Individual exploitation plans

MICHELIN: Industrial/commercialization strategies: **MICHELIN will (i) develop new process solutions & design rules to sort, pre-treat and integrate the wasted PET from composites into valuable materials** that is a "ready to be enzymatically depolymerized PET": these materials and associated process and rules will be deployed in different industries involved in recycling of composites (target 500 kt/y). First perimeter of application will be the tyre recycling industry. (ii) **Design and produce new tyre ranges including these sustainable materials** derived from the project, such as recovered PET: these tyres will equip a wide variety of vehicles especially personal cars. MICHELIN will enhance the reuse of ELT and increase the content of r-PET in its tyre lines: 70 kt of PET per year will move from a fossil source to a recycled source for direct internal use. The strategy is to upscale and massively deploy the pre-industrial processes by 2030 both in new commercial product launches and in existing products.
Main clients: Automotive OEMs, car drivers, agricultural tractor OEMs, and associated aftermarket.
Delivery channels: Tyre dealers.

MANDALS: Industrial/commercialization strategies: Hoses with r-PET will be sold to already existing markets worldwide, such as water transport, firefighting, dewatering, pipe rehabilitation, hydraulic fracturing, compressed air, agriculture, construction, and raisers in wells. **The r-PET hoses will be integrated in commercial launches of premium hoses.** MANDALS will also develop a service that will recover end-of-life products.
Main clients: construction companies, firefighter authorities, general public, farms, industrial companies, etc.
Delivery channels: Direct sales

[50] Global Polyethylene Terephthalate Market Size By Application (Food & Beverages, Sheets & Films, Consumer Goods), By End Use Industry (Packaging, Electrical & Electronics, Automotive, Construction), By Geographic Scope And Forecast
Report ID: 41941 | Published Date: Jul 2021 | No. of Pages: 202 | Base Year for Estimate: 2020

CARBIOS: Industrial/commercialization strategies: Broad replication will occur via licensing CARBIOS's technology to PET producers who will build and operate industrial plants. CARBIOS has built an operational demonstration plant. CARBIOS will use this demonstrator platform during and after WHITECYCLE to facilitate the replication of its process for a large variety of sectors and companies. CARBIOS will gather data from the process to prepare a Process Design Package that is a blueprint and guide enabling one to build r-PET production lines based on enzymatic recycling. It will allow precise determination of quality and price levels of the recycled monomers as well as environmental impact. CARBIOS will "sell" this blueprint to PET producers with the following exploitation roadmap: (a) **CARBIOS licences the technology by 2026 to 3 major PET producers, who will build (b) one operational plant by 2026 and (c) 10 by 2031**. Licensing will be done with the support of Technip, a large engineering project management company who will bring its expertise in licensing contracts and its network of PET producers while CARBIOS will bring IP and know-how on the technology. CARBIOS will receive fees for assistance and technical support. CARBIOS' proprietary enzyme will be produced by Novozymes, the global enzyme producer in Denmark. CARBIOS will directly sell the enzymes required for depolymerization to PET producers.
Main clients: Chemical companies, PET manufacturers, PET recyclers.
Delivery channels: Direct delivery of Process Design Package, consulting services and enzymes

KORDSA: Industrial/commercial strategy: **KORDSA is aiming at being able to produce industrial grade yarns from recycled feedstocks**. Commercialization of recycled PET fibres for demanding applications are not mature yet. As soon as the recycled PET fibres are produced, mainstream products with r-PET will be manufactured.
PET flakes are widely available in the industry mainly supplied from China. Any establishment of an EU-based cost competitive PET flake supply will increase competitiveness. Therefore, commercial strategy to be evolved on enrichment of KORDSA product portfolio that will serve both environment and business sustainability.
Main clients: Textile industry, tyre industry, hose industry – more largely any industries that use PET-based yarns.
Delivery channels: Direct sales of r-PET yarns.

ESTATO: Industrial/commercial strategy: **The new sorting machine will be upscaled and installed to improve the existing material flow and quality and separate the PET fibre fraction from rubber**. Rubber and PET fractions will be marketed separately.
Main clients: Chemical companies, PET manufacturers and plastic recyclers.
Delivery channels: Direct sales

IPOINT: Industrial/commercial strategy: After completion of the project, **IPOINT plans to integrate the results into its own software products for LCA**, which will be customized through existing channels and customer relationships. In addition, IPOINT expects to develop completely new target groups and, in the medium term (2-5 years after the end of the project), to achieve significant additional annual growth in turnover and staff thanks to additional revenue opportunities in sustainability consulting and in license sales of the Umberto software.
Main clients: Companies and consortia that need to carry out extensive LCA analyses.
Delivery channels: Licenses are primarily offered as rental licenses or software-as-a-service

SYNERGIES: Industrial/commercial strategy: **SYNERGIES aims to implement the multilayer sorting technologies developed by IRIS its future industrial plant**. This will allow SYNERGIES to sort complex clothing feedstocks that are not recycled now in time to meet stricter regulations. SYNERGIES will then sell these sorted materials to the textile, tyre, plastics, building, and automotive industries. SYNERGIES business model is to sell treatment of end-of-life clothing, professional and industrial textiles and to secondary raw material preparation.
Main clients: Sales of waste management services to textile industries and brand-owners; sales of sorted textile waste to be compliant with recyclers' specifications/requirements.
Delivery channels: Direct sales of services and sorted/treated textiles to textile industries and brand-owners.

IRIS: Industrial/commercial strategy: **IRIS aims to license its optical-based technologies for in-line textile waste characterization, detection, and sorting to waste managers and textile companies**. JV will be considered.
Main clients: Recyclers, waste managers, textile companies.
Delivery channels: Direct licensing, joint venture.

INDITEX: Industrial/commercial strategy: INDITEX manages very large amounts of textile waste every year that are, for the most part, not recycled. INDITEX aims to implement technologies at industrial scale that could turn textiles into a fully circular product. INDITEX is contemplating investment in technological components that are being developed through WHITECYCLE to retrieve PET contained in the INDITEX clothing waste. **INDITEX will be selling new lines of garments that contain r-PET.**
Main clients: General public, companies.
Delivery channels: Direct sale of clothing containing recycled PET

The revenues from exploitation phase for the exploitation partners are presented in Figure 13.

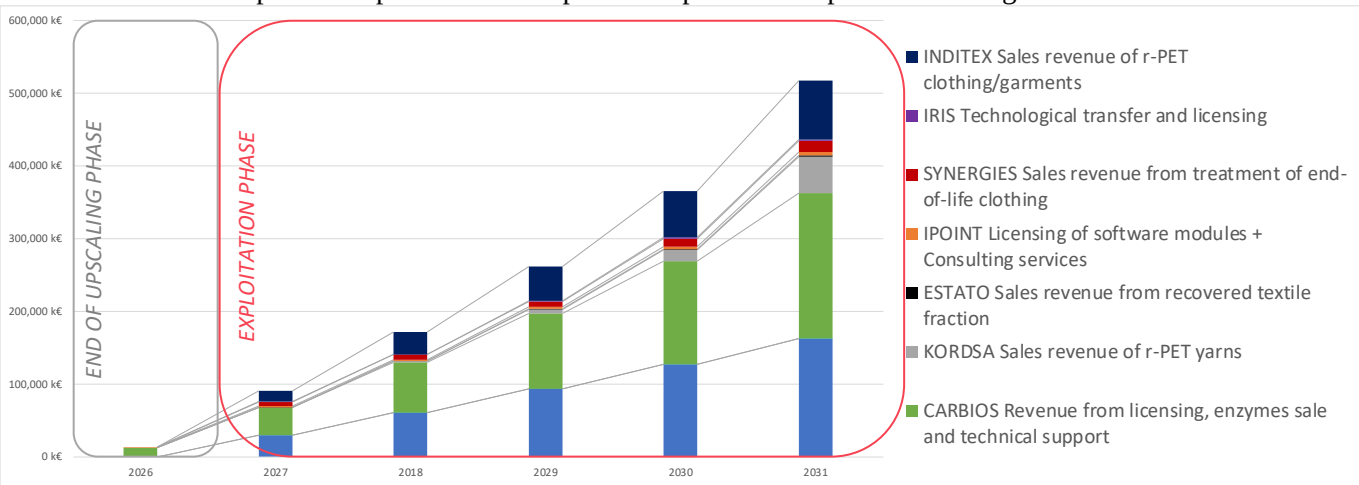

Figure 13 : Total expected revenues from the exploitation partners

The total revenues of the 9 exploitation partners of WHITECYCLE will reach 500 M€ by 2031.

2.2.3 Intellectual property rights

Intellectual Property Rights protection activities are managed under WP0. The overall IPR strategy of the project is to ensure that partners can fully exploit their market position, involving patent protection. The partners will sign a Consortium Agreement (CA) prior to the signature of the Grant Agreement (GA) with the EC in which all the rules for ensuring confidentiality (procedure for granting publication of the results), ownership regime and IPR protection (background included and excluded, ownership, general licensing terms, etc.) are detailed. The CA will be based on the IP background and foreground provided below and on the EC portal, which will be the starting point for IPR Protection and Exploitation Plans.

	Background IP brought to the project
MICHELIN	Patents relating to (i) plasma-treatment of PET reinforcement to amorphize (EP2855133 A1, EP3074199 B1), (ii) hybrid textile cords made of aramid and PET or hybrid cords usable for mono carcass ply of passenger car tyre (WO2020/058632), (iii) carcass PET-based ply of car tyre suitable for run flat use (EP3010731 B1).
CARBIOS	Expertise, and patents on PET recycling technology adapted to bottle feedstocks and pure PET textile. • PET pretreatment (WO 2017/198786), • PET enzymatic depolymerization (WO 2014/079844, WO 2020/094661, WO 2020/094646), • Efficient PET depolymerase (WO 2018/011284, WO 2018/011281, WO 2020/021116, WO 2020/021117, WO 2020/021118) • Monomers purification process
KORDSA	Knowledge on key process parameters related to the twisting, weaving dipping processes of v-PET based yarns
MANDALS	Work on PET plasma-treatment to chemically activate yarn - enhance bonding btw weave and hose cover layer
PPRIME	Two patents related to the electrostatic separation of granular materials
IRIS	Knowledge on in-line monitoring systems based on photonic technologies (e.g., NIR, HSI, LIBs…) improved by AI techniques (e.g., chemometrics, machine learning, data fusion) for the recognition of different kinds of materials (including some plastics/polymers) and coupled with in Process Analytical Technologies (+ own software platform). Patents in NIR or HSI based devices (VISUM)

2.3 Summary

KEY ELEMENT OF THE IMPACT SECTION

SPECIFIC NEEDS	EXPECTED RESULTS	D & E & C MEASURES
Viable **technologies enabling companies to: Meet mandatory 2025 and 2030 circularity targets** which limit landfilling & incineration, require separate sorting of textiles, and require minimum content of recycled plastics in packaging,	**Successful pilots on the 3 waste streams**: complex textiles, tyres, and hoses waste **r-PET high quality production for pilot series** of tyres, clothing, hoses **KPIs** that meet industry needs in terms of yields, processing time, cost, etc. (page 6) **Process Design Kits** to enable construction of sorting and r-PET production plants after the project **IT Decision-Support Platform** for waste managers	**Dissemination** to convince and involve stakeholders via direct outreach, process design kits, IT decision-support platform, guidelines, blueprints, intelligence bulletins, newsletters, workshops, conferences, etc. Involvement in public-private partnerships Synergies with other EU funded projects: symbiosis meetings 10 Publications in peer-reviewed journals **Exploit.:** The 9 exploitation partners expect to generate ~500M€ of revenues by 2031.

Reduce GHG emissions to meet mandatory Fit-for-55 climate targets, **Meet growing demand for r-PET** which already exceeds production capability via mechanical recycling	**FAIR data** **LCA and LCC** for the 3 use-cases **12 brand-owners, 3 PET manufacturers, and 3 large-scale waste managers / recycling equipment manufacturers sign agreements** for technology adoption	**Communication** to the general public, scientific, research and industrial communities via social media, public website, press releases, video, interviews in popular media, transfer of data and biographies of two partners to training programmes.

TARGET GROUPS	OUTCOMES	IMPACTS
Waste management companies **PET manufacturers** **Recycling equipment manufacturers** **Brand-owners** whose products contain PET **Researchers** Circular Economy **Training programmes**	**165 direct jobs** created as a first exploit. step Licence to **3 major PET producers by 2026** **1 industrial scale enzymatic r-PET production plant by 2026** and **10 by 2031** **Sorting technologies** implemented in at least **3 waste managers by 2030** **2 Mt r-PET produced by 2030** to meet needs of new EU regulations Full circular value chains in Europe	**Prosperous, sustainable, and vibrant EU circular economy** **2.06 Mt $CO2_{eq}$ emissions** avoided **1.80 Mt PET landfilling** avoided **Ability to meet 2025 and 2030 EU targets** and to set **more stringent targets leading up to 2050** for the full circularity of plastics and climate neutrality.

3 Quality and efficiency of the implementation

3.1 Work plan and resources

Table 3.1a List of work packages

WP	Work Package Title	Lead	Lead #	Person-month	Start Month	End month
0	Project and data management and reporting to the EC	MICHELIN	1	86.05	1	48
1	PET identification and separation from complex waste (clothing, hoses, and ELT)	IRIS	1	185.65	1	36
2	Pre-treatment of separated PET from complex waste	IFTH	8	75.75	1	36
3	Enzymatic depolymerization and monomer purification	CARBIOS	3	59.50	1	48
4	Production of r-PET-based tyre-cord fabrics, lay-flat hose grade and technical clothing grade yarns	KORDSA	4	120.10	12	34
5	Technical textile containing products validation at demo and industrial scale – tyres, lay-flat hoses, and garments	MICHELIN	1	45.00	23	48
6	LCA, social-LCA, LCC, Market studies, business modelling and digitisation	HVL	12	130.45	1	48
7	Dissemination, Communication, Strategic Intelligence, Outreach and Replication	AXELERA	9	53.50	1	48
			Total PM	756.00		

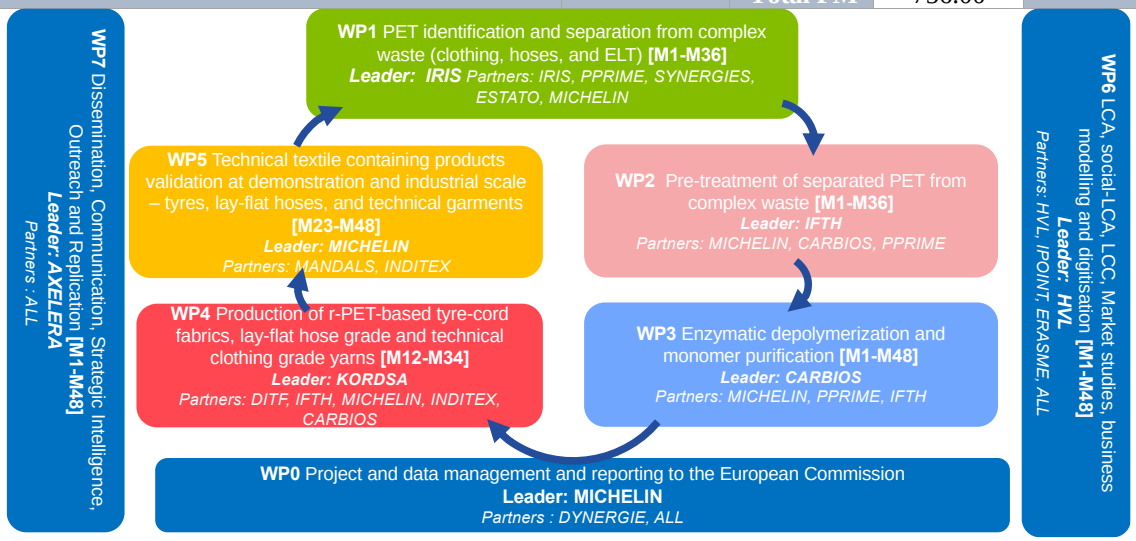

Figure 14 : PERT diagram

	Workpackage & Task Titles	Leader	Start	End
WP0	Project and data management and reporting to the European Commission	MICHELIN	1	48
Task 0.1	Technical management	MICHELIN	1	48
Task 0.2	Administrative and financial management	DYNERGIE	1	48
Task 0.3	Development and implementation of the FAIR data management plan	HVL	1	48
Task 0.4	Quality and risk control	DYNERGIE	1	48
Task 0.5	Advisory Board Management	MICHELIN	1	48
Task 0.6	IPR safeguarding	DYNERGIE	1	48
WP1	PET identification and separation from complex waste (clothing, hoses and ELT)	IRIS	1	36
Task 1.1	LIBS NIIR and hyperspectral technologies and methodology development for the characterization of clothing waste feedstock, reaching TRL 3-5	IRIS	1	8
Task 1.2	Exploratory testing of other available technologies for the chemical characterization of key materials contained in clothing waste feedstock	MICHELIN	1	6
Task 1.3	TRL increase of the processes for the identification and sorting of complex textile waste feedstocks – Reach TRL3-5	SYNERGIES	5	14
Task 1.4	Pilot scale process loop activities – PET concentration increases from the complex PET-containing waste feedstock through sorting and separation processes	SYNERGIES	12	14
Task 1.5	TRL increase of the processes for the separation and sorting of PET-containing complex waste feedstocks – Reach TRL 6-8	MICHELIN	14	36
WP2	Pre-treatment of separated PET from complex waste	IFTH	1	36
Task 2.1	Pre-treated analytical sample furniture for WP3	MICHELIN	1	12
Task 2.2	Pilot-scale study of amorphization of PET sample containing contaminants (rubber, textile coating etc.) from WP1 and for WP3	IFTH	13	15
Task 2.3	Preindustrial upscale of the shaping and amorphization processes of PET waste to prepare it for the depolymerization process	MICHELIN	14	36
Task 2.4	Characterization (transversal task) of PET-based wastes to determine their properties and amorphous and crystalline state according to the main transformation steps	IFTH	1	32
WP3	Enzymatic depolymerization and monomer purification	CARBIOS	1	48
Task 3.1	Impact analysis of the presence of impurities in complex clothing and ELT waste feedstocks on the enzymatic depolymerization step and first purification steps at lab scale	CARBIOS	1	17
Task 3.2	Pilot-scale loop: enzymatic depolymerization of the PET contained in 0.6 t (ELT feedstock) and 0.4 t (clothing waste feedstock) at pilot scale of pretreated PET waste from WP2 and monomer purification	CARBIOS	15	24
Task 3.3	By-products management at pilot scale	MICHELIN	12	48
Task 3.4	Pre-industrial scale up of the enzymatic depolymerization and monomer purification processes - on simplified waste feedstocks (High PET content waste)	CARBIOS	21	22
WP4	Production of r-PET-based tyre-cord fabrics, lay-flat hose grade and technical clothing grade yarns	KORDSA	12	34
Task 4.1	Pilot-scale production of r-PET polymer	DITF	20	26
Task 4.2	Pilot-scale production of r-PET yarn and test and characterization (after spinning)	IFTH	12	30
Task 4.3	Pilot-scale production of single end cord via cabling twisting and dipping for tyre application	KORDSA	24	25
Task 4.4	Pre-industrial scale production of r-PET polymer	MICHELIN	23	27
Task 4.5	Pre-industrial scale production of r-PET yarn and test and characterization (after spinning)	KORDSA	27	30
Task 4.6	Pre-industrial production of single end cord via cabling twisting and dipping for tyre application, and production of weave fitted for hose application	KORDSA	28	34
WP5	Technical textile containing products validation at demonstration and industrial scale – tyres, lay-flat hoses, and technical garments	MICHELIN	23	48
Task 5.1	Pilot-scale use-case: Making circular tyres, hoses and clothing	MICHELIN	23	33
Task 5.2	Pre-industrial-scale use-case: Manufacturing of tyre, hose and clothing and physical and behavioural performance assessment of the three targeted market products	MICHELIN	33	48
WP6	LCA, social-LCA, LCC, Market studies, business modelling and digitisation	HVL	1	48
Task 6.1	Stock-taking of data to prepare the evaluation of social, economic and environment dimensions	HVL	1	36
Task 6.2	Assessment of environmental dimension (Life-Cycle Assessment)	IPOINT	1	48
Task 6.3	Assessment of economic dimension (Life Cycle Cost, market's structure, Circular business models)	ERASME	1	48
Task 6.4	Assessment of social dimension (Societal impacts and social acceptance)	ERASME	13	36
Task 6.5	Stock-taking of assessment results (Impacts Cross-Matrix of Complex System)	HVL	1	48
Task 6.6	Development of an online IS solution enabling an automatic calculation of a feedstock's environmental impact and its compatibility on CARBIOS recycling process	MICHELIN	1	48
WP7	Dissemination, Communication, Strategic Intelligence, Outreach and Replication	AXELERA	1	48
Task 7.1	Strategic Intelligence bulletins	AXELERA	1	48
Task 7.2	Identification and engagement of targeted early adopters, end-users, and other stakeholders	AXELERA	1	48
Task 7.3	Update of the project dissemination plan	AXELERA	15	48
Task 7.4	Scientific publications and presentations	AXELERA	6	48
Task 7.5	Training development	AXELERA	6	48

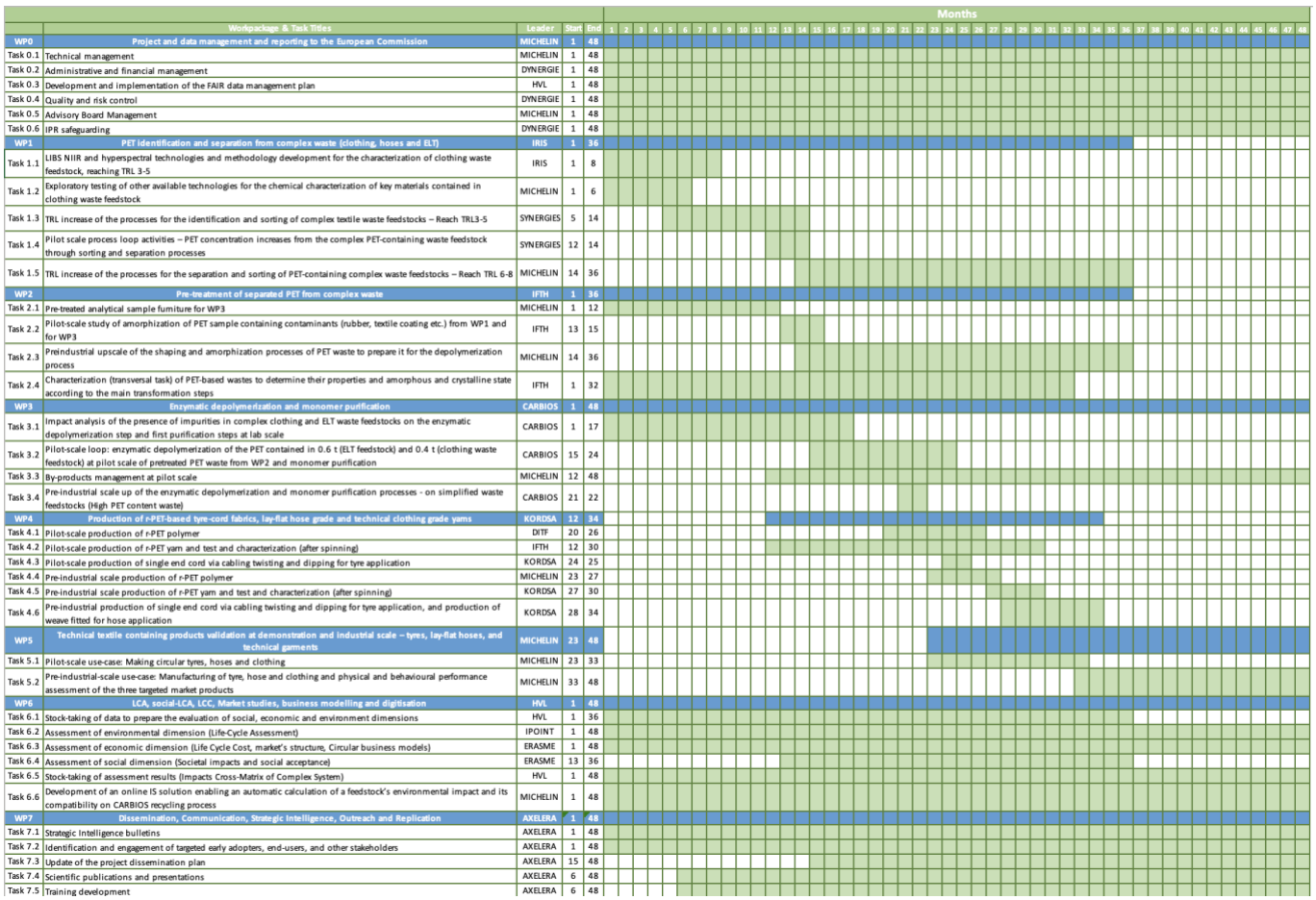

Figure 15 : GANTT chart

3.2 Capacity of participants and consortium as a whole

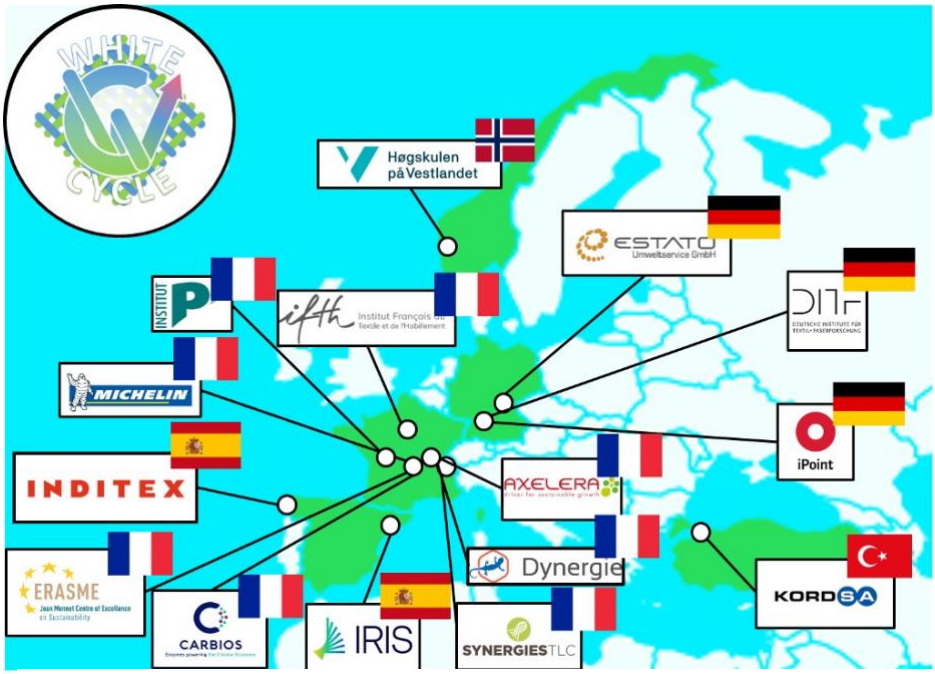

Figure 16 : The project provides a good coverage of the EU territory

The WHITECYCLE consortium consists of 15 partners of different types, namely recycling companies, brand-owners, final product manufacturers, end-user companies, digital companies, deep-tech and technology provider companies, a public-private eco-tech innovation networks, universities, and public research organisations. They provide good geographic coverage across 5 Member States or associated countries: France, Germany, Norway, Spain, and Turkey.

The consortium is built on the will to develop a circular economy of the PET textile contained in tyres, multilayer textiles, and lay-flat hoses. Each partner is expected to bring a key contribution toward this objective and strongly collaborate with the rest of the consortium to ensure the success of the project.

MICHELIN, the coordinator, is a well-known international tyre manufacturer who has already been involved in 10 EU-funded consortium projects including 1 as coordinator, the previously mentioned BLACKCYCLE. MICHELIN will bring its knowledge in tyre manufacturing, collection, and recycling but also its scientific excellence in the fields of PET recognition on complex textile waste, textile waste shaping processes, fibre and matrix electrostatic separation

and PET pre-treatment). MICHELIN will also provide key infrastructure to the project, such as material transformation process (extrusion, treatment, calendering) tyre manufacturing, material and tyre test machines and material analysis (MEB, SEM, IR…). Its core strategy includes transferring the results of BLACKCYCLE and WHITECYCLE to their factories to use currently impossible to recycle fractions of old tyres (PET, carbon black, etc.) to make new tyres on a large scale before 2030.

MANDALS, a leading lay-flat hose manufacturer, has already been involved in 1 FP7-funded consortium project. The company will oversee the use of recycled PET-textile in the production of lay-flat hoses and will especially study the adhesion behaviour of the outer layer made of thermoplastic or rubber material on the recycled PET-textile. MANDALS's core strategy is to make these composite hoses out of end-of-life materials on a large scale.

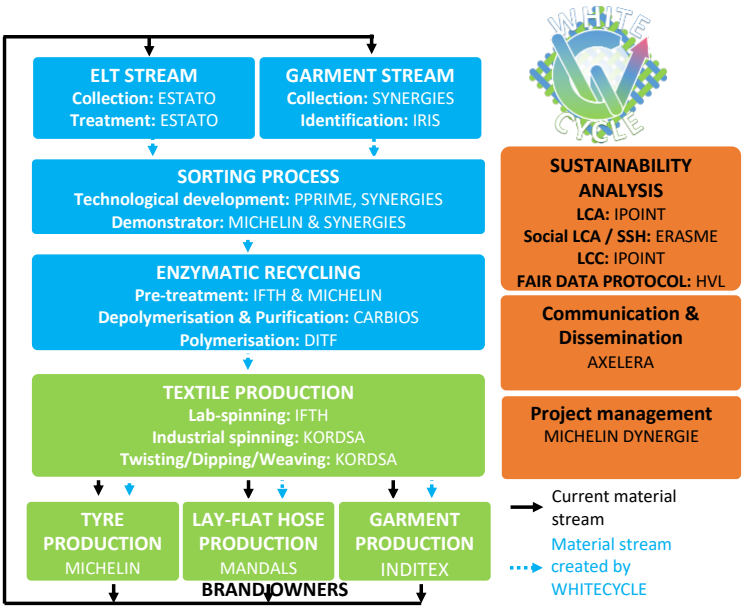

Figure 17 : WHITECYCLE circular value chains to boost 3 brand-owners to become circular by 2030 plus replication

INDITEX, the third brand-owner, is a European clothing manufacturing company with 8 brands, including ZARA, Pull&Bear and ZARA home. The company will provide key infrastructure to the project, such as industrial scale weaving, knitting equipment. Its strategy is to ensure design for recycling of its clothing and to use recycled textile materials in its new clothing lines. INDITEX will also provide known production wastes to the WP1 to be recycled.

CARBIOS provides a key missing link needed to make the PET value chain circular. Its enzymatic recycling process for PET plastic and textile waste is a world first and led to a publication in Nature followed by various industrial partnerships with global leaders such as NOVOZYMES, L'OREAL and NESTLE WATERS. This most innovative SME "Unicorn" was granted 11 patents in 2020 including seven in the United States and now owns 38 patent families. It is turning its disruptive innovation into bankable solutions. It's successful capital increase of 27 million euros (July 2020) is further evidence of this. The company will adapt its depolymerization and monomer purification processes for composite textile and complex textile clothing waste considering the higher amounts of impurities in theses feedstocks than in plastic packaging waste. CARBIOS will also provide key infrastructure to the project, such as a preindustrial unit (2 t of PET waste treated per week in a 20 m^3 reactor, TRL 7) inaugurated in Sept. 2021. Equipment for pre-treatment, depolymerisation purification of terephthalic acid (TA), purification of mono ethylene glycol (MEG) and purification of coproduct will be installed at the same site. Its business model has begun bearing fruit and is based on licensing its technology to PET producers who will build/operate industrial plants to produce r-PET first from simpler PET waste streams and after WHITECYCLE, even from the most complex PET wastes.

KORDSA, the leading manufacturer of industrial nylon and polyester yarn, tyre cord fabric and single end cord, has already been involved in 2 H2020-funded consortium projects. The company will lead WP4, in collaboration with **DITF**, to convert the monomers produced by CARBIOS into yarns and cords delivered to MICHELIN, MANDALS and INDITEX. The company will be responsible to produce r-PET based tyre cords and other use-cases but also work in close collaboration with DITF for the polymerization and IFTH for the lab-spinning to ensure the requested quality of the material used in this production.

ESTATO is a certified (ISO 9001 and 50001) waste disposal specialist with numerous regional and (inter)national customers who has already been involved in the EU-funded consortium project BLACKCYCLE with MICHELIN as coordinator. ESTATO will supply the PET waste from tyres and will be involved in the development of the new sorting technologies and its implementation for retrieving PET from ELT shreds in collaboration with MICHELIN. They plan to upscale and broadly implement tyre separation and sorting to provide secondary raw materials to PET manufacturers who will use CARBIOS's enzymatic recycling process.

SYNERGIES is a company working alongside collectors and sorters of textiles, for the sorting and recycling of end of life clothing waste, ultimately moving toward a full circular economy for textiles and footwear. SYNERGIES as the end-user of the multilayer sorting technology developed by IRIS will collaborate with this company to adapt the equipment to meet the industrial standards. Concomitantly, SYNERGIES is investing into an industrial-scale line for the separation of mono-layer clothing textile waste. SYNERGIES plans to upscale and broadly implement EoL clothing and footwear sorting to provide secondary raw materials to PET manufacturers who will use CARBIOS's enzymatic recycling process.

IRIS is a deep-tech and technology provider company which has already been involved in 102 EU-funded consortium projects including 39 as coordinator. The company will partner with SYNERGIES TLC to develop the automatic identification and dedicated recognition process for the multilayer textiles. They plan to upscale and equipment to the companies that will perform large-scale plastic and textile separation and sorting to provide secondary raw materials to PET manufacturers who will use CARBIOS's enzymatic recycling process.
IPOINT is a leading provider of software and consulting for environmental and social product compliance, and sustainability having tens of thousands of client companies around the globe. This organisation has already been involved in 2 EU-funded consortium projects including 1 as coordinator. IPOINT will bring its expertise and tools to the consortium for the LCC and LCA of the complete value chain.
DITF, founded in 1921, is the European largest textile research centre and performs public research, industrial projects and services for all organizations involved in fibres and textile in many sectors thanks to its research staff of ca. 300 employees. DITF has already been involved in 23 EU-funded consortium projects including 3 as coordinator. The research centre will study the complete polymerisation process using the monomers produced by CARBIOS and then supply IFTH and KORDSA in polymers. So DITF will work in close collaboration with these 3 entities.
IFTH is an industrial technical centre dedicated to support progress in the textile and clothing sectors. This establishment has already been involved in 10 EU-funded consortium projects including 1 as coordinator. IFTH will develop the amorphization process vital for the further enzymatic depolymerization performed by CARBIOS. IFTH will also be responsible for the r-PET cords production by melt spinning at pilot scale in collaboration with KORDSA.
PPRIME is a research unit of the CNRS created in 2010 and composed of more than 600 people whose research themes concern the Sciences for Engineering and Materials Physics. The laboratory will partner with MICHELIN to develop a fibre and matrix electrostatic separation process relying on vibratory, aerodynamic, and electrostatic properties for the material stream from ESTATO and SYNERGIES.
ERASME is an interdisciplinary programme, labelled in recognition of its expertise in sustainable development and co-financed by the European Union as part of the Jean Monnet program of Erasmus +, that brings together scientific and civil society expertise to advance sustainable development in the EU. This research organisation will oversee the social acceptance analysis, market studies and business modelling.
HVL is a university and research organization dedicated to solving the societal challenges of sustainability. HVL has already been involved in 11 EU-funded consortium projects including 1 as coordinator. It will perform the social acceptance analysis and will lead the consortium in setting-up, implementing and sharing FAIR data protocol.
AXELERA is the French cluster at the crossroads of the chemical and environmental sectors gathering 367 members. This cluster has already been involved in 8 EU-funded consortium projects. The cluster links innovators and helps them to de-risk and implement disruptive ecotechnologies, build new relationships and communicate effectively across sectors and for a general audience. AXELERA acquired expertise in terms of communication, dissemination and networking and will bring these competencies into the consortium.
DYNERGIE is a SME specialized in managing European R&D project, alongside the coordinator MICHELIN. DYNERGIE will organize the project consortium meetings, the reporting to the EC and the overall governance of the project. DYNERGIE has participated in 2 H2020 projects.

Table 3.1b: Work package description

WP number	0							Lead beneficiary							MICHELIN
WP title	Project and data management and reporting to the European Commission														
Part. number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Short name of part.	MICHELIN	CARBIOS	KORDSA	ESTATO	SYNERGIES	DITF	IFTH	AXELERA	PPRIME	ERASME	IRIS	HVL	IPOINT	DYNERGIE	INDITEX
PM per part.	23.00	6.00	2.00	3.00	8.00	2.00	2.00	4.60	0.45	1.00	3.00	5.00	3.00	20.00	3.00
Start month	M1				End month			M48							

Objectives: The WP0 aims to develop an effective and comprehensive technical, administrative, and financial management to ensure the successful execution of the WHITECYCLE project. Specific objectives will be:

- To ensure the achievement of all project's objectives on time, quality, and costs.
- To manage and coordinate smoothly and efficiently the WHITECYCLE partners, ensuring the achievement of the specific objectives from each of the WPs.

- To communicate a high-quality project reporting to the EC.
- To manage the different stakeholders through the Advisory Board.

Task 0.1: Technical management – Duration: [M1-M48] - Leader: MICHELIN - Part.: ALL

The Project Management will be conducted to coordinate the technical efforts and outputs of the partners. Technical management meetings will be held on a regular basis to ensure the coordination between the related WPs and activities. The project progress, i.e., risk management, technical results, and deliverable preparation, will be monitored to comply with technical milestones, intermediate outputs and eventually project objectives. MICHELIN will set up a portal with the templates for all project's related documents such as deliverables, technical follow-up tracking actual versus planned project progress or slide show presentations (D8.1).

Task 0.2: Administrative and financial management – Duration: [M1-M48] - Leader: DYNERGIE - Part.: ALL

The objective of this task is to ensure that the contract terms are followed and when necessary updated. The governance structure, as well as the communication flows and methods will be presented in the kick-off meeting. DYNERGIE will define templates that allow participants to provide inputs for the reporting in an efficient way while maintaining details, accuracy, and completeness. After each review meeting, an internal report on the project's progress will be prepared and distributed to all partners (D8.2).

DYNERGIE will act as the main contact point and monitor any contract amendment which might be requested by the Consortium for submission to the EC. DYNERGIE will deliver a progress report on a semi-annual basis and a detailed report at the end of every period. Moreover, MICHELIN will distribute EC funds amongst partners according to the Grant agreement.

Task 0.3: Development and implementation of the FAIR data management plan - Duration: [M1-M48] - Leader: HVL - Part.: ALL

The objective is to ensure overall data governance adhering to the FAIR principles at implementation level, i.e., enabling machine-actionability and human-readability. This includes the identification of all relevant data, the organization of data processing standards, and the assignment of responsibilities. The workflow in brief is the following. Findability: Creating an inventory of all project data and organize data along with content into data objects. Assign data objects with persistent identifiers. Accessibility: Define access procedures and execution rights. Interoperability: Describe data and metadata standards for the data objects using existing standards, taxonomies, and ontologies (all tagged with persistent identifiers). Reusability: Assign licenses for reusing data. Ensure dissemination of data to different stakeholders. Technical realization: RDF, SPARQL. A blueprint of the FAIR implementation protocol will be developed for re-use in similar industrial context (upload to the EOSC platform as an example; D0.7). This includes reference implementation files for data sets based on existing industrial standards, classifications and ontologies for chemical substances, industrial processes, and societal use. This also embraces the linking of data with open vocabularies and standards.

Task 0.4: Quality and risk control – Duration: [M1-M48] - Leader: DYNERGIE - Part.: ALL

A quality policy will be validated by the Steering Committee covering all aspects related to quality assurance and practical management of the project. The main issues addressed will include technical and non-technical risk management (including critical path and contingency plans), project SWOT, communication and quality control activities, gender, societal, and ethical issues (D.0.4). For instance, it will clearly state internal processes related to standards of documents, proofreading, clear validation workflow for the approval of the deliverables, reports, dissemination tools like the platform. The monthly follow-up of the Key Performance Indicators will allow to evaluate the performance of the project and its impact. MICHELIN will check on a regularly basis (at list every quarter) the fit of the deliverables with the Gantt of WHITECYCLE project and be guarantor of the corrective action plan of all WPs (in hand with WP leaders and partners involved) to deliver at the right time.

Task 0.5: Advisory board – Duration: [M1-M48] - Leader: MICHELIN - Part.: ALL

An Advisory Board (AB) will be set-up and composed of consumer associations, European tyre recycling associations, such as ALIAPUR, brand owners in different sectors using textiles and plastics, a PET manufacturer, two universities (WESTERN SYDNEY UNIVERSITY, UNIVERSIDAD DE LEON), one consumer association, one public authority responsible for circularity regulations, one respected think-tank making recommendations for circularity policies, such as the OECD. Each member of the AB will have to sign a Non-Divulgation Agreement. This group of experts will be invited to at least one General Assembly per year, to give their feedback about the project achievements (technical, economic, social aspects) and will provide their feedback about local needs and market potential in their respective countries. Report of the AB is integrated to the progress technical deliverable (D0.2 through D0.5).

Task 0.6: IPR safeguarding – Duration: [M1-M48] - Leader: DYNERGIE - Part.: ALL

Aside from the IPR issues to be covered by the CA, additional work needs to be done in this area to make the most of project results. Main activities will be the support and assessment for Licensing, Partnering, and Joint Exploitation Agreements; Helping in the negotiation and drafting of the bi-lateral or multi-lateral agreements necessary to exploit project results; Assessment of strategies on 3rd party licensing, first rights of refusal, minimum production quotas, etc.; and patenting of project results. A detailed list of each partner background will be included in the exploitation plan and updated in further iterations. These IPR activities will be carried out by DYNERGIE.

WP number	1	Lead beneficiary								IRIS					
WP title	PET identification and separation from complex waste (clothing, hoses, and ELT)														
Part. number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Short name of part.	MICHELIN	CARBIOS	KORDSA	ESTATO	SYNERGIES	DITF	IFTH	AXELERA	PPRIME	ERASME	IRIS	HVL	IPOINT	DYNERGIE	INDITEX
PM per part.	35.00			17.50	36.00				37.15		60.00				
Start month	M1				End month	M36									

Objectives: he objectives of this WP1 are to:

- Carry out methodology development to characterise key materials contained in clothing waste feedstocks and detect those that are rich in PET through HIS, LIBS, NIR, hyperspectral technologies, etc.
- Preliminary **process upscale to reach TRL5** with the design & implementation of pilot-scale sorting system:
  - On complex clothing waste recognition and sorting: to develop a semi-automatised identification and separation system to constitute 80% PET content waste.
  - On PET composite fibre/matrix separation to produce sorted PET from ELT, ELH, coated fabrics.
- To test the whole demonstration loop, whereby 2 t of clothing waste, 0.3 t of ELH and 10 t of ELT are processed to retrieve 0.5 t from clothing preparation and 0.8 t from electrostatic separation of sorted PET.
- **Process upscale to reach TRL6 up to 8** with the development of pre-industrial scale process
  - On the clothing sorting system consisting of identification and preparation system, capable to handle 20 kt/year of clothing. This system will be tested at SYNERGIES waste sorting facilities.
  - On the physical sorting unit able to produce 60 kg/h of sorted PET from dirty textile extracted from complex composites (ELT, hoses).
- To define the downstream process for unselected non-PET and low PET content materials to improve LCA.

Task 1.1: LIBS NIIR and hyperspectral technologies and methodology development for the characterization of clothing waste feedstock, reaching TRL 3-5 – Duration: [M1-M8] Leader: IRIS - Part.: SYNERGIES

SYNERGIES will provide complex clothing waste through manual sorting. The purpose of this task is the identification and content measurement complex multilayer clothing textiles such as stuffed clothing, coated textile, and membranes among others. The monitoring prototype applied in T1.5 will encompass HSI and LIBS technologies with an innovative advanced chemometrics approach and AI modelling strategy to define fast and accurate models to elaborate data and provide real-time identification textile composition to identify PET based clothing textiles. Two main optical and spectroscopic techniques will be assessed by IRIS (i) Vis-NIR and SWIR HSI and (ii) LIBS. Different configuration modes will be tested as absorption, diffusion reflectance, interactance and transflectance.

Additionally, offline mono- and multilayer reference samples will be created by SYNERGIES, chemically characterized by a lab, and used by IRIS (i) to assess the accuracy in the materials identification by the technologies and the required data processing for each case and (ii) to develop preliminary AI hybrid models based on chemometric and machine learning principles to provide compositional information. Special attention will be given to PET as the exact content of this material is required for the discrimination. The production of model samples can be considered to fill the gaps in the range of PET content.

Task 1.2: Exploratory testing of other available technologies for the chemical characterization of key materials contained in clothing waste feedstock – Duration: [M1-M6] Leader: MICHELIN - Part.: IRIS

Through bibliography and testing at MICHELIN or via subcontractors, the other technological paths such as Terahertz imaging, X-ray fluorescence spectroscopy are positioned vs. LIBS technology pursued as the main recognition technology. Their efficiency will be studied on the 3 feedstocks (coated, multilayer and stuffing).

If they are relevant for the PET identification, they will either be considered as back-up (in case LIBS doesn't give satisfaction) or be used as an additional system to reinforce LIBS analysis (reducing risk of wrong sorting).

Task 1.3: TRL increase of the processes for the identification and sorting of complex textile waste feedstocks – Reach TRL3-5 – Duration: [M5-M14] - Leader: SYNERGIES - Part.: IRIS, MICHELIN, PPRIME

Subtask 1.3.1: *Manual-based process upscale of complex clothing waste recognition and sorting* - ***leader:*** *IRIS* - ***Part.:*** *SYNERGIES* - Based on the results obtained in T1.1, the hybrid models will be finalised, and a selection of hardware and software specifications will be established. The optical systems, the enclosure and optomechanical parts will be designed using mechanical CAD tools with input from ray-tracing software (e.g., Zemax OpticStudio). The core devices will be optimized to identify the main target materials in terms of spectral and dimensional resolution, lighting system, scanned size, speed. Additionally, a preliminary software platform will be developed to acquire, model the data, and display results. In this sense, a TRL5 test bench (hardware and software) will be designed and delivered to be tested (offline and manually) in SYNERGIES. Trials using characterised samples provided by SYNERGIES will be used for the validation of the models. Part of the PET feedstock that will be provided to WP2 will be sorted during these trials.

Subtask 1.3.2: *Upscale of the shaping processes of the sorted PET-containing textile waste* - ***leader:*** *SYNERGIES* - ***Part.:*** *MICHELIN, PPRIME* – SYNERGIES will be designing a set of processes to be applied on the multilayer clothing textile waste to prepare it for the PET identification/characterization phase. These processes will be tested on the actual waste feedstock to assess their capability to separate all the different forms of clothing items present in the considered waste (stuffing, fabric, contaminants e.g., zippers, buttons, etc.). Based on test results, SYNERGIES will invest in the different processing machines.

*Subtask 1.3.3: TRL increase of the electrostatic processes of fibre matrix waste sorting for hoses and tyres – **leader: PPRIME - Part.:** MICHELIN* - On the existing PPRIME lab scale sorting equipment, the operating conditions will be optimized to reach the maximum theoretical flow rate which is 5 to 15 kg/h. Some additional stages (sieving, decompressing, or shaking operations) will be tested before the electrostatic treatment to maximize the quantity of PET recovered. ESTATO process will also be tested with the MANDALS hoses to recover the PET. The separation performance to separate the PET from the matrix will be evaluated and compared to ELT. The process parameters will be optimised to maximise the PET recovery.

Task 1.4: Pilot scale process loop activities – PET concentration increases from the complex PET-containing waste feedstock through sorting and separation processes – Duration: [M12-M14] - Leader: SYNERGIES - Part.: MICHELIN, ESTATO, PPRIME

SYNERGIES will provide 2 t of used clothing containing >80% of PET:

- The developed LIBS technology will be used to sort 0.5 t complexes cloths.
- 1.5 t will be manually sorted (based on the label tag) or using NIR detection.

ESTATO will produce of 2 t of dirty textile from ELT and 300 kg from granulated hoses.

- Optimization of a lab equipment for tribo-electrostatic separation of PET; production of 600 kg of PET.
- Adaptation of electrostatic technique to recover PET from granulated ELH; if possible, production of 100 kg of PET.
- Feasibility tests of tribo-electrostatic separation of PET from certain classes of clothing textiles. If it works, production of 100 kg of PET fibres.

Task 1.5: TRL increase of the processes for the separation and sorting of PET-containing complex waste feedstocks – TRL 6-8 – Duration: [M12-M36] Leader: MICHELIN - Part.: ESTATO, SYNERGIES, PPRIME, IRIS

Subtask 1.5.1: *Design, development, and implementation at SYNERGIES's facilities of an industrial-scale recognition, sorting and shaping system for complex clothing feedstocks – **Duration**: [M12-M24] **Leader:** SYNERGIES - **Part.:** IRIS* - The final monitoring prototype for PET recognition will be a system including a conveyor to work in inline conditions and will provide a real-time identification of multilayer textile clothes. Based on the requirements and specifications of SYNERGIES, IRIS will redesign and upgrade the TRL5 system gathered after T1.4 and operational parameters (hardware and software) for the upscale conditions and will deliver it to be integrated into a SYNERGIES dedicated line for validation. The physical stabilisation and presentation of the sampled object to the sensing device will be addressed as well as the uniformity of the optical path-length, the fluctuations in the atmosphere due to thermal gradients and the presence of dust and undesired particles. The signal-to-noise (SNR) of the instrument will be optimized through mechanical, electrical and sensor stabilisations.

In collaboration with SYNERGIES and the monolayer sorting equipment provider, tests will be run to adapt IRIS recognition system to the plant sorting system. Data generated by the monitoring device will be made available to the sorting system, ensuring that the monitoring unit latencies are compatible and that both are synchronized. The option of an independent all-in-one system including input flow stream, material identification and recognition and physical sorting will be studied as well.

Subtask 1.5.2: Design, development, and implementation at ESTATO's facilities of a pre-industrial-scale sorting system for ELT feedstock – **Duration:** [M12-M36] **Leader:** MICHELIN - **Part.:** ESTATO, PPRIME,

Pre-industrial upscale of the sorting system defined in task 1.2.2 to reach sorting capacity of 30 to 60 kg/h:

- Technical specifications will be written by PPRIME with input from ESTATO for the implantation and MICHELIN for industrial processes. The design study of the equipment will be done by MICHELIN.
- The equipment will be realized by a subcontractor managed by MICHELIN with the assistance of PPRIME
- The equipment will be commissioned at the subcontractor location by MICHELIN and PPRIME and installed at one of the ESTATO facilities. MICHELIN will provide support during initial equipment operations.
- Process parameters are tuned with a first trial with real waste coming from ESTATO facilities. Key process parameters (fibres content, throughput) are measured.
- 1 t of PET fibres are sorted from 10 t of ELT.

WP number	2						Lead beneficiary								IFTH
WP title	Pre-treatment of separated PET from complex waste														
Part. number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Short name of part.	MICHELIN	CARBIOS	KORDSA	ESTATO	SYNERGIES	DITF	IFTH	AXELERA	PPRIME	ERASME	IRIS	HVL	IPOINT	DYNERGIE	INDITEX
PM per part.	36.00	0.10					36.00		3.65						
Start month	M1				End month			M36							

Objectives: The objectives for this WP2 are for the ELT and clothing PET waste to:

- Pilot-scale study of the amorphization process of PET-based wastes containing pollutants (rubber, textile coating, non-PET fibres…) (TRL 5).
- Define and implement an industrial-scale process for the pre-shaping and amorphization of PET-based waste to be compatible for enzymatic depolymerization (TRL 7)

- Develop an analytical protocol to characterize amorphous/crystal ratio of PET to validate the efficiency of the amorphization process.

Task 2.1: Providing pre-treatedanalytical sample furniture for WP3 – Duration: [M1-M12] - Leader: MICHELIN - Part.: IFTH

Pre-screening of amorphization process with neat PET samples from clothing and fabrics from ELT (resp. clothing and technical grade) will be amorphized by the mean of the 3 different paths considering intrinsic properties of polymers i.e., IV and textile shape. 1 kg of analytical sample (from each waste) should be provided for depolymerization and preliminary purification trials in WP3. To secure the 1 tonne scale of the WP2, the work should be carried out at different scales (1 kg to 10 kg for pre-screening model). The robustness, efficiency and environmental impacts of the 3 methodologies will be compared.

Subtask 2.1.1: Pre-shaping process (from ELT & garment PET) - For each amorphization process, it could be necessary to convert the fibres into a suitable shape allowing to withstand the selected process. The shaping process must be defined and tested. It could be done using densification for extrusion purpose or non-woven process on the IFTH pilot's installation to have a continuous and cohesive material surface.

Subtask 2.1.2: Lab Scale activated gas path (IFTH) - Highly energetic species and high temperature gas is known to lead to both functionalization and polymorphism (including amorphization) mechanisms. Amorphized morphologies will be studied using a combination of time exposure and cooling. Post shaping as cutting of the treated material will be done before the following step of depolymerization.

Subtask 2.1.3: Lab Scale extrusion path (IFTH / MICHELIN) - The shaped mixture will be amorphized with extrusion screw technology with existing equipment. It will be micronized and supplied to CARBIOS to assess the depolymerization efficiency.

Subtask 2.1.4: Lab Scale heat/quench path (MICHELIN) – Thermal route could also be conducted through combination of heating (infrared, radiation...) and consequent quenching.

Task 2.2: Pilot-scale study of amorphization of PET sample containing contaminants (rubber, textile coating etc.) from WP1 and for WP3 – Duration: [M13-M15]- Leader: IFTH - Part.: MICHELIN

Subtask 2.2.1: Validation of the amorphization processes (of neat PET tested in 2.1) with PET sample containing contaminants from WP1 - Several samples, of about 100g, containing more than 80% PET should be provided to WP3 to evaluate enzymatic depolymerization efficiency. Amorphisation will be tested on samples of different purities and shapes (clothing stuffing; coated textile and fabric parts of ELT): 80% PET – 20% Rubber/ 90% PET – 10% Rubber/ 95% PET – 5% Rubber. This leads to selection of the more suitable process for demonstration-scale.

Subtask 2.2.2: Pilot up-scale of the selected amorphization process - The lab-scale processes and process parameters defined in task 2.1. and 2.2.1 are deployed to a pilot scale on dedicated installation either on IFTH facilities (non-woven, plasma treatment, extrusion) or on subcontractor facilities. 1 ton of amorphized PET (a-PET) will be available for WP3.

Task 2.3: Preindustrial upscale of the shaping and amorphization processes of PET waste to prepare it for the depolymerization process – Duration: [M14-M36] - Leader: MICHELIN - Part.: IFTH, CARBIOS

The shaping and amorphization technologies selected in the task 2.2 will be deployed in a new preindustrial scale equipment in MICHELIN's facilities, able to run at 30 to 60 kg/h. This will be done based on the different industrial equipment and facilities selected during the previous tasks.

Task 2.4: Characterization (transversal task) of PET-based wastes to determine their properties and amorphous state according to the main transformation steps – Duration: [M1-M32] - Leader: IFTH - Part.: MICHELIN

A protocol using complementary analysis methodology is developed to characterize amorphous/crystalline ratio of PET mixes based on differential scanning calorimetry, infrared spectroscopy (IR, Raman) and microscopy (numerical, MEB) on samples. Indirect methodology using standard PET dyeing recipe may also be used to assess PET amorphous state.

WP number	3				Lead beneficiary				CARBIOS						
WP title	Enzymatic depolymerization and monomer purification														
Part. number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Short name of part.	MICHELIN	CARBIOS	KORDSA	ESTATO	SYNERGIES	DITF	IFTH	AXELERA	PPRIME	ERASME	IRIS	HVL	IPOINT	DYNERGIE	INDITEX
PM per part.	4.00	55.00					0.50								
Start month	M1				End month	M48									

Objectives: The objectives for this WP3 are:

- Determine the maximum rate of impurities in composite waste (tyres & hoses) and complex clothing to be provided by WP1 thanks to samples prepared in WP2 to achieve a sufficient enzymatic depolymerization yield (75-80%).
- Perform enzymatic depolymerization of PET contained in ELT and clothing waste feedstock at pilot scale (1m^3 reactor) by processing 0.6 tonne (ELT/ELH) and 0.4 tonne (clothing waste) of pre-treated PET from WP2.
- Develop a purification process suitable to produce high purity monomers from complex mixed waste with numerous impurities. Produce 200kg of purified monomers from each feedstock at pilot scale for WP4.

- Assess best end-of life options for impurities filtered out of the reaction mixture at pilot scale
- Provide 1-2 t of purified monomers from high PET content waste for WP4 to achieve validation of recycled PET quality for high demanding applications (tyres) at preindustrial scale

Task 3.1: Impact analysis of the presence of impurities in complex clothing and ELT waste feedstocks on the enzymatic depolymerization step and first purification steps at lab scale– Duration: [M1-M17] - Leader: CARBIOS - Part.: MICHELIN, IFTH

The aim of this task will be to find the right equilibrium between fineness of sorting/separation performed in WP1, accessibility of PET after pre-treatment performed in WP2, and enzymatic efficiency. For each feedstock, CARBIOS will perform depolymerization tests at lab scale (50 g in 500 mL reactor) on samples with various PET/impurities ratio to determine the optimal sorting parameters and validate the depolymerization parameters. The parameters optimised for high PET content textile waste will be used at first: 60 °C, 200 g PET / L, 2 mg enzyme / g PET to target at least 75% depolymerization for ELT feedstock and 80% depolymerization for clothing waste feedstock (based on preliminary test results with a pre-treated sample containing 70% PET and 30% rubber from ELT). Depending on the results obtained, these parameters can be optimised such as the ratios of enzyme / waste / water and temperature. For each feedstock, 2 series of 3-4 samples will be treated to determine the sorting/pre-treatment combination leading to the highest depolymerization. For each feedstock, one sample will be selected, and steps of purification will be validated on selected sample by producing 70g of precipitated crude terephthalic acid. (CTA). The produced CTA will be characterized in CARBIOS lab (DSC, CIELab color, HPLC). 2 months are necessary for the validation of the purification steps.

Task 3.2: Pilot-scale loop: enzymatic depolymerization of the PET contained in 0.6 t (ELT feedstock) and 0.4 t (clothing waste feedstock) at pilot scale of pre-treated PET waste from WP2 and monomer purification – Duration: [M15-M24] - Leader: CARBIOS - Part.: MICHELIN,

After validation of sorting/separation and pre-treatment in WP1 and WP2, scale-up of depolymerization will be performed (TRL6: 100 kg waste in 1m3 reactor). CARBIOS will use the procedure determined in task 3.1 for the depolymerization. CARBIOS will treat the two feedstocks in series for depolymerization and purification. For each feedstock, CARBIOS will treat the waste in batches of 80 to 100 kg e.g., 6-7 batches for ELT feedstock and 4-5 batches for clothing waste feedstock. For each feedstock, the first batches will be used to validate the depolymerization scale-up and optimize the purifications steps. The purification scheme used for high PET content textile waste will be tested at first and optimized if necessary. CARBIOS will then produce 85-100 kg of TA to produce 100 kg of PET in WP4. Subsequently, 85 to 100 kg of TA will be crystalized at once. After purification of TA, the filtrate will be treated, 20 kg of MEG will be recovered. 2 extra months per feedstocks will be necessary for MEG recovery. Process simulation will be conducted on ProSimPlus to assess mass balance and energy performances to prepare optimized scale up.

Task 3.3: By-products management at pilot-scale Duration: [M12-M48] - Leader: CARBIOS - Part.: MICHELIN

CARBIOS will perform analysis on the Na2SO4 salt, subproduct of the CARBIOS process, to assess the possibility to purify it through the purification scheme previously developed for high PET content waste. Solid residues extracted after the depolymerization process will be characterized by MICHELIN through IR, DSC, and microanalysis on electronic microscopy. Environmental consequences of various scenarii of valorization will be evaluated (solid recovered fuel, cement-plant, pyrolysis) especially using knowledge developed through BLACKCYCLE project. All the by-product generated will be valorized through the best environmental path.

Task 3.4: Pre-industrial scale up of the enzymatic depolymerization and monomer purification processes - on simplified waste feedstocks (High PET content waste) Duration: [M21-M22] - Leader: CARBIOS

CARBIOS built a pre-industrial unit (TRL7) in Clermont-Ferrand for PET enzymatic recycling and began operations in Sept. 2021 with high PET content waste, especially coloured bottles. In the framework of WHITECYCLE, CARBIOS will produce 2 t of monomers via its process in order that the consortium can validate its quality for high demanding applications.

A batch of 2 t plastic waste will be depolymerized in a 20 m^3 reactor using enzymes as catalyst and water as solvent/reagent. After 24 h at a temperature of 60-72°C both monomers (TANa2 form and MEG) are found in solution, with the remaining enzymes, unreacted PET, and impurities (labels, caps, polyamide present in some multilayer bottles, paper…). Yield of depolymerization will be checked by HPLC assay of TA and MEG and a mass balance after solid residual separation.

TA and MEG will then be purified. The reaction mixture will be filtered. The filtrate containing TA and MEG will undergo discoloration. TA will be precipitated, filtered out, crystalized and dried. The purified TA will be characterized in CARBIOS lab (sublimation temperature by DSC, CIELab color, HPLC) and by external parties to determine its purity and granulometry, compared to commercial TA. The MEG solution will be neutralized before being distilled. The salt, byproduct of the enzymatic reaction, will be recovered and purified to be valorised.

WP number	4	Lead beneficiary							KORDSA						
WP title	Production of r-PET-based tyre-cord fabrics, lay-flat hose grade and clothing grade yarns														
Part. number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Short name of part.	MICHELIN	CARBIOS	KORDSA	ESTATO	SYNERGIES	DITF	IFTH	AXELERA	PPRIME	ERASME	IRIS	HVL	IPOINT	DYNERGIE	INDITEX
PM per part.	2.00	0.10	59.00			31.00	23.00								5.00

Start month	M12	End month	M34

Objectives: The objectives of this WP4 are to produce and process **r-PET from WP3 monomers that are compliant with the technical requirements** of the reinforcement frame of tyres, hoses, and technical garments

At pilot-scale 160 kg of r-PET (IV > 0.95 comparable with commercial PET technical grade),130 kg of r-PET yarn (tenacity > 7 cN/dtex), 60 kg of tyre cord quality reinforcements (through twisting and dipping processes), 20 kg of twisted HT cords (compliant with lay-flat hose manufacturing purposes), 60 kg PET (IV > 0.65), 40 kg of r-PET garment yarns, 30 kg of weaved textile (compliant with technical garment manufacturing purposes).

At pre-industrial scale:1,800 kg of r-PET (through a solid state polycondensation), 250 kg to produce textile grade yarns, 1,500 kg to produce tyre grade yarns, 600 kg of tyre cord fabric (dedicated baby roll), 210 kg yarn (to produce lay-flat hose), 400m² of weaved multicomponent fabric (clothing).

Task 4.1: Pilot-scale production of r-PET polymer – Duration: [M20-M26] - Leader: DITF - Part.: KORDSA, CARBIOS

Subtask 4.1.1 1 Esterification and polycondensation (DITF, KORDSA, CARBIOS)

The existing PET synthesis process, consisting of esterification of TA to BHET and polycondensation of BHET to PET, will be adapted to novel recycled monomers. To eliminate the deviation caused by putative impurities, new synthesis recipes will be jointly developed by KORDSA and DITF. In the current case, DITF has a PET synthesis capacity from 1 kg to 25 kg per batch and the ability to synthesize PET from TA polymerization route methods. New synthesis recipes will jointly be developed by KORDSA and DITF to eliminate the potential drawbacks caused by the impurities remaining after the monomer purification. For this purpose, new additives will be adapted, such as types and amounts of catalyst, processing aids, nucleating/coupling agents, chain extenders. Thus, the negative effects of acid residues and unknown contamination will be neutralized, long-term thermal stabilization, mechanical and rheological properties will be improved. The new additive will enable r-PET possessing virgin-like features during polycondensation.

Subtask 4.1.2 Solid state polycondensation (SSP) (DITF, KORDSA) - Performed to increase the molar mass and obtain technical grade, the objective will be to obtain an r-PET resin featuring equivalent properties to those of virgin PET. KORDSA and DITF will be conducting SSP studies with 1-25 kg capacity for lab-scale studies.

Subtask 4.1.3 Testing and characterization (before esterification, after polycondensation, after SSP) (DITF, KORDSA) - DITF and KORDSA will test and characterise the recycled monomers to define their chemical grades before starting polycondensation tasks. Included are Head Space GC-MS analysis, melting point for TA and boiling point for EG determinations. Polymer properties after polycondensation and SSP will be analyzed by viscosity (IV), End group determination, rheology, GPC, NMR, DSC, TGA, XRD, MFI, moisture and color CIELab analysis.

Task 4.2: Pilot-scale production of r-PET yarn and test and characterization (after spinning) – Duration: [M12-M30] - Leader: IFTH - Part.: KORDSA, MICHELIN, DITF, INDITEX

The melt spinning behaviour of the SSP-made r-PET will be studied by the mean of a semi-industrial melt spinning line at IFTH. The main parameters studied will be spinnerets, extrusion temperatures profile, throughput, drawing, relaxation. The aim is to prototype 20 kg to 50 kg of multifilament yarn for tyre, hoses and clothes applications at demonstrator scale and the characterization of the yarn compared to required specifications: tensile / shrinkage properties.

To optimize process parameters for r-PET spinning, commercial PET with IV from 0.65 to 0.95 will be first studied.

Subtask 4.2.1 Melt spinning at demonstration-scale (KORDSA, INDITEX, IFTH)

- Study of the thermal and rheological behaviour of the r-PET including comparisons with commercial grades for HMLS, technical and clothes grades.
- Study the spin-ability of commercial PET and the r-PET by the mean of a semi-industrial melt spinning line and prototyping of 100g to 50 kg of multifilament yarn.
- Characterization compared to required specifications (HMLS, high tenacity and clothes applications)
- Validation of the process parameters for the upscaling at task 4.5 (textile grade for clothes)

Sub-Task 4.2.3 Testing and characterization (after spinning) (KORDSA, INDITEX, IFTH, MICHELIN):

Thermo-mechanical properties like breaking strength, elongation, shrinkage, stiffness.

Task 4.3: Pilot-scale production of single end cord via twisting and dipping for tyre application – Duration: [M24-M25] - Leader: KORDSA - Part.: IFTH

Subtask 4.3.1. Cabling / Twisting (KORDSA) - The cabling of 1440 and 1100 dtex yarns will be performed. Cabling/Twisting process involves bringing together two or more plies of multi-filaments and turn around each other to make up a twisted yarn. Depending on yarns, propositions twisting is important for determination of the tyre cord strength and fatigue performances. The r-PET will be used to produce prototype yarn at IFTH, and this yarn will be utilized in lab scale twisting. The pilot scale work package consists of:

- Study on twisting with reference raw yarns that produced with commercial PET chips by IFTH for tyre grade.
- Study on twisting with r-PET yarn yarns –twist level, and speed- optimization for tyre applications.
- Visual observation – mechanical quality.

Subtask 4.3.2: Tyre cord via dipping (KORDSA, IFTH) - After twisting, r-PET cords will be treated with waterborne adhesive formulations to afford tyre cord. Cords will be treated twice, first with a waterborne activation adhesive formulation, then with resorcinol-formaldehyde-latex (RFL) adhesive formulation. After dipping, adhesive formulations are heat cured at temperatures above 200°C. During these two steps of adhesive treatment, appropriate tensions are applied during dipping and heat curing steps. KORDSA will work on the determination of the exact adhesive formulations, number of tensions and the exposure time for heat curing. The demonstration-scale production will be conducted in laboratory-dipping unit.

Study on SEC dipping in laboratory scale: tension, temperature, and exposure time optimization with reference raw and r-PET twisted yarn for tyre applications.

Subtask 4.3.3. Testing and characterization (after twisting and after dipping) - Adhesive, tensile properties (breaking strength, elongation at break, elongation at a specified load, shrink force, thermal shrinkage, and stiffness) of the twisted yarns and dipped SEC, and fatigue of the final tyre cords will be measured.

Task 4.4: Pre-industrial scale production of r-PET polymer – Duration: [M23-M27] - Leader: MICHELIN - Part.: CARBIOS

This task is carried out by Technip Zimmer (through subcontracting). The repolymerization will take a month. 1,3 t of produced PET will go through an additional solid state polycondensation to increase the IV to 0.95 and make it suitable for tyre applications. Feedstock and (by-)products are analytically monitored. Relevant parameters are e.g., colour, viscosity, carboxylic end groups, DEG content.

Task 4.5: Pre-industrial scale production of r-PET yarn and test and characterization (after spinning) – Duration: [M27-M30] - Leader: KORDSA - Part.: IFTH, INDITEX

The pre-industrial scale production of HMLS consist of drying the r-PET chips with high IV, extruding the supplied chips, followed by melt region, spinning line – faceplate, and winding. During the pre-industrial scale production not only the product properties, but also the operational efficiency, namely the spinnability, will be studied. At first, the process conditions and operational activity optimizations will be studied with commercial high IV PET chips with the semi-industrial studies conducted by IFTH used as baseline. The optimization studies with r-PET will cover following parameters: drying conditions of r-PET chips, extrusion pressure and temperature profile, polymer throughput, spinnerets, drawing region speed and temperature profile, and operational program. The main aim prototyping of 200 to 700 kg of 1100 or 1440 dtex HMLS yarn for tyres and hoses applications.

*Subtask 4.5.1: Pre-Industrial scale spinning - HMLS production at KORDSA -*Detailed study on mechanical and melt-rheology behaviour of r-PET will investigate melt region process. And Operational method will be developed for highly efficient transition between commercial and r-PET production. Spinnability study will be realized on commercial high IV PET and high IV r-PET by the mean of a pre-industrial scale melt spinning line and production (HT yarn and then HMLS). Prototyping 200 to 900kg multifilament yarn will be done. Operational and process conditions will be reported and used for possible commercial productions.

Subtask 4.5.2: Testing and characterization (after spinning) - The test and characterization of HMLS yarn includes: number of broken filament measurement with offline fray analyser, tensile properties (breaking strength, elongation, shrinkage, linear density measurement), morphological and mechanical analysis with XRD, GPC, DSC, DMA instrumental analysis (IV, CEG).

Subtask 4.5.3: 200kg textile grade production at INDITEX facilities - INDITEX will produce 200 kg of textile grade yarns.

Task 4.6: Pre-industrial production of single end cord for tyre application, and production of weave for hose application and garments – Duration: [M28-M34] – Leader: KORDSA – Part.: MANDALS, MICHELIN, INDITEX

Subtask 4.6.1: Twisting dipping on baby roll at KORDSA - The cabling of 1400 or 1100 dtex yarns supplied by IFTH will be performed after twisting, r-PET cords will be treated with waterborne adhesive formulations to afford tyre cord. In this work, KORDSA will work on the determination of the exact adhesive formulations, number of tensions and the timespan for heat curing. During the studies, the inputs from Task 4.3.1 and 4.3.2 will be used as a baseline.

- Study on twisting parameters optimization (twist level and speed), dipping parameters optimization (temperature profile, tension profile and exposure time)
- Study on operational efficiency as number of breaks and broken filaments during r-PET twisting and dipping.
- Prototyping 600kg of HMLS baby roll product from 700 kg r-PET yarn for tyre applications.

Subtask 4.6.2: Testing and characterization (after twisting and dipping) - Adhesive, tensile properties (breaking strength, elongation shrinkage, and stiffness) of the twisted yarns and fatigue of the final tyre cords will be measured. The additional test at third parties will conducted for further chemical, physical and morphological analysis.

Subtask 4.6.3: Weaving done at INDITEX, 400 m² of representative weaved fabric will be produced by INDITEX subcontractors. Mechanical performance, dyability will be measured to validate technology applicability.

WP number	5				Lead beneficiary						MICHELIN				
WP title	Technical textile containing products validation at demonstration and industrial scale – tyres, lay-flat hoses, and technical garments														
Part. number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Short name of part.	MICHELIN	CARBIOS	KORDSA	ESTATO	SYNERGIES	DITF	IFTH	AXELERA	PPRIME	ERASME	IRIS	HVL	IPOINT	DYNERGIE	INDITEX
PM per part.	27.00														18.00
Start month	M23				End month	M48									

Objectives: The objectives of this WP5 are to prove that r-PET is a relevant raw materials for three use cases:

- At pilot-scale:
- highly technical HMLS r-PET tyre: 25 prototype tyres building trial will meet TRL5 requirement level (building process, geometrical sizing, and long-lasting endurance performances).
- technical r-PET lay flat hoses by extensive testing of 25 m hose in which at least 40% of v-PET is replaced by r-PET.
- as r-PET technical garment by extensive testing of 20 m^2 produced at full scale.

At pre-industrial-scale use:

- Manufacture 100 tyres containing r-PET. Requirement tests according to the EU regulations on tyres.
- Manufacture 1,500 m of lay-flat hoses through different weave batches and extrusion.
- Coat 400 m^2 of multicomponent fabric and manufacture of 4 lines of technical garments.

Task 5.1: Pilot-scale use-case: Making r-PET tyres, hoses, and clothing – Duration: [M23-M33] - Leader: MICHELIN - Part.: MANDALS, INDITEX

Subtask 5.1.1: Manufacture 25 tyres containing r-PET (MICHELIN) [M27-M33] Using first deliveries of few kgs of r-PET, lab evaluations will be launched in MICHELIN and in cooperation with evaluations driven by IFTH and KORDSA in WP4. The purpose of these crossed studies will be to maximize the reliability of assessment of new recycled tyre cord for tyre building and tyre use. Using r-PET supplied by WP4 in M27, rubber coated fabrics will be built on laboratory drums to demonstrate tyre performances and assess tyre building process performance. 25 Passenger Car tyres will be built in the R&D pilot shop and tested in static and dynamic conditions to check their real usage properties and performances. (Geometrical sizing, and endurance tests on drums). Depending on gap analysis between v-PET and r-PET, thermomechanical lab evaluation of r-PET could be conducted. They would allow tyre calculations of sizing and multi axial stiffnesses.

Subtask 5.1.2: Manufacture 25 m of lay-flat hoses (MANDALS) [M29-M33] The first deliveries of few kg of r-PET will be used for bench scale tests of yarn characteristics relevant for hose performance. This includes pre-treatment of yarn, the weaving operation, extrudability and compatibility between weave and TP or rubber cover. The data will be used to decide on full scale hose model system where the hose containing r-PET will be compared to similar hoses with 100% v-PET. At least 25 m of r-PET based hose will be produced in full-scale production. Perform full scale test at static and dynamic conditions to check real usage properties and performances (burst pressure, adhesion, stress-strain fatigue creep, chemical resistance).

Subtask 5.1.3: Manufacture of 20m^2 of multicomponent/coated textiles for technical garments (INDITEX) [M29-M33] Coated and multilayered fabrics will be made on proto scale looms and coating lines and tested. The process and final product performance are compared to v-PET based products.

Task 5.2: Pre-industrial-scale: Manufacturing of tyre, hose and clothing and physical performances assessment of the 3 targeted markets – Duration: [M33-M48] - Leader: MICHELIN - Part.: MANDALS, INDITEX

Subtask 5.2.1: Manufacture 100 tyres containing r-PET (MICHELIN) [M33-M48] Using second deliveries of higher volume of r-PET, semi-industrial evaluations will be launched in MICHELIN. 100 passenger car tyres will be built in industrial shops and tested. The industrial feasibility will be evaluated along the whole processing.

Subtask 5.2.2: Manufacture 1,500 m of lay-flat hoses (MANDALS) [M33-M48] Using second deliveries of higher volume of r-PET, semi-industrial evaluations will be launched in MANDALS shops. At least 1,500 m of lay flat hoses will be produced by varying weave and hose typed to prepare the introduction at a full scale of r-PET.

Subtask 5.2.3: Coating of 400 m^2 of multilayers textiles and manufacture 4 lines of technical garment (INDITEX) [M33-M48] - Using second deliveries of higher volume of r-PET, semi-industrial evaluations will be launched in INDITEX shops. 400 m² of coated multi-layered fabrics will be produced by varying weave and coating type to prepare the introduction at a full scale. 4 lines of technical garments will be manufactured and tested.

WP number	6			Lead beneficiary						HVL					
WP title	LCA, social-LCA, LCC, Market studies, business modelling and digitisation														
Part. number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Short name of part.	MICHELIN	CARBIOS	KORDSA	ESTATO	SYNERGIES	DITF	IFTH	AXELERA	PPRIME	ERASME	IRIS	HVL	IPOINT	DYNERGIE	INDITEX
PM per part.	9.00	0.20	7.00	3.50	1.00	1.00	1.00		0.25	43.00	2.00	31.00	27.50	3.00	1.00
Start month	M1				End month			M48							

Objectives: WP6 combines system thinking methods, sustainability evaluation, and econometrics to develop transferable circular economy tools, prototypes, and business models. The overall objective is to progress and anchor concepts for sustainable industrial transformation, which foster social inclusion and transparency along with economic sustainability, while minimizing adverse impacts on the environment and society. The result is a demonstration on how to develop indicators for SDG reporting at corporate level. The use cases that are explored technically in WHITECYCLE (WP2-WP5), are thereby

enlarged with a comprehensive sustainability assessment. Generic components of these specific investigations are identified and assessed for their blueprint potential in related industries.

Task 6.1: Stock-taking of data to prepare the evaluation of social, economic and environment dimensions – Duration: [M1-M36] - Leader: HVL - Part.: ERASME, MICHELIN, IPOINT

The usefulness of evaluation results rests upon data availability and data quality. This task ensures that all partners have a joint understanding about data needs, data coverage, potential data gaps, and data quality. Starting with a survey to WP6 task leaders, T6.1 identifies input requirements and user rights for the chosen methods of T6.2-6.5. Building on this information, metadata schemes are developed, linking to the project's FAIR DIP (WP0), international reporting standards, and developing a joint ontology across WP6 tasks. The final part of T6.1 is dedicated to developing standards for transparent reporting of data and results. This covers the tracking of the provenance of data and checkpoints for the transparent communication of WP results within the project and beyond (in compliance with GDPR and IPR). A first delivery will be done to WP0 by M5 for inclusion in the DMP by M6. A final version will be provided by M36 for inclusion in the final DMP.

Task 6.2: Assessment of environmental dimension (Life-Cycle Assessment) – Duration: [M1-M48] - Leader: IPOINT - Part.: ALL

This task performs a comprehensive environmental life cycle assessment (LCA) of the WHITECYCLE solutions and compares them with conventional alternatives, accounting for the complexity of the materials (incl. potential recyclability of end-of-life components) and criteria for technical performance (e.g., impact on the durability of materials). The LCA follows a cradle-to-gate framework and will focus on the following environmental impact categories: GHG emissions (carbon footprint, in particular biogenic carbon), primary energy demand (energy footprint), water use and consumption (water footprint), land use, as well as impacts on human health and biodiversity. Results will be used in an iterative approach along the project to i) guide the development of the investigated value chains towards more sustainable solutions, ii) quantify the potential benefits of the WHITECYCLE solution compared to alternative technologies (baseline), and iii) help future suppliers and customers make more informed decisions.

Task 6.3: Assessment of economic dimension (Life Cycle Cost, market's structure, Circular business models) – Duration: [M1 - M48] - Leader: ERASME, IPOINT, - Part.: ALL

This task analyzes causal relationships between cost-benefit analysis, market structures, and firm-level economic performance to support circular economy Framework oriented innovation (Blomsma & al., 2019). We start with an R-framework (4R to 10 R, Kirrchherr & al., 2017) to map the existing and proposed Circular Economy strategies and to foresee how they influence the viability of value chains for the WHITECYCLE innovation cycle. Next, we cross the economic analysis with the analysis at three spatial scales (micro, meso and macro) to elicit generic interactions. Thirdly, we develop economic indicators (such as variables providing relevant information for decision making, company-level reporting of SDGs), differentiating between end-product production (economic efficiency), consumption, waste management and production of secondary raw materials (EC, 2015), and by adjusting them to the different properties of innovation process (Input to T.6.5). A focus on creating value for the different actors in the value chain will be designed to ensure acceptance of the circular technologies by the full value chain. Life Cycle Costing (LCC) and economic analysis of selected WHITECYCLE processes will be carried out. Overall costs of the process related to the production, use and end-of-use will be assessed.

Task 6.4: Assessment of social dimension (Societal impacts and social acceptance) - Duration: [M13-M36] - Leader: ERASME - Part.: HVL – Part.: ALL

This task assesses social impacts of WHITECYCLE innovations and potential issues for social acceptance. We define social aspects as all parameters, indicators or issues related to the social relations between individuals and between individuals and society (Diemer, 2019), and we apply social aspect classifications established in the Social Life Cycle Assessment (SLCA) methodology (Jorgensen & al. (2008), Dempsey & al. (2011) and Ellen McArthur Foundation (2015). WHITECYCLE use cases are investigated in the light of SLCA indicators and Sustainable Development Goals. The analysis is informed by the following empirical methods: At micro-level, we establish focus groups and communication channels within the MICHELIN, MANDALS and INDITEX companies to analyse the degree of social acceptance and diffusion of innovation. At a meso- and macro-level, we identify key stakeholders through an influence-interest matrix, eliciting stakeholder insights through semi-structured interviews and participatory scenario building exercises with the aim of creating value for citizens and consumers. As an input to T.6.5, the task contributes with suggestions for indicators of the social dimension.

Task 6.5: Stock-taking of assessment results (Impacts Cross-Matrix of Complex System) - Duration: [M1-M48] - Leader: ERASME - HVL - Part.: HVL, IPOINT

It integrates the results from environmental, social, and economic evaluations and maps the interactions between flows and stakeholders. Systems thinking and system dynamics modelling are used to identify systemic behaviour patterns, leverage points of intervention (Nedelciu, Diemer, 2020) and to develop company-level SDG reporting. Moreover, a systemic analysis of different circular business models (Diemer, 2020) is launched to lay open potential controversy between technological centralization of global chains and territorial de-centralization priorities.

Task 6.6: Development of an online IS platform enabling an automatic calculation of a feedstock's environmental impact and its compatibility on CARBIOS recycling process **– Duration: [M1-M48] - Leader: MICHELIN - Part.:** IRIS, SYNERGIES

Development of an IS solution able to automatically give a feedstock an environmental score based on LCA calculation made by IPOINT and the ability to be used on the CARBIOS process. The algorithm will directly use the data from the IRIS' online characterization system and will generate an environmental score according to its compatibility with CARBIOS recycling process. The IS system will be connected to the RECYCLE platform (from Refashion) which already makes the inventory of different recycled feedstock and recycling solutions available. This task is about: developing the algorithm and release a stable

and user-friendly solution on RECYCLE platform. The solution will be able to environmentally evaluate a recycled PET feedstock and make the data available for the customer.

WP number	7	Lead beneficiary													AXELERA
WP title	Dissemination, Communication, Strategic Intelligence, Outreach and Replication														
Part. number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Short name of part.	MICHELIN	CARBIOS	KORDSA	ESTATO	SYNERGIES	DITF	IFTH	AXELERA	PPRIME	ERASME	IRIS	HVL	IPOINT	DYNERGIE	INDITEX
PM per part.	4.00	0.50	9.00	6.00	1.00	1.00	1.00	11.00	1.00	4.00	5.00	2.00	2.00	3.00	3.00
Start month	M1				End month	M48									

Objectives: The objectives of this WP7 are:

- To engage targeted early adopters: 3 PET manufacturers, 3 waste managers / recycling equipment manufacturers and 12 brand owners
- To promote WHITECYCLE process design kits for the construction of industrial-scale sorting and recycling plants
- To promote project objectives, events, opportunities, and results via B2B meetings, large-scale events, workshops, the project website, the 6-monthly newsletter and social media
- To gather key strategic information (such as competing technologies, market evolution and overviews) and share it with partners in a user-friendly, exploitable format
- To ensure dissemination to the research and training networks in science / engineering
- To update and implement the dissemination and exploitation plans and ensure high impacts.

Task 7.1: Strategic Intelligence bulletins – Duration: [M1-M48] - Leader: AXELERA - Part.: ALL

AXELERA will gather information and data about competing technologies, complementary initiatives, relevant meetings, markets, and interesting potential end-users. AXELERA will convert this information and data into actionable intelligence to enrich strategic and managerial decisions. The process of filtering, distilling, and analysing the data will be done by AXELERA who will then propose succinct, visual, clear intelligence to the coordinator and other partners to help them make their decisions. The resulting intelligence bulletins may be consulted when "on-the-go" via newsletters. The bulletins will be released every 3 months starting on M6. All partners will participate in this task to provide needs and specifications, to read the bulletins and to provide feedback to AXELERA for the improvement of the content, accessibility, and presentation of the bulletins.

Task 7.2: Identification and engagement of targeted early adopters, end-users, and other stakeholders – Duration: [M1-M48] - Leader: AXELERA - Part.: ALL

The WP7 meetings will be open to each partner's main technical and communications contact(s). The committee will meet once every two weeks during M2 and M3 to ensure that communication tools are rapidly put into place. The initial meetings will enable AXELERA to gather partners' specifications regarding the newsletter and intelligence bulletins, the content of initial press release, the website, logo, flyer, etc. After M3, the committee will meet regularly during the months before communication and dissemination actions are due to provide partners' needs and to optimize communication efficiency and results.

Subtask 7.2.1 GRPD compliance. To build the contact database for WHITECYCLE, AXELERA and other partners will collect personal contact data of stakeholders. AXELERA confirms that all the data we intend to process is relevant and limited to the purposes of the research project (in accordance with the 'data minimization principle' of the GRPD regulation). The purpose is to ensure the dissemination and broad uptake of the results of the WHITECYCLE project. A description of the technical and organizational measures that will be implemented to safeguard the rights and freedoms of the data subjects will be submitted as a deliverable by M2 of the project.

Subtask 7.2.2 Dissemination tools. AXELERA will prepare the logo and visual identity of the project as well as the public website (by M6). The website will be regularly updated by AXELERA (at least 3-monthly) with relevant, public content from the intelligence bulletins, newsletters, project results, upcoming events. A project flyer will be prepared by AXELERA by M6 in an electronic version. An updated version will be prepared in preparation for the workshop. AXELERA will prepare and circulate to the project mailing list, a 6-monthly newsletter. All partners will invite contacts to join the project mailing list. At the time of each website update and newsletter mailing, AXELERA will push content via social media, including LinkedIn and Twitter. Partners will be encouraged to repost this content. Each partner will inform the other partners in advance whenever a major publication, a presentation, event, or other milestone is planned linked to WHITECYCLE. For the key communication stages of the project, the first and the last press release: the project coordinator will write the text of the press release with support from partners. AXELERA will prepare 3 communication kits during the project (the first one by M6 and updates at M18 and M40). These kits will contain all communication materials available and will be shared on the project's shared online space. AXELERA will also prepare 2 videos in interview format during the project.

Subtask 7.2.3 Identifying and engaging stakeholders. Based on partners own networks and intelligence as well as the 6-monthly project intelligence bulletins, AXELERA will propose a list of targeted organizations and people to approach and engage. The proposed list will be submitted to the consortium for approval before engagement. **CARBIOS, ESTATO, IRIS, DITF, KORDSA, MICHELIN and SYNERGIES will seek agreements with at least 3 PET manufacturers, 3 waste managers / recycling equipment manufacturers and 12 brand owners** whose products contain PET to uptake project technologies by promoting the project's KPIs, meeting with them to understand their resources and needs and ultimately **disseminating process design kits** that meet their needs. (See section 2.2.1.)

Once agreements have been obtained with these stakeholders, they will continue to deepen relationships with these stakeholders to move toward full industrial deployment and market uptake just after the project. AXELERA will organize two workshops targeting the participation of 40 to 60 participants. The workshops will serve to strengthen relationships with the broader circle of stakeholders, to obtain their needs and feedback, to explain how the results could be useful for the stakeholders, etc.

To facilitate cooperation with related European projects in the field, with each intelligence bulletin, the list of relevant projects will be updated by AXELERA. The coordinators and partners of all these projects will be added to the project mailing list. Each project will also be invited to join all public project events. If representatives from other projects attend these events, a dedicated, face-to-face "project symbiosis" session will be organized. The WP leaders of all complementary projects will be invited to attend 6-monthly "symbiosis" conference calls with WHITECYCLE WP leaders to discuss common issues and opportunities. Dissemination to key stakeholders will whenever relevantly include links to access and reuse exploitable data published during the project.

ERASME partner will also cover dissemination towards the large public (section 2.2.1) by hosting debates, podcasts on circular economy. Partners' policy departments will contribute to policy discussions to support the emerging circular economy.

Task 7.3: Update of the project diss. plan – Duration: [M15-M48] - Leader: AXELERA - Part.: ALL

AXELERA will update the project's dissemination plan for each periodic report and the final report with inputs from all partners. These updates will meet the EC requirements.

Task 7.4: Scientific publications, presentations – Duration: [M6-M48] - Leader: AXELERA - Part.: ALL

The research institute partners plan to produce 10 major open-access project publications. 3 related "follow-up" publications will be prepared as easy-to-understand summaries by AXELERA. These boost the impacts of the major publications among non-specialists. AXELERA will obtain approval by the authors of the original publications before submission of the follow-up publications to Horizon, the EU Research & Innovation Mag., partners' websites, the project website, local news media, trade journals. Academic partners will organize conferences to promote their results among the scientific community (section 2.2.1).

Task 7.5: Training development – Duration: [M1-M48] – Part leader: AXELERA – Part: ALL

AXELERA will develop links and help build strong partnerships between the WHITECYCLE consortium and training programs by twinning with European training centers and networks like C-Planet and CAPTURE (funded European training programs). WHITECYCLE will put forward case studies, data, needs concerning future scientists and engineers (section 2.1.4).

Table 3.1c: List of Deliverables

Del. n°	Deliverable name	WP	Lead part.	Type	Diss. lvl	Delivery date
D0.1	Collaborative portal	0	DYNERGIE	OTH	SEN	M3
D0.2	Minutes of official meetings, set up of corrective action & their follow-up	0	DYNERGIE	R	SEN	Every 6M
D0.3	Data Management Plan	0	HVL	DMP	PU	M6/48
D0.4	WHITECYCLE's ethics requirements	0	DYNERGIE	ETHICS	PU	M3/42
D0.5	Final Report	0	MICHELIN	R	SEN	M48
D0.6	IPR activities report	0	DYNERGIE	R	SEN	M48
D0.7	Blueprint of a FAIR Implementation Protocol use cases (upload to EOSC)	0	HVL	R	PU	M48
D1.1	Report on LIBS, NIIR and hyperspectral technologies adequacy as recognition system of clothing textile waste	1	IRIS	R	SEN	M8
D1.2	Report on efficiency of other technologies for chemical PET textile identification vs. LIBS	1	MICHELIN	R	SEN	M6
D1.3	TRL5 recognition system	1	IRIS	DEM	SEN	M14
D1.4	Report on the shaping processes upscale for multilayer clothing textile waste	1	SYNERGIES	R	SEN	M12
D1.5	Report on the key parameters for the electrostatic sorting of PET from ELT	1	PPRIME	R	SEN	M12
D1.6	Report on the pilot scale process loop activities	1	SYNERGIES	R	SEN	M14
D1.7	Design, development, and implementation of a pre-industrial sorting system for clothing textile waste feedstock	1	SYNERGIES	DEM	SEN	M24
D1.8	Design, development, and implementation of a pre-industrial sorting system for ELT waste feedstock	1	MICHELIN	DEM	SEN	M36
D2.1	1 kg of analytical sample (from each feedstock) for depolymerization and preliminary purification trials in WP3	2	MICHELIN	DEM	SEN	M12
D2.2	1 tonne of amorphized PET	2	IFTH	DEM	SEN	M18
D2.3	Report on upscaling activities of the selected pre-industrial processes	2	MICHELIN	R	SEN	M36
D2.4	Protocol of characterisation of amorphous/crystalline ratio of treated/untreated PET waste	2	IFTH	R	SEN	M32
D3.1	ELT feedstock: production and characterization of CTA at lab scale from a selected sample	3	CARBIOS	R	SEN	M14

D3.2	Clothing waste feedstock: production and characterization of CTA at lab scale from a selected sample	3	CARBIOS	R	SEN	M17
D3.3	Production of 85-100 kg of monomer from ELT feedstock	3	CARBIOS	OTH	SEN	M19
D3.4	Production of 85-100 kg of monomer from clothing waste feedstock	3	CARBIOS	OTH	SEN	M24
D3.5	Report on analysis performed on the salt M24	3	CARBIOS	R	SEN	M24
D3.6	Report on the various opportunities of by-product valorization, comparison in term of environmental impact	3	MICHELIN	R	SEN	M48
D3.7	1-2 t of monomers from high PET content waste	3	CARBIOS	OTH	SEN	M22
D4.1	Process parameters to achieving IV > 0.95 comparable with commercial PET technical grade on lab scale (8 Kg)	4	DITF	R	SEN	M15
D4.2	160 Kg PET (IV > 0.95) and 60 kg PET (IV > 0.65) comparable with commercial PET technical grade	4	DITF	OTH	SEN	M22
D4.3	40 kg of HMLS yarn from commercial PET	4	IFTH	OTH	SEN	M18
D4.4	50 kg of r-PET HMLS yarn for tyre application	4	IFTH	OTH	SEN	M23
D4.5	25 kg of r-PET HMLS yarn for hoses and 40 kg of textile grade for clothing	4	IFTH	OTH	SEN	M27
D4.6	200 kg of r-PET textile grade for clothing textile	4	INDITEX	OTH	SEN	M30
D4.7	20 kg of twisted HMLS yarn from HMLS yarn with commercial PET	4	KORDSA	OTH	SEN	M21
D4.8	40 kg of r-PET HMLS single end cord for tyre application	4	KORDSA	OTH	SEN	M25
D4.9	20 kg of r-PET HMLS twisted yarn for hoses application & 910 kg of r-PET HMLS yarn for tyre and hoses application	4	KORDSA	OTH	SEN	M30
D4.10	600 kg of r-PET HMLS tyre-cord baby roll	4	KORDSA	OTH	SEN	M34
D5.1	Tyre demo of r-PET with at least the same performance as ref.	5	MICHELIN	DEM	SEN	M33
D5.2	Hose demo of r-PET with at least the same performance as ref.	5	MANDALS	DEM	SEN	M33
D5.3	Clothing demo of r-PET with at least the same performance as ref.	5	INDITEX	DEM	SEN	M33
D5.4	r-PET tyre scale-up validation with at least the same performance as ref.	5	MICHELIN	DEM	SEN	M48
D5.5	r-PET hose scale-up validation with at least the same performance as ref.	5	MANDALS	DEM	SEN	M48
D5.6	r-PET clothing scale-up validation with at least the same performance as ref.	5	INDITEX	DEM	SEN	M48
D6.1	Update of the project data inventory with WP6 data (input to the FAIR DIP) / WP6 metadata schema and ontological agreement / WP6 data transparency and communication checkpoints	6	HVL	DATA	SEN	M12/24/30
D6.2	Screening LCA for hotspots identification of environmental impacts/Comprehensive LCA report	6	IPOINT	R	SEN	M12/24/48
D6.3	Life Cost Cycle - Market Analysis	6	IPOINT	R	SEN	M18
D6.4	SWOT Matrix Impact - Circular Economy R Framework Report	6	ERASME	R	SEN	M24
D6.5	Circular Business Models Typology	6	ERASME	R	SEN	M42
D6.6	Social Life Cycle Assessment Methodology / Report on Social Impacts - Social Acceptance Process / Social Acceptance Survey's Analysis	6	ERASME	R	SEN	M24/36/42
D6.7	System Dynamic Model (loops' design) / Stakeholders Mapping - Participatory System Dynamics Modelling / Company Level SDGs Report	6	ERASME	R	SEN	M24/36/44
D6.8	Information system solution for feedstock environmental impact + recycling process compatibility assessment up and ready	6	MICHELIN	OTHER	SEN	M48
D7.1	GRPD compliance report	7	AXELERA	R	PU	M2
D7.2	Dissemination, exploitation and communication plan	7	AXELERA	DEC	PU	M6
D7.3	Summary of main benefits provided by WHITECYCLE's strategic intelligence reports	7	AXELERA	DEC	PU	M24
D7.4	Summary of Follow-up publications	7	AXELERA	DEC	PU	M48
D7.5	Workshops	7	AXELERA	DEC	PU	M48

Table 3.1d: List of milestones

MS #	Milestone name	WP	Due date	Means of verification
MS0.1	Pre-financing distributed	0	M1	Payments are received by the partners
MS1.1	Separation of 500kg of PET from complex textile clothing	1	M14	500 kg of high PET containing textile material extracted from clothing waste and characterized
MS1.2	Separation of 800kg of PET from composite wastes (ELT, hoses, textiles)	1	M14	800 kg of high PET containing fibrous material extracted from ELT and characterized
MS1.3	Separation of 1 tonne of PET from ELT wastes	1	M36	1 tonne of high PET containing fibrous material extracted from ELT and characterized
MS2.1	Impact of PET waste pollutant on amorphization/purification is characterized and understood	2	M18	Impact assessment report on the pollutants D2.1
MS2.2	The amorphization process is compatible with the enzymatic depolymerization	2	M24	D2.4 report
MS3.1	ELT feedstock: Validation of 75% depolymerisation at lab scale for ELT feedstock	3	M12	Characterization results exhibits at least a 75 % depolymerisation level for ELT
MS3.2	Clothing waste feedstock: Validation of 80% depolymerisation at lab scale for clothing waste feedstock	3	M15	Characterization results exhibits at least a 80 % depolymerisation level for clothing textile waste

MS3.3	Report on the characterization of the various by-products from the different streams	3	M48	Characterization results exhibiting best by-product recycling options along value chain. Analysis of their implementation potential.
MS4.1	Enough information has been gathered to prove feasibility to produce twisted multifilament HMLS yarn with suitable specifications for hoses application at pre-industrial scale line	4	M28	D4.5
MS4.2	Enough information has been gathered to prove feasibility to produce multifilament HT yarn with suitable specifications for hoses application at pre-industrial scale line	4	M30	D4.9
MS4.3	Enough information has been gathered to prove feasibility to produce multifilament HMLS single end cord with suitable specifications for tyre application at pre-industrial scale line	4	M30	D4.10
MS5.2	The physical and chemical properties of r-PET are suitable at pilot scale to launch industrial production of tyres, lay-flat hoses, and garments	5	M33	Characterization results on the physical and chemical properties of r-PET compared to v-PET.
MS5.3	The physical and chemical properties of r-PET are suitable at industrial production scale for tyres, lay-flat hoses, and garments	5	M43	Upscale validation results on the physical and chemical properties of r-PET compared to v-PET. D5.4 D5.5 D5.6
MS7.1	Project workshop held	7	M42	Minutes, documentation, photos

Table 3.1e: Critical risks for implementation

Description of risk (indicate level of (i) likelihood LL, and (ii) severity S: Low/Medium/High)	WP	Contingency & Mitigation measures
Clothing textile waste id. technology not efficient enough to identify PET from multilayered feedstocks due to PET being in intermediate layers **MS 1.1 - LL: Medium / S: High**	1	A combination of back-up optical methods for id. of chemical composition of clothing textile waste feedstock will be tested and implemented in WP1 as contingencies (terahertz etc.)
Processes and technologies for sorting of the PET fraction from ELT not efficient enough to provide high PET content (>80%) **MS 1.2/1.3 - LL: Medium / S: Medium**	1	Extension of the period regarding the piloting activities to find the best process parameters to reach the best purity level – mix with higher PET content waste for subsequent work
Processes and technologies implemented for amorphization of the PET not efficient enough to reach less than 15% crystallinity for the PET **MS 2.1 - LL: Medium / S: Medium**	2	Further preliminary studies will be conducted to find a set of process parameters that enables IFTH to reach <15% crystallinity. Sorting improvement.
Enzymatic depolymerization not efficient enough **MS 3.1 - LL: Medium / S: Medium**	3	Pre-treatment improvement, enzyme optimization, tyre, and textile eco-design to facilitate depolymerisation step
Difficulties in monomer purification to achieve high quality required with an acceptable environmental and economic impact **MS 3.2 - LL: High / S: Medium**	3	Pre-treatment improvement to remove most of impurities before depolymerization, tyre and textile eco-design, dilution of complex waste by adding simpler waste containing more PET to dilute impurities
The contaminants may make polymerisation and high-speed multifilament spinning processability difficult and cause some broken filament formation **MS 4.1 - LL: High / S: Medium**	4	Blending the SSP-ed r-PET flakes with virgin monomer-based PET at different proportions will be the main strategy for developing the uniformity of SSP-ed PET during high-speed melt spinning.
r-PET based tyres may not exhibit good enough performances to pass requirements to commercialization **D 5.3 - LL: Medium / S: High**	5	Adapt design tyre scope or use different mixes of r-PET/v-PET in the pilot tyres to reach the required performances
Overall LCA results for the whole value chain may not exhibit expected CO_2 reductions and/or energy efficiency D6.2 - **LL: Low / S: High**	6	Adapt the overall concept to reduce the CO2 emission by including the long-term impacts of by-product recycling management
Inadequate commitment by partners in the value chain (e.g., brand-owners, waste managers…) D7.2 - **LL: Low / S: High**	7	Discussions to be held with at least 18 companies in the first half of the project. In M42 when the opportunity leaflet, process design kits and LCC/LCA KPIs are confirmed, additional companies will be approached to garner their interest to adopt project results.

Table 3.1f: Summary of staff effort

Partner	WP0	WP1	WP2	WP3	WP4	WP5	WP6	WP7	Total PM
MICHELIN *(incl. MANDALS)*	**24.00**	35.00	36.00	4.00	2.00	**27.00**	9.00	4.00	140.00
CARBIOS	6.00		0.10	**55.00**	0.10		0.20	0.50	61.90
KORDSA					**59.00**		7.00	9.00	77.00
ESTATO	3.00	17.50					3.50	6.00	30.00
SYNERGIES	8.00	36.00					1.00	1.00	46.00
DITF	2.00				31.00		1.00	1.00	35.00
IFTH	2.00		**36.00**	0.50	23.00		1.00	1.00	63.50
AXELERA	4.60							**16.10**	20.70
PPRIME *(incl. UNIV. POITIERS)*	0.45	37.15	3.65				0.25	1.00	42.50
ERASME	1.00						43.00	4.00	48.00
IRIS	3.00	**60.00**					2.00	5.00	70.00
HVL	5.00						**31.00**	2.00	38.00
IPOINT	3.00						27.50	2.00	32.50
DYNERGIE	20.00						3	3	26.00
INDITEX	3.00					5.00	18.00	1.00	30.00
TOTAL	86.05	185.65	75.75	59.50	120.10	45.00	130.45	53.50	761.10

Table 3.1g: 'Subcontracting costs' items

01/02 - MICHELIN/MANDALS SUBCONTRACTORS	Costs (€)	Justification
To be disclosed	€ 63,800.00	Task 1.3: Upscale of the shaping process
To be disclosed	€ 250,000.00	Task 1.5: TRL increase for separation and sorting for ELT + ELH
To be disclosed	€ 218,000.00	Task 2.3: Upscale on PET amorphization from garment + ELT + ELH
TECHNIP ZIMMER	€ 260,000.00	Task 4.4: Polymerization and solid state polycondensation
Total	€ 791,800.00	
03 - CARBIOS SUBCONTRACTORS	**Costs (€)**	**Justification**
ITERG/processium	€ 80,000.00	Task 3.4: MEG purification
VTT	€ 25,000.00	Task 3.4: Enzyme production
To be disclosed	€ 1,200.00	Task 3.4: Salt analysis
FILAB	€ 2,000.00	Task 3.4: Monomer analysis
Total	€ 108,200.00	
04 - KORDSA SUBCONTRACTORS	**Costs (€)**	**Justification**
CHERTEC	€ 15,000.00	Task 4.3: Service procurement for GC-MS testing, number of samples are 10.
CHERTEC	€ 15,000.00	Task 4.3: Service procurement for GPS - MS Test. Number of tests are 10.
ODTÜ: MIDDLE EAST TECHNICAL UNIVERSITY	€ 15,000.00	Task 4.3: Service procurement for NMR Test. Number of tests are 10.
ODTÜ: MIDDLE EAST TECHNICAL UNIVERSITY, SASA; SASA	€ 15,000.00	Task 4.3: Elemental Analysis. Number of analysis are 10.
To be disclosed	€ 1,500.00	Task 4.5: Other potential characterizations (boiling point, flash point, partisol, NMR)
UTA GmbH - RWTH Aachen University	€ 5,000.00	Task 4.5: Aachen Characterization Tests
Hacettepe	€ 15,000.00	Task 4.5: Maldi Toff Tests * 10 samples
Total	€ 81,500.00	

06 - SYNERGIES SUBCONTRACTORS	Costs (€)	Justification
Alpes TLC, Vosges TLC	€ 80,000.00	Task 1.4: Manual sorting trial
RMIS	€ 120,000.00	Task 1.5 : On site process design and implementation
Total	€ 200,000.00	
08 - IFTH SUBCONTRACTORS	**Costs (€)**	**Justification**
Trev Lab	€ 2,000.00	Task 4.5 : analyses IV
Science et Surface	€ 4,000.00	Task 2.4 : XPS
Total	€ 6,000.00	
10 - PPRIME SUBCONTRACTORS	**Costs (€)**	**Justification**
CITF/IDEATEC/SINEX/... SME	€ 10,000.00	Task 1.5 : Adaptation module de labo Neutralisation électrostatique/Tamisage
CITF/IDEATEC/SINEX/... SME	€ 20,000.00	Task 1.5 : Adaptation module de laboratoire Séparation électrostatique
CITF/IDEATEC/SINEX/... SME	€ 10,000.00	Task 1.5 : Adaptation module de laboratoire Purification électrostatique
CITF/IDEATEC/SINEX/... SME	€ 5,000.00	Task 1.5 : Amélioration module de labo Neutralisation électrostatique / Tamisage
CITF/IDEATEC/SINEX/... SME	€ 10,000.00	Task 1.5 : Amélioration module de laboratoire Séparation électrostatique
CITF/IDEATEC/SINEX/... SME	€ 5,000.00	Task 1.5 : Amélioration module de laboratoire Purification électrostatique
Total	€ 60,000.00	
12 - ERASME SUBCONTRACTORS	**Costs (€)**	**Justification**
CERDI	€ 90,470.00	Task 6.4 : social acceptance and social impacts
Total	€ 90,470.00	

Table 3.1h: 'Purchase costs' items (travel and subsistence, equipment and other goods, works and services)'

01/02 - MICHELIN/MANDALS	Costs (€)	Justification
Travel	€ 61,300.00	Travels at subcontractors for trials (€21,300.00), Travels for Dissemination and consortium meetings purposes (€40,000.00)
Equipment	€ -	
Other goods and services	€ 290,500.00	Industial processing slot (tissue) (€10,000.00), Industial processing slot (tire) (€40,000.00), Evaluation (cord) (€10,000.00), Evaluation (tissue) (€30,000.00), Evaluation (tire) (€100,000.00), Recognition tools evaluation (€42,500.00), Process upscale (€32,000.00), IS tool (€20,000.00), Audit costs (€6,000.00), Phase1: Production of 150m test weave to be extruded and tested (€2,200.00), Phase 2: Production of 1500m of test lengths in various hose sizes (€27,000.00), Final stage: Testing regularity/ reproduceability - 2x200m hoses (€4,800.00)
Total	€ 351,800.00	
03 - CARBIOS	**Costs (€)**	**Justification**
Travel	€ 14,000.00	1 conference per year during 4 years (WP7 AMA,MST) (€4,000.00), GA meetings (€10,000.00)
Equipment	€ 149,960.43	6 reactors 500 mL (immo 00360000) (€5,470.67), 2 reactors 5L (immo 00361000) (€1,939.67), Microplate spectrophotometer (immo 00366000) (€985.00), Mettler Toledo Dsc3 (immo 00338000) (€1,266.67), Sampler Mettler (immo 00699000) (€690.00), 2 HPLC Thermo dionex (immo 00353000) (€2,196.10), Rotavapor (immo 00345000) (€257.09), Pilot equipments (€17,548.10), Pilot equipments (lease-back financing) (€72,023.81), Demo Plant (€43,125.00), Cristallization (€4,458.33)
Other goods and services	€ 46,000.00	Pilote (€40,000.00), Audit cost (€6,000.00)
Total	€ 209,960.43	
04 - KORDSA	**Costs (€)**	**Justification**
Travel	€ 58,500.00	Project meetings every 6 months (5 trip / 3 person) (€22,500.00), Partners will be visited for superfisition, technical discussion and new recipes (DITF, IFTH - 4 trip / 2 person) (€12,000.00), 3 Conference (3 person) / Dissemination (€12,000.00), 3 Conference (3 person) / Dissemination (€12,000.00)
Equipment	€ 105,000.00	MS, head-space and component library for GC (€11,666.67), XRF (€35,000.00), Pre-industrial scale production line revision - improvement (Extruder spinpack) (€58,333.33)
Other goods and services	€ 189,000.00	Solvent, chemicals, catalysts, coating agents, paper tubes etc. (€80,000.00), Calibration costs for Kordsa's current analysis tools for polymer and tire cord (€20,000.00), Pre-Industrial scale optimization trials with regular petrosourced PET (€74,000.00), IPR Cost: 2 FTO Analysis (Başta kabaca analiz sonda analiz) (€6,000.00), Patent (Patent pre-research) (€2,000.00), 2 Patent (Patent writing and archive) (€2,000.00), Audit (€5,000.00)
Total	€ 352,500.00	
05 - ESTATO	**Costs (€)**	**Justification**
Travel	€ 16,000.00	participation project GA / meetings (including travel and accomodation for 5 trips with 2000 € each) (€9,000.00), attending conferences & fairs (based on identification at dissemination section, 2.000 € per trip) (€7,000.00)
Equipment	€ -	
Other goods and services	€ 19,000.00	Sample production consumables (mainly energy, working materials) (€10,000.00), Audit costs (CFS - certificate on financial statement) (€5,500.00), Sampling costs and shipment / logistics (€3,500.00)
Total	€ 35,000.00	
06 - SYNERGIES	**Costs (€)**	**Justification**
Travel	€ 45,000.00	WP meeting and experimentations (8 trips * 2 persons) (€24,000.00), Project meetings (5 trips * 2 persons) (€15,000.00), Dissemination travels to fairs and conferences (€6,000.00)
Equipment	€ 84,000.00	Part of standard sorting equipment (€60,000.00), Conveyors (€24,000.00)
Other goods and services	€ 5,000.00	Financial audit (€5,000.00)
Total	€ 134,000.00	

07 - DITF	Costs (€)	Justification
Travel	€ 22,000.00	Project Meetings (1-2 persons x 5 times) (€12,000.00), Attending of conferences or fairs (1-2 person; 1 each year) (€10,000.00)
Equipment	€ -	
Other goods and services	€ 35,000.00	Chemicals (€10,000.00), Consumables and supplies (€10,000.00), IP costs (€10,000.00), Audit costs (€5,000.00)
Total	€ 57,000.00	
08 - IFTH	**Costs (€)**	**Justification**
Travel	€ 20,000.00	12M Meeting (€15,000.00), Congress (€5,000.00)
Equipment	€ 101,200.00	Fibers agglomerator (€4,500.00), bobinoir auto (€19,200.00), Microscope (€40,000.00), hydroliage (€7,500.00), rheotester (€30,000.00)
Other goods and services	€ 20,000.00	spinning and nonwovens consummables (Filter, sealing washers, bobbins...) (€10,000.00), commercial polymer,fibres and nonwovens with pure PET (€5,000.00), AUDIT (€5,000.00)
Total	€ 141,200.00	
09 - AXELERA	**Costs (€)**	**Justification**
Travel	€ 12,000.00	Travel (€12,000.00)
Equipment	€ -	
Other goods and services	€ 92,280.00	Workshop (€30,000.00), Visual Identity (€6,360.00), Website (€24,720.00), 2 videos (€7,200.00), Minor services with Axelera's partner clusters across Europe (€24,000.00)
Total	€ 104,280.00	
10 - PPRIME	**Costs (€)**	**Justification**
Travel	€ 28,500.00	IEEE/IAS (€6,000.00), Joint Conference on Electrostatics (€6,000.00), IEEE/IAS (€6,000.00), Project General Assembly (GA) meetings (€10,500.00)
Equipment	€ -	
Other goods and services	€ 6,600.00	Vis plastiques (€300.00), Plaques PMMA, PVC et Teflon (€600.00), Grilles de tamis (€700.00), Audit (€5,000.00)
Total	€ 35,100.00	
13 - IRIS	**Costs (€)**	**Justification**
Travel	€ 40,500.00	Project General Assembly (GA) meetings (€22,500.00), 4 Technical meetings (€12,000.00), 3 Dissemintation activities (€6,000.00)
Equipment	€ -	
Other goods and services	€ 270,600.00	Audit Cost (€3,600.00), Patents and IPR assessment (€8,000.00), Conference registration and publications fees (€6,000.00), Shipment and insurance costs for the prototype (€8,000.00), Materials for building of monitoring system prototype: e.g. cameras HSI-based and LIBs-based, optomechanical parts for sampling systems and others, IT Servers for hosting and deployment digital system, screen, Process Unit Laptop for Data Management and other components (€195,000.00), Sample characterization analysis (€30,000.00), Certification (eg. ATEX, CE) (€20,000.00)
Total	€ 311,100.00	
15 - I-POINT	**Costs (€)**	**Justification**
Travel	€ 48,000.00	Conferences and fairs travels (€20,000.00), Consortium meetings (€28,000.00)
Equipment	€ -	
Other goods and services	€ 14,400.00	Audit costs (€5,000.00), Software and cloud storage capacity (€9,400.00)
Total	€ 62,400.00	
16 - DYNERGIE	**Costs (€)**	**Justification**
Travel	€ 24,000.00	Steering Committees (8 meetings - 2 persons) (€20,000.00), Attendance to conferences (2 events - 1 person) (€4,000.00)
Equipment	€ -	
Other goods and services	€ 5,000.00	Audit (€5,000.00)
Total	€ 29,000.00	

Table 3.1i: 'Other costs categories' items (e.g., internally invoiced goods and services) – not applicable

Table 3.1j: 'In-kind contributions' provided by third parties – not applicable

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